

공학박사 학위논문

Improving On-site Reservoir Operation Policies by Robust and Adaptive Optimization Models

강건 · 적응형 최적화 모형을 통한
실제 저수지 운영 정책의 개선

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서울대학교 대학원

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Abstract

Improving On-site Reservoir Operation Policies by Robust and Adaptive Optimization Models

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Effective and sustainable water resources management under non-stationarity and deep uncertainty of climate change is hardly achievable with the traditional water resources management techniques. Thus, various novel decision making approaches for successful water resources management utilizing optimization algorithms and the status quo policies have been developed and applied in climate change adaptation in water resources. In this context, this dissertation aims to present adaptive and robust reservoir operation schemes applicable to the Boryeong multipurpose reservoir, located in South Korea.

Primarily, a comparative analysis including multiple optimization models from the two families of prominent optimization algorithms, dynamic programming (DP) and

evolutionary multi-objective direct policy search (EMODPS), was composed. The DP models were provided with a different type of forecast information, and the shape and the parameters of the approximation function of the EMODPS models were differentiated. One of the optimization models was then coupled with the onsite hedging rule in order to enhance the practicality of the solutions. The onsite hedging rule was then reformed by reflecting the recent multi-year drought of the target basin with long memory time series. The alternative policies were then utilized adaptively and continuously updated with novel information. Finally, the concept of both feasibility and solution robustness was incorporated into a stochastic DP model. The robust policies were evaluated under a wider range of inflow series which were synthetically generated to reflect the deep uncertainty of climate change. The performances of the models were quantified in terms of both traditional and regret-based robustness metrics.

Among the optimization models, the EMODPS model with the Gaussian shape of approximating function performed the best despite the fact that it had the least number of parameters to be optimized. Moreover, both coupling the onsite hedging rule with the optimally derived policy and the inclusion of long memory during the hedging rule generation process both yielded favorable outcomes in terms of storage at the expense of frequent demand deficits. Especially, the coupling of the hedging rule with the EMODPS-Gaussian model reduced the maximum duration of failure by 45.59%. Adaptive reservoir operations with alternative hedging rules also reduced the frequency of critical failures in terms of demand by 17.7%. Finally, the robust policy which included the concept of solution robustness outperformed both the conventional optimization model and the robust policy only with feasibility robustness when measured with respect to regret-based robustness metrics. Ultimately, since the

adaptive and robust reservoir operation schemes presented in this dissertation advances the results from typical decision making techniques in water resources exceptionally under extreme climate conditions, a shift towards novel decision makings under climate change holds promise in the future.

Keywords: optimization, reservoir operation, on-site hedging rule, adaptive operation, robust optimization, climate change

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Chapter 1. Introduction

Planning and management of water resources to ensure the safety of the public have been considered to be a longtime goal for mankind since the beginning of civilization. Established in 2015, the United Nation's 17 Sustainable Development Goals, which include the key interlinked goals for a better and sustainable Earth's future, also aims to provide universal access to clean water and sanitation. Consequently, today's water resources systems including reservoirs, pipelines, pumping stations, and wastewater treatment plants are still advancing towards achieving the ultimate objective: to assure reliable freshwater sources in terms of both quantity and quality to all nearby consumers. Nevertheless, because the boundary conditions and system characteristics in water resources problems can never be accurately defined, solving the aforementioned objective still dwell as one of the most challenging aspirations humanity is currently facing despite the vast technological advances. Multiple initiatives to evaluate the status of the field of water resources and to compile scientific agendas regarding the subject took place at international levels (Montanari et al. 2013; Thompson et al. 2013). Recently, innumerable complications regarding hydrology and water resources have been synthesized into the 23 problems which were then classified into the following categorizations: time variability and change, space variability and scaling, the variability of extremes, interfaces in hydrology, measurements and data, modeling methods, and interfaces with the society (Blöschl et al. 2019). This dissertation aims to tackle few of the 23 problems in water resources by focusing on the most urgent issues the current human society is facing.

1.1 Current Challenges in Water Resources Systems Management

Among countless candidates, a few elements have been narrowed down which contribute the most to the extremely challenging nature of water resources management problems. These include competition among conflicting stakeholder groups, the regional disparity in available freshwater sources, increasing globalization, the unexpected change in supply and demand, and climate change-induced water stress (Cosgrove and Loucks 2015). Moreover, the inherently complex, black-box nature of water resources systems, in which the sequential procedures during the analysis ultimately leading to the final decision are vague and hardly revealed (Quinn et al. 2019) ultimately hinders even the experts with a higher level of knowledge from clearly explaining the results from the analysis. These attributes of the problem consequently lead to limited uptake of the optimal set of decisions by the decision makers onsite, the key stakeholder group responsible for actual operations (Pianosi et al. 2020). Finally, climate change, which can be characterized by non-stationarity (Hurst 1951; Milly et al. 2008) and deep (or Knightian) uncertainty (Lempert 2014; Dittrich et al. 2016) of hydro-climatic conditions, has forced the shift of paradigm in traditional decision making techniques regarding water resources.

In this dissertation, the water resources infrastructure optimization problem is tackled by focusing on a few of the foregoing factors: the innate complexity of water resources management problems, aversion of solutions from the optimization models by the practitioners onsite, and a notable increase of hydro-climate irregularities primarily induced from the recent change in climate conditions. Especially, uncertain

events induced from climate change led to the intensification of unexpected hydro-climate disasters across the globe which eventually may lead to lethal failures. Thus, this dissertation aims to reduce the risk of water resources infrastructures during the decision making process by providing means for utilizing contemporary decision making techniques which were designed with advanced technologies. In other words, this dissertation endeavors to contribute both to the field of academia and also in practice by linking the theoretically optimal solutions with the actual policies that are currently being implemented in onsite decision making in order to confront the challenges in the field of water resources systems management.

1.1.1 Intricacy in Water Resources Systems Analysis

In general, water resources systems analysis (WRSA) refers to the immense field of study that solves the complex nature of water resources-related problems (Brown et al. 2015). Whereas the previous WRSA primarily focused on developing optimal planning of water resources infrastructures (Maass et al. 1962; Cohon and Marks 1975), the present-day WRSA has been expanded to include broader themes related to water resources problems: coupled human-natural interactions (Liu et al. 2007; Quinn et al. 2017a,b), hydro-economic modeling (Harou et al. 2009), game theory (Madani 2010), and advances in optimization algorithms (Castelletti et al. 2010; Giuliani and Castelletti 2016). Although the inclusion of these subjects to the traditional WRSA problems has compelled a higher level of modeling and computing techniques of related experts, a more comprehensive inspection of the system needs to be conducted. In this dissertation, optimization models of various kinds are to be utilized for the analysis of complex water resources systems.

1.1.2 Limited Collaboration with Stakeholders

Despite the extensive technological advancements especially regarding the optimization techniques adopted in WRSA, they are infrequently implemented in actual decision making. Few reasons behind the discrepancies not only include the aforementioned black-box nature of the original problem but also the lack of expert knowledge among related stakeholders (Quinn et al. 2019) contribute to the disparity indeed. Given the paramount importance of collaborative decision making techniques in WRSA (Palmer et al. 2013; Langsdale et al. 2013), closing the gap between the theoretically optimal solutions and the implementable policies which the practitioners may favor is considered as another key research topic to be explored. Thus, this dissertation aims to improve on-site water resources infrastructure policies by collaborating with related stakeholders.

1.1.3 Climate Change-Induced Stresses Affecting WRSA

Climate change and its consequences on Earth's precipitation and temperature have been constantly addressed by multiple studies regarding water resources systems management (Němec and Schaake 1982; Lettenmaier et al. 1999; Christensen et al. 2004). In this dissertation, climate change refers to the notable deviation from the recorded hydro-climatic statistics either induced from natural climatological variability or inclusive anthropogenic activities (Kolokytha et al. 2017). The two representative characteristics of climate change, non-stationarity, in which the historical statistical properties are not kept constant (Razavi et al. 2015), and deep (Knightian) uncertainty, in which even the probabilistic distribution of future states of the world is unknown, are focused

in this dissertation. Consequently, alternative decision making techniques have emerged in the field of WRSA in order to either mitigate or adapt to the adverse impacts caused by climate change. In this dissertation, adaptive and robust optimization models are introduced to combat climate change impacts.

1.2 Dissertation Objectives

This dissertation contributes to the array of scientific literature regarding the advancement of the WRSA techniques using optimization models in twofold: by providing alternatives that are applicable in practice and by considering the novel decision making approaches chiefly induced from climate change. Moreover, since the main problem formulation procedures fundamentally follow those from traditional optimization problems in WRSA, a thorough overview over both traditional and contemporary optimization algorithms is also conducted in this dissertation.

In order to achieve the main objective of the dissertation, it is partitioned and specified into the following specific objectives, which are conducted throughout the dissertation:

- (a) To generate theoretical optimal solutions to complex water resources systems problems using diverse optimization algorithms, and to investigate how different algorithms or stochasticity of the conditions affect the quality of generated solutions,
- (b) To narrow the gap between the theoretical solutions (optimization) and the actual operations (onsite), so that even the practitioners with relatively low expertise can understand and utilize them,
- (c) To utilize adaptive and robust decision making techniques, by mainly considering the effects of climate change, into planning and management procedures of water resources infrastructures.

1.3 Scope and Organization of the Dissertation

This dissertation is composed of seven chapters. In Chapter 2, the array of literature related to (1) prominent optimization tools used to explore water resources systems, (2) decision making techniques generally adopted in the field of WRSA, and (3) the shifted decision making theories due to the recent surge in extreme climate events, mainly due to climate change, from conventional decision making strategies are introduced. In Chapter 3, the target basin and the reservoir is briefly introduced along with the overview of the three discrete yet interconnected studies of the target basin. In Chapter 4, results from the comparative study among different optimization models and efforts to couple them with onsite hedging rules are introduced. Chapter 5 aims to reform current drought response measures in reservoirs based on the zone-based hedging rules with adaptive operations by considering the recent multi-year drought. Then, Chapter 6 couples the robust approach with the optimization model commonly adapted during reservoir operation problems as a measure to adapt to climate change and provides a robust alternative in reservoir operation problems. Finally in Chapter 7, a comprehensive conclusion of the dissertation and the topics for potential future studies are introduced.

The main body chapters of this dissertation are interconnected with each other in the following manners. Chapter 4 first makes use of diverse theoretical optimization algorithms in deriving the solutions in a reservoir operation problem, and also tests the applicability of the policy that is being adopted in the the practical world. Then in Chapter 5, alternative methods that reform the hedging rule with adaptive measures are presented with the intent to cope with climate change-induced multi-year drought. The key methodology presented in this chapter is the adaptive measures, which is one

of the two pillars this dissertation presents as means to cope with climate change. Finally, in Chapter 6, the theoretical optimization algorithm is paired with the concept of robustness, which is suggested as the other pillar coping with climate change, and the proposed concept is evaluated under a wider range of future uncertain inflow scenarios. Collectively, this dissertation aims to contribute to the literature with adaptive and robust reservoir operation schemes based on optimization models in the era of climate change, which is characterized by non-stationarity and Knightian uncertainty.

As a comprehensive summary, Table 1.1 provides a brief overview of the main themes, methodological approaches, and the utilized tools corresponding to the specific studies of this dissertation. Shortly, dynamic programming and evolutionary multi-objective direct policy search models are the theoretical optimization algorithms utilized in Chapter 4 for a comparative analysis. The stochastic versions of the dynamic programming model were also adopted in Chapter 6, which were coupled with robust optimization in order to derive robust alternatives. In Chapter 5, autoregressive fractionally integrated moving average model was used to derive a set of inflow series with long memory. Finally, k-nearest neighbor disaggregation method was employed to the derived inflow series in order to divide annual inflow series into shorter time steps in Chapter 5.

Table 1.1: Specific themes, methodologies, and utilized tools adapted in the bodies of this dissertation

Ch.	Main Themes	Methodological Approaches	Utilized Tools
4	• Theoretical optimization • onsite policy	• Optimization • Cross validation	• Dynamic programming (DP) • Evolutionary multi-objective direct policy search (EMODPS) • K-fold cross validation
5	• onsite policy • Climate change	• Time series generation & disaggregation • Adaptive operation	• Autoregressive fractionally integrated moving average model (ARFIMA) • K-nearest neighbor (k-NN) disaggregation
6	• Theoretical optimization • Climate change	• Optimization • Robust optimization (RO)	• Stochastic dynamic programming (SDP) • Feasibility & solution robustness • Robustness metrics

Chapter 2. Literature Review

Given the excessive history regarding the field of WRSA and related decision making theories, keen scrutinization of the entire theme is exceedingly demanding. Furthermore, although studies regarding climate change has a relatively shorter records than the other two topics, extensive researches regarding climate change also have been conducted especially in the last 30 years. Alternately, this chapter aims to focus on a few representative topics belonging to the main themes by narrowing down the scope on the few concepts that are primarily utilized throughout this dissertation.

The rest of this chapter is organized as follows: section 2.1 introduces the optimization models that have been frequently adopted in the problems of water resources systems focusing on the two concepts of optimality: Bellman and Pareto optimality; section 2.2 classifies the conventional decision making schemes and introduces their advancement towards contemporary concept of collaborative decision making. Finally, section 2.3 briefly highlights how climate change has brought the inevitable shift from traditional decision making theories into novel concepts, mainly by focusing on the concept of adaptive and robust decision making strategies.

2.1 Recent Trends in Water Resources Systems Optimization

According to the general definition, water resources system refers to the collective structure of hydrologic, infrastructure, ecologic, and anthropogenic factors that involves water (Brown et al. 2015). Especially, since the level of complexity inherited inside water resources systems is far more intricate than the results from the analytical techniques developed so far, approximated models are often utilized to explore the system in general. Moreover, the complete inclusion of the collective relationship between physical, biochemical, ecological, social, legal, and political processes (Loucks and Van Beek 2017) flawlessly inside a collective model is extremely challenging. Consequently, multiple studies in the past have focused on the development of realistic water resources models by making few inevitable simplifications of the real system (Belaineh et al. 1999; Simonovic 2002; Razavi et al. 2012).

Among the various classifications of the models, this section categorize the utilized models into the fundamental basis of the models: either optimization-based or simulation-based. In brief, simulation-based models may hold promise under the circumstances in which the experts are able to make decisions based on their intuition, experience, and the level of understanding of the system (Eum 2007). The use of the simulation-based models may result in better consequences under situations in which rapid decision making is mandatory, such as an outbreak of extreme flood. However, this dissertation aims to focus on introducing the intrinsic concepts of traditional optimization-based WRSA models and their recent advancements in accordance with the technological advances, given the excessive studies and research regarding the optimization models adapted in WRSA in the rest of the section. Among the mani-

fold of optimization models, this dissertation categorized the following model-based optimization approaches, linear and nonlinear programming (LP and NLP), dynamic programming (DP), genetic programming (GP), particle swarm optimization (PSO), and multi-objective evolutionary algorithms (MOEAs), into two criteria. Traditional optimization models include LP, NLP, and DP; the latter part of this subsection focus on the programming models based on evolutionary algorithms (EAs) including GP, PSO, and MOEAs.

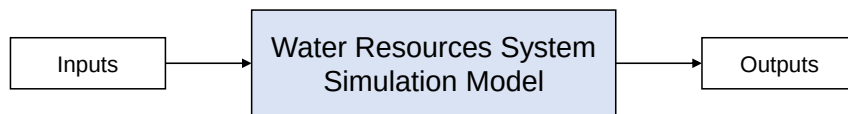
2.1.1 Advances in the Traditional Optimization Models

Development of the optimization models, in which the main objective is to search for the optimal values for the set of parameters (decision variables) that maximizes/minimizes the prespecified objective function, has been extensively developed and advanced in WRSA studies. Although few simulation-based models, in which the primary goal is to downscale the actual water resources system with reasonable assumptions and simplifications relying on trial and error processes, have been utilized for WRSA, theoretical developments concentrated on advancing the various optimization models. The structures of typical WRSA problems are thoroughly related and can be transformed to those of mathematical programming models with ease, backing up the reasons behind extensive research that focused on the advancement of the mathematical programming models.

As in Figure 2.1, the notable difference between simulation models and optimization models is in the inclusion of the objective function, constraints, state variables, and decision variables during the calculation. The ideal outcome of running an optimization model is the set (or sets) of optimal decision variables corresponding to the prespecified

definition of optimality stipulated prior to the iterative calculation process. Among the optimization models, this subsection aims to introduce the constrained optimizations, or mathematical programming, approaches which involve finding the optimal values of decision variables given specified relationships that have to be satisfied (constraints) (Loucks and Van Beek 2017).

(a) Simulation



(b) Optimization

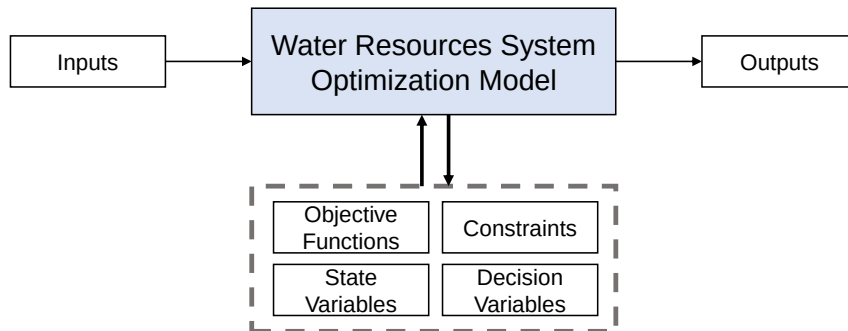


Figure 2.1: Disparities between (a) simulation-based and (b) optimization-based models in WRSA, redrawn from Loucks and Van Beek (2017)

Linear Programming

Due to its wide applicability and the extensive development of solving technologies, the LP model and its variabilities have been broadly adopted in deriving optimal solutions in water resources infrastructure operation problems (Loucks and Dorfman 1975; ReVelle and Gundelach 1975; Houck et al. 1980). LP is applicable in problems where both the objective functions and the constraints are exclusively provided in linearly shaped functions. For instance, the structure of an LP model with objective function $F(\mathbf{X})$, decision variable vector $\mathbf{X} = (x_1, x_2, \dots, x_j)$, and constraints $g_i(\mathbf{X})$ can be expressed into Eq. 2.1 and Eq. 2.2 (Loucks and Van Beek 2017):

$$\text{Maximize or minimize } F(\mathbf{X}) = \sum_j P_j x_j \quad (2.1)$$

subject to

$$g_i(\mathbf{X}) = \sum_j a_{ij} x_j \leq \text{or } \geq b_i \quad \text{for } i = 1, 2, 3, \dots, m \quad (2.2a)$$

$$x_j \geq 0 \quad \text{for } j = 1, 2, 3, \dots, n \quad (2.2b)$$

where P_j = prespecified weighting factor for linear combination of decision variables x_j ; a_{ij} and b_i = constants, respectively.

LP models have been frequently adapted in solving multiple WRSA problems, since the tools for solving the LP models are prevalently provided in many computer-based software. Moreover, since expressing the system states by just utilizing linear shaped function was feasible, development of LP models in WRSA problems were conducted. In theory, any type of WRSA problem that can be expressed in the general structure of

an LP model can consequently be solved using the array of computer based LP-solvers.

Recent advancements regarding LP in WRSA include: expansion to multi-objective model with interval parameters with interactive compromising algorithms (Han et al. 2011), coupling with the fuzzy concept in the problem formulation process (Jairaj and Vedula 2000), and expanding the model to include system uncertainties (Miao et al. 2014). Despite the distinctive shortcoming of the LP models which can only handle linear functions, applications and advancements regarding them still hold multiple theoretical values and can be applied to providing the primary set of candidate solutions in many WRSA problems.

Non-Linear Programming

On the contrary, NLP model is an extension from the LP model, where the shapes of the objective function and the constraints are not specifically limited to linear shapes. Instead, the NLP model is able to capture a wider range of problem formulation, regardless of the shapes of the function constituting the objective function and the constraints. For illustration, similarly from the previous example, an NLP model with the decision variable vector $\mathbf{X}=(x_1, x_2, \dots x_j)$ can be formulated utilizing the following equations:

$$\text{Maximize or minimize } F(\mathbf{X}) \quad (2.3)$$

subject to

$$G_i(\mathbf{X}) \leq \text{or} \geq b_i \quad \text{for } i = 1, 2, 3, \dots, m \quad (2.4a)$$

$$x_j \geq 0 \quad \text{for } j = 1, 2, 3, \dots, n \quad (2.4b)$$

where $F(\mathbf{X})$ and $G_i(\mathbf{X})$ = given functions (objective function and constraints) regarding the j decision variables, not strictly limited in their shapes.

Although the solving techniques for the NLP models are not widely spread as those from the LP models, few representative techniques, such as the use of calculus, the hill climbing methods, and the Lagrangian multiplier method, are utilized in deriving solutions of the NLP models (Loucks and Van Beek 2017). Alongside with the advancement of the LP models, the NLP models have also incorporated the fuzzy multi-objective natures of the problem (Li et al. 2017a), stochasticity (Higgins et al. 2008), and the interval concepts (Nikoo et al. 2013) to yield the appropriate set of solutions of the target problem.

Dynamic Programming

Before the introduction of specific formulations regarding the dynamic programming (DP) models, the underlying concept behind DP, Bellman's principle of optimality, (Bellman 1957) is to be explained. Bellman's principle of optimality is the concept in which the all models belonging to the DP family are based on. According to Bellman (1957), "an optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision." In other words, given the current state

and stage, the optimal policy that is to be determined is independent from the previous set of decisions that are already determined. This principle can be implemented into the backward DP model, commonly adopted in WRSA problems, in which the goal is to find the set of optimal solutions that minimize the objective function as Eq. 2.5 (Figure 2.2):

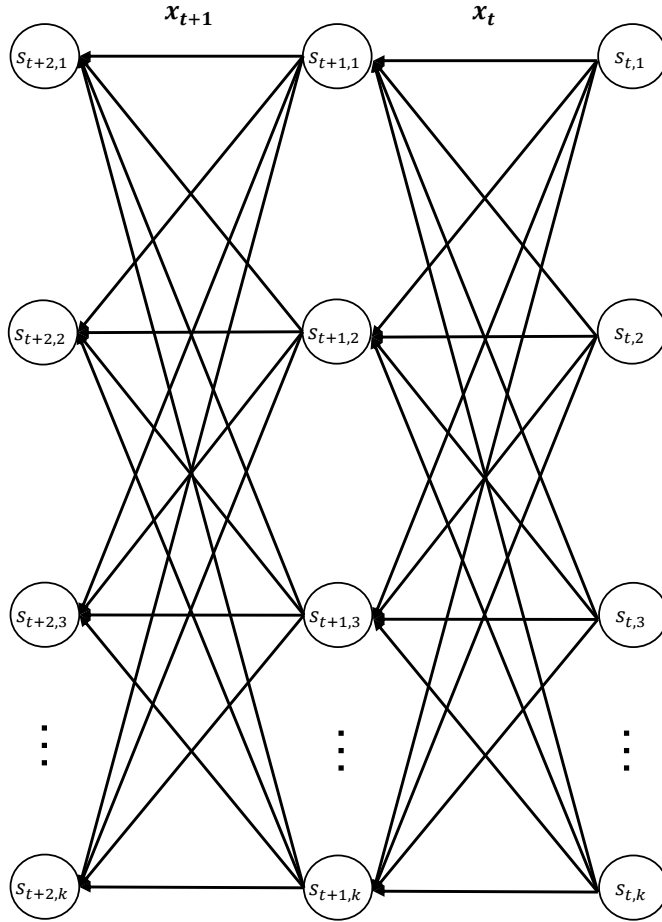


Figure 2.2: An illustration of the concept of Bellman optimality adopted in a backward-moving DP model. Each value of the discretized state variable is calculated as the cumulative value of the cost function calculated until the present time period

$$f_t^*(s_t) = \min_{u_t^*} [E\{Z_t(s_t, x_t) + f_{t+1}^*(s_{t+1})\}] \quad (2.5)$$

where t = time index, $t = 1, 2, \dots, T$; $f_t^*(\cdot)$ = the optimal value of the cumulative cost function until the time period t ; s_t = the state variable at stage t ; u_t^* = the value of the optimal decision variable at stage t ; $Z_t(\cdot)$ = the value of the single cost function at stage t ; x_t = the decision variable at stage t .

Unlike LP, in which the shapes of the functions constituting the model are forced to be linear, the applicability of DP is not bounded to problems composed of only linear functions. Moreover, due to its relatively wide applicability in general WRSA problems (especially in reservoir operation optimization problems), not only DP (Yakowitz 1982), but also the models belonging to the DP family, such as the stochastic DP (SDP) (Stedinger et al. 1984; Kim et al. 2007; Eum and Kim 2010), the gradient DP (GDP) (Foufoula-Georgiou and Kitanidis 1988), and discrete differential dynamic programming (DDDP) (Heidari et al. 1971) have gained popularity. Nevertheless, since DP also requires a time-separable, commensurated cost function ($Z_t(\cdot)$), alternative optimization algorithms in which the foci are in discovering the relationship between conflicting objectives have emerged.

The summary of literature regarding the traditional optimization models, including the review papers by Yeh (1985) and Labadie (2004), with the main research question and the key findings is organized as Table 2.1.

Table 2.1: Literature summary of the traditional optimization models utilized in WRSA

Authors (Year)	Optimization Tool	Target Problem (Main Research Question)	Key Findings (Conclusions)
Yeh (1985)	LP, NLP, DP models & real time operations	Reservoir operation problem (review paper on optimization schemes)	Active incorporation of optimization models in actual operations
Labadie (2004)	LP, NLP, DP, Stochastic models, Heuristic programming models	Reservoir operation problem (state-of-the-art review paper)	Suggests the inclusion of heuristic programming methods
Stedinger et al. (1984)	Stochastic DP (SDP)	Use of best inflow forecast as the hydrologic state variable & Application to Nile River Basin	Suggests the use of best inflow forecast instead of proceeding period's inflow
Tejada-Guibert et al. (1995)	Stochastic DP (SDP)	How the differently selected hydrologic state variables affect the performance of reservoir operation problem in Shasta-Trinity system?	Depending on the type of the benefit function, the value of hydrologic state variables differs
Kim & Palmer (1997)	Stochastic DP (SDP), Bayesian-SDP (BSDP)	How BSDP and alternate SDP models affect maximization problem from a hydropower system with different seasonal forecasts?	The inclusion of seasonal forecasts into the BSDP model leads to better performance
Kim et al. (2007)	Sampling Stochastic DP (SSDP)	Inclusion of ensemble streamflow prediction (ESP) with SSDP in reservoir operation problem during the drawdown period	Explicit inclusion of inflow uncertainty and updating the system with ESP forecasts are beneficial

2.1.2 Optimization Models based on Evolutionary Algorithms

The mathematical programming models including LP, NLP, and DP models are built from the underlying assumptions that the specific details regarding the target system are able to be expressed into the model formulations. This approach in WRSA was once considered as the mainstream of the research before the active utilization of EAs. Models based on the EAs, classified as data-driven models, take the observed data more into consideration and derive an optimal set of solutions with simulation-based models. These heuristic, data-driven procedures include various types of EAs during their search procedures, such as genetic algorithms (GA), GP, and MOEA that focus on capturing the multi-objective nature of the problem. This subsection aims to introduce three of the most frequently adapted models and concepts in WRSA that are based on evolutionary algorithms: GP, PSO, and MOEA.

Genetic Programming

GP, which can be classified as one of the machine-learning techniques, is an expanded concept that differs from GA. Whereas GA only improves the values of the parameters, GP includes the structure of the model also to evolve towards the better objective values than those from the previous generation (Savic et al. 1999). Figure 2.3 represents a sample application of GP drawn in a parse tree.

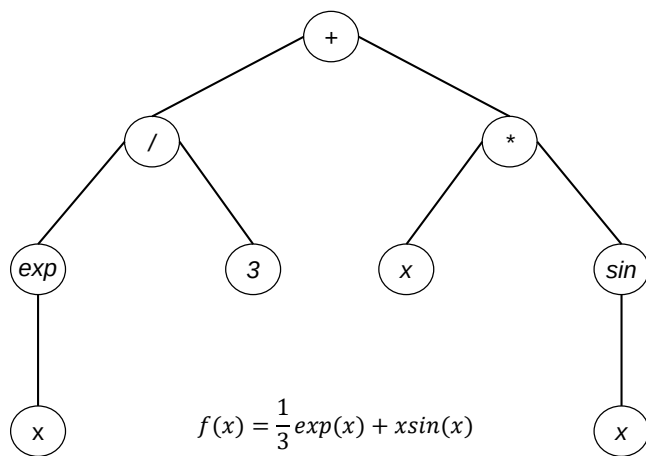


Figure 2.3: An example of GP mathematical expressions, redrawn from Savic et al. (1999)

With a predefined set composed of the combinations of arithmetic operators (e.g., +, −, ×, /), Boolean operators (e.g., and, or), and mathematical functional operators (e.g., sin, cos), GP searches for the best possible combination of operators that maximizes or minimizes the objective function. Searching procedures utilized in GP are primarily based on those from natural selection: crossover and mutation. The specific computing procedures adapted in the application of the GP model differs on the type of the EA selected during the iterative calculations.

GP, first introduced by Cramer (1985), have been adapted by multiple WRSA problem solving procedures, primarily owing to its ability to capture the complexity present in most of the actual WRSA problems. Few cases in which the GP models are incorporated include the following: application into real-time decision making during reservoir operations (Fallah-Mehdipour et al. 2012), rainfall-runoff modelling (Savic et al. 1999), and assessing the impacts from climate change with a multi-objective model (Ashofteh et al. 2015).

Particle Swarm Optimization

PSO is also classified as a type of approach that utilizes the evolutionary algorithms during its search procedures. Since the first introduction of the algorithm by Eberhart and Kennedy (1995), PSO has been selected in solving multiple WRSA problems due to its advantage over GP especially regarding the computing time. Moreover, due to its nature, solution searching procedures using the PSO algorithm is especially effective in problems that are both non-linear and non-convex. Since the algorithm does not heavily depend on the initial population, the risk for convergence to the local optima is relatively lower than the other substitutes. Ultimately, the PSO has been proved to be effective in

finding the global optimum from previous studies (Parsopoulos and Vrahatis 2002).

PSO, in which the fundamental concept of the algorithm is inspired from the social behavior of bird flocking, is classified as a population-based heuristic search technique (Eberhart and Kennedy 1995; Kumar and Reddy 2007). Similarly to any of the procedures from the GA, PSO also starts with a initial population of random candidate solutions. However, instead of relying on the concept of natural selection, each candidate solution in the PSO exchanges information to improve its performance. Therefore, throughout the search process, each individual particle advances towards the current best point occupied by the other member in the set of particles. Ideally, after the iterative search process, all individual particles will have advanced to the global optimum.

Since the first introduction of PSO, PSO has been adapted in multiple WRSA problems due to their efficiency and efficacy. Few of the applications of the PSO into the WRSA problems include: multipurpose reservoir operation problem by the inclusion of elitist-mutation mechanism (Kumar and Reddy 2007), expansion to include multi-objective nature of WRSA (Baltar and Fontane 2008), and a comparative research with multiple conventional EAs (Jahandideh-Tehrani et al. 2020). The results of these studies highlight the aforementioned strengths of the PSO: rapid computational time, diversity of the solutions, and accuracy of the converged solutions.

Many-Objective Evolutionary Algorithms

Finally, MOEA is a decision support tool in which its primary goal is in the explicit representation of the trade-off between multiple objectives (Coello Coello et al. 2007). Contrasting from the DP models based on Bellman optimality, MOEAs select the

concept of Pareto optimality in order to emphasize the multi-objective nature of the target problem. Pareto optimality, or Pareto efficiency, is a concept named after a prominent economist, Vilfredo Pareto, and is widely adopted across multiple disciplines including engineering, economics, and social sciences. By its definition, Pareto optimal (or "nondominated") solutions are not dominated in every aspect to all other feasible solutions (Woodruff et al. 2013). In other words, in a multiobjective problem, improving an objective without degrading any of the other objectives in a Pareto optimal solution is impossible. This specific concept of optimality is especially powerful in problems with conflicting multiple objectives where the alternatives are selected later in the decision making process after the generation of possible solution candidates (Coello Coello et al. 2007).

For illustration, consider a maximization problem in which the objective is to maximize the two objectives: f_1 and f_2 (Figure 2.4). Utilizing the optimization algorithms based on the Pareto optimality ideally will yield the set of Pareto optimal solutions to converge to the Pareto Front, which is the collective set of all Pareto efficient solutions. During the calculation processes, dominated alternatives, such as C in Figure 2.4, eventually progress towards to the Pareto Front. Therefore, derivation of the Pareto front by improving the dominated alternatives efficiently is the key aspect in deciding the adequateness of the optimization algorithms built on the concept of Pareto optimality.

Mathematically, the concept of Pareto optimality can be expressed by comparatively defining a Pareto optimal solution (X) to a dominated (not Pareto optimal) solution Y in a optimization problem maximizing i objectives ($f_1, f_2, \dots, f_i$). Solution X is considered Pareto dominant when compared to solution Y when it satisfies the following two conditions: (1) for all j , $F_j(X) \geq F_j(Y)$; and (2) for at least one j among i objectives,

$F_j(X) > F_j(Y)$. According to this definition, solution X correspond to the point A or B in Figure 2.4, while Y correspond to the point C, respectively.

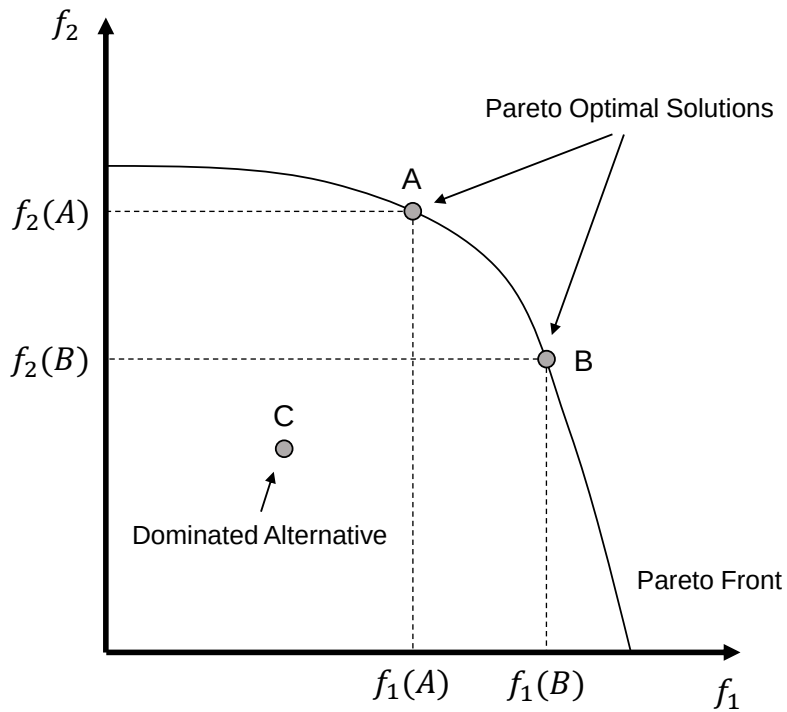


Figure 2.4: A graphical illustration of Pareto optimality concept in a two-objective maximization problem. Solution A and B are considered as the Pareto optimal solutions since the improvement of an objective is infeasible unless degrading the other objective

Specifically, MOEAs are the optimization algorithms, based on Pareto optimality, in which the goal is to find the best approximation of the Pareto set, while maintaining convergence (evolving to the Pareto front) and solution diversity (distributing solutions on the full extent of the Pareto front) (Reed et al. 2013). MOEAs achieve this goal by the key concepts identical to those from the EAs: iterative mating, mutation, and selection process to improve a population of solutions (Coello Coello et al. 2007). An ideal MOEA would yield its approximated Pareto front which is close to the known true Pareto front within tolerated calculation periods (Goldberg 2000).

Since the majority of the problems regarding WRSA comply with the concept of Pareto dominance defined in the Pareto optimality framework, multiple studies have utilized MOEAs during their solving processes. Giuliani et al. (2016a) coupled the MOEAs with the parameterization-simulation-optimization (PSO) approach, first introduced in the field of control theory by Koutsoyiannis and Economou (2003), in their evolutionary multi-objective direct policy search (EMODPS) framework utilized it in the searching process in multi-objectives WRSA problems. Kollat and Reed (2006) adopted the use of MOEA in solving long-term groundwater monitoring problem and proved the efficacy of the algorithm even when the size and the scope of the target problem is enlarged. Wang et al. (2015) applied the algorithm in solving the optimal design problem of a water distribution system with the conflicting objectives between cost and performance. Nevertheless, the application of MOEAs is clearly not exclusively limited in the WRSA problems, given its extensive advantage in dealing with complex problems in which the solutions are selected after the generation process. Table 2.2 provides a summary of literature regarding the optimization models based on evolutionary algorithms.

Table 2.2: Literature summary of the optimization models based on evolutionary algorithms utilized in WRSA

Authors (Year)	Optimization Tool	Target Problem (Main Research Question)	Key Findings (Conclusions)
Savic et al. (1999)	Genetic Programming	Rainfall-runoff modelling (flow prediction) and comparison with artificial neural network approach	Data-driven approaches more consistent than conceptual models
Fallah-Mehdipour et al. (2012)	Genetic Programming	Real-time decision making during reservoir operations with simultaneous inflow predictions	Proposed rule with deterministic variables is more effective
Ashofteh et al. (2015)	Genetic Programming (Multi-objective)	Climate change impact assessment in a multi-objective problem model using genetic programming models	29-32% improvements of models those account climate change than baseline models
Kumar & Reddy (2007)	Particle Swarm Optimization (Standard and Elicit-mutation)	How does the standard and elicitation PSO affect the performance of solution sets in a multi-objective reservoir operation problems?	PSO model with elicitation yields better solutions with less number of iterative calculations
Giuliani et al. (2016)	Multi-Objective Evolutionary Algorithm (MOEA)	multipurpose reservoir operation problems with conflicting objectives	Solutions from the EMODPS surpass those from the SDP models
Quinn et al. (2017)	Multi-Objective Evolutionary Algorithm (MOEA)	Socio-ecological system with irreversible tipping points with the introduction of direct policy search	DPS is a promising tool for managing problems under deeply uncertain conditions

2.2 General Decision Making in WRSA

Although the generation of optimal set of solutions via optimization algorithms is considered as the primary objective when utilizing the optimization algorithms introduced in section 2.1, execution of the selected alternatives in WRSA involves few more complex procedures. The complications which have to take place before the implementation of the selected alternative is largely due to the fundamental nature of WRSA problems which involves boundless subjects across disciplines. Therefore, in this section, a broader explanation regarding the general decision making framework commonly adopted in WRSA for analysis and implementation is to be described.

Generally, since the consequences of the decision making procedures regarding WRSA problems generally lead to irreversible, influential outcomes, the measures includes the involvement of multiple stakeholder groups each with diverse backgrounds. Although the specifics regarding the details of each stages may differ by each case study, Loucks and Van Beek (2017) proposed a general framework for analysis and implementation in WRSA problems. The framework by Loucks and Van Beek (2017) is composed of five stages: (1) inception, (2) situation analysis, (3) strategy building, (4) action planning, and (5) implementation, and at the same time includes iterative feedback loops which enables the decision makers to go back and forth between the stages. Ultimately, the related decision makers together aim at achieving the agreed common objective under this framework.

The first step in the decision making framework is the inception stage, in which the fundamental objectives and the constraints of the target problem is revealed to both the group of stakeholders and decision makers. During the discussion process, consensus

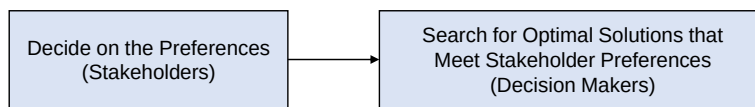
on the objectives and the criteria (or the constraints to be met) of the WRSA problem is met, and the group is now ready for the second process, the situation analysis stage. During the second stage, multiple investigations are made by utilizing the tools used in WRSA problems. Specifically, data collection and modelling processes, which many research consider to be decisive, happen during this stage. The participants should have a better understanding regarding the problem they are dealing with after the second stage of the framework.

Given the detailed description of the problem, the latter portion of the decision making framework focuses more on strategy building and action planning. During the third and the fourth stage, all actions aim at successful preparing for implementation of the preferred alternative. In the third stage of strategy building, a primary strategy is selected among the candidates of possible strategies through the discussion among the stakeholders and decision makers. Then, during the action planning stage, the selected strategy is prepared for final implementation. As a result, a detailed plan for implementation of the specific strategy including environmental impact assessments results will be available for the decision makers. Finally, the selected alternative is implemented in order to solve the target WRSA problem. As previously mentioned, the decision makers may choose to go back to the previous stages whenever necessary, enabling interactive feedback loops between the stages.

Since the ultimate objective of utilizing WRSA techniques is in the execution of best possible set of decisions that maximizes or minimize the predefined objective, WRSA is thoroughly connected to the general theories in adopted in the field of decision making. In this context, WRSA problems have been adapted frequently as examples for illustrating the practicality of decision making theories in actual systems

(Loucks and Van Beek 2017). In this context, the rest of section 2.2 of this dissertation aims to introduce general decision making theories focusing on the concepts of *a priori* and *a posteriori* preference articulation (Coello Coello et al. 2007), first classified by Cohon and Marks (1975) (Figure 2.5). These concepts are to be introduced through mathematical formulations commonly adopted in multiobjective optimization problem solving models, as introduced in section 2.1. Finally, today's efforts towards collaborative decision making, in which both the *a priori* and *a posteriori* techniques are incorporated into the procedures, regarding the WRSAs are explored with onsite application examples.

(a) *A priori* preference articulation



(b) *A posteriori* preference articulation

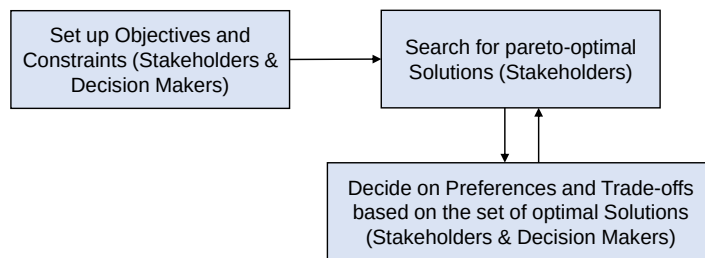


Figure 2.5: Key concepts and differences between (a) *a priori* and (b) *a posteriori* preference articulation methods

2.2.1 *A priori* Decision Making

A priori is a Latin phrase commonly adopted in the field of philosophy to describe the type of knowledge that is irrelevant from the past experiences of the subject. In the context of decision making theories, *a priori* decision making technique generally refers to making decisions before the searching process ("first choose then generate") (Cohon and Marks 1975). This is possible through the decision maker to make extremely strong assumptions of expertise or have a built familiarity with the given problem. In a multi-objective problem, arranging the objectives according to the decision maker's priorities is essential in order to approach the problem with *a priori* decision supporting tools.

For instance, an example of the *a priori* decision making technique can be demonstrated through the general form of the Global Criterion Method, in which the objective is to minimize the distance from the ideal objective vector (Eq. 2.6) (Coello Coello et al. 2007):

$$f(\mathbf{x}) = \sum_{i=1}^k \left(\frac{f_i^0 - f_i(\mathbf{x})}{f_i^0} \right)^p \quad (2.6)$$

where f_i^0 = i^{th} component of the ideal objective vector; $f_i(\mathbf{x})$ = i^{th} objective value given decision vector $\mathbf{x}$; and p = exponent that defines the distance norm to be used.

Since Eq. 2.6 assumes identical importance across the i objectives in a given value of p , few alternatives which belong to the *a priori* family have emerged in order to grant different weights to each vector consisting the objective vector $\mathbf{x}$. One of the most classical approach include compromise programming (Duckstein 1984), generalized goal programming (Ignizio 1981), and the weighted sum approach. All three approaches, in general, aim to seek for the ideal set of solutions with predefined weight terms. Specifically, the weighted sum approach weighted sum approach requires

the introduction of a new set of parameters for weighting, w_i to commensurate the objective function into a single one as in Eq. 2.7. Although many actual problems utilize the weighted sum approach due to its easy implementation to a wide range of problems, this is only valid under the assumption that the decision space is convex.

$$Z(\mathbf{x}) = \sum_{i=1}^k w_i f_i(\mathbf{x}) \quad (2.7)$$

Although many assumptions or prior knowledge regarding the target problem should be proceeded, application of the *a priori* decision making techniques is in fact feasible in many of actual problems (Kumar et al. 2012). The aforementioned DP model, which is frequently adopted as the optimization model in WRSA problems, is a typical example since the cost function $Z_t(\cdot)$ in Eq. 2.5 should be commensurated for calculation. Furthermore, *a priori* decision making models are widely utilized in numerous onsite examples including the prominent utility theory (Fishburn 1970), which is to be described briefly regarding the application in the field of water resources management.

Utility Theory

The utility theory tries to provide reasons behind people's specific decisions by making assumptions on the people's preferences. Since the emergence of this theory in the 19th century, the utility theory is considered as a representative example of *a priori* decision making theory and has been utilized in the fields of mathematics and economics. The multiattribute utility theory, or happiness attribute, includes the following two strong core assumptions: preference independence and utility independence. In this

context, preference independence refers to the fact that the attribute is monotonic in utility, and that it is independent from other attributes. Utility independence is when the preference under uncertainty for an attribute is independent from the uncertainty of another attribute. Given these two assumptions, utility theory provides theoretical reasons behind people's preference, and consequently behind their final decisions.

Depending on a decision maker's preference towards risk, all decision makers can be classified into one of the following when making decisions under uncertain conditions: risk-averse, risk neutral, and risk lover. Figure 2.6 represents the utility function, or a graph that measures the level of satisfaction as a function of the decision, for a risk-averse decision maker. Risk aversion, which is a trait that the decision maker is willing to pay a positive amount of money to eliminate risk, has been proven to exist especially regarding decision making in water resources (Bredehoeft and Young 1983; Li et al. 2017b). Since water resources management problems may lead to crucial consequences than other issues, risk-averse decision makers on water resources prefer to eliminate risk (or variance) by disbursing sufficient compensation, namely risk premium (R in Figure 2.6) .

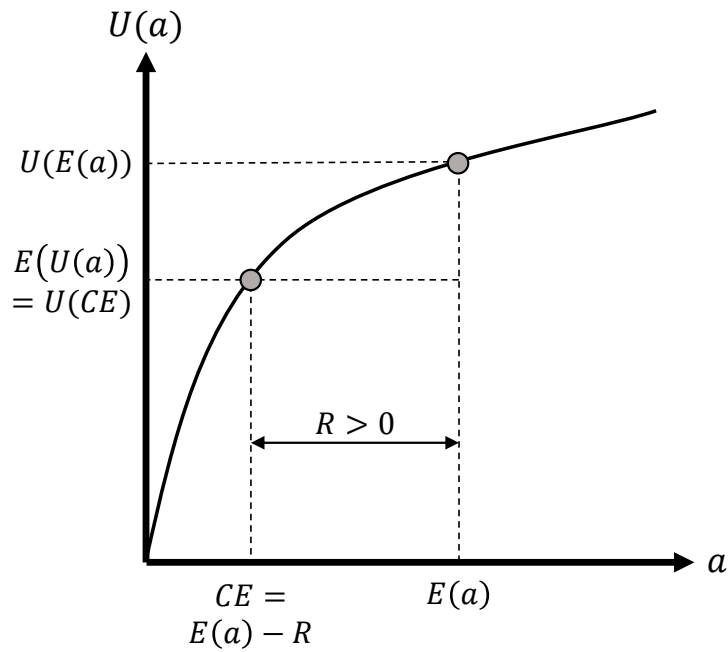


Figure 2.6: Graphical illustration of the utility function for a risk-averse decision maker. The notations on the figure are as follows: a = random variable; $U(\cdot)$ = utility function; CE = certainty equivalent; R = risk premium (shadow cost of risk bearing); $E(\cdot)$ = expectation operator based on the subjective probability distribution of a ; and $U(E(a))$ = utility of the expected value of the uncertain variable a ; $E(U(a)) = U(CE)$ = expected value of the utility of the uncertain variable a (utility of the certainty equivalent)

2.2.2 *A posteriori* Decision Making

Contrarily from the *a priori* concept, *a posteriori* knowledge refers to the knowledge based on the empirical evidence, such as scientific observations or personal experience. Consequently, the application of *a posteriori* decision making techniques induces the subject to make decisions after perceiving the range of alternatives to choose from. Therefore, *a posteriori* techniques focus on deriving the complete set of possible alternatives regardless of the decision maker's preference, rather than seeking for a few possible candidates among all possible solutions. As in Figure 2.5(b), since the decision maker's preferences are elicited after the solution searching process under *a posteriori* approach, they are provided with more complete set of solutions. Consequently, the decision makers under the *a posteriori* approach are provided with the opportunities to go over the sensitivity or thresholds followed by their actions.

Due to its characteristics, the *a posteriori* approach has been frequently adapted in control problems, in which the goal is in the correct design of the implementable policy that correctly include both observed and projected information in the problem formulation process (Herman et al. 2020). Since the generation of nondominated set of solutions is essential in the *a posteriori* approach, novel computational frameworks have emerged. According to Herman et al. (2020), the frameworks can be classified into either robust or dynamic planning frameworks (Maier et al. 2016; Kwakkel and Haasnoot 2019) depending on how they handle uncertain information. Whereas the robust framework focuses on discovering the solutions that perform relatively well across a wider range of scenarios, policies revealed by the dynamic framework selects actions which updates their decisions with the introduction of novel information. Since

the concepts of robust and dynamic planning frameworks are also widely utilized regarding coping with climate change, their philosophies and applications are to be explained in more detail in section 2.3.

2.2.3 Collaborative Decision Making

In this dissertation, collaborative decision making technique refers to all the decision making processes which collectively include a wide range of stakeholder groups during the decision support process. For instance, the constructive decision aiding concept (Tsoukiàs 2008), composed of the four stages (Table 2.3) is a prototype of a collaborative decision making strategy, since the aim of the framework is in making decisions by reflecting the conceptions from the participants. Moreover, as both the *a priori* and *a posteriori* decision support techniques may include the stakeholders during the processes, the collaborative decision making should not be considered separately from the predefined decision making techniques. Additionally, since the recent surge in hydro-climatic disasters induced from climate change requires the collective actions from related stakeholders, the collaborative decision making strategies are also firmly interconnected with both adaptive and robust decision making, which are to be introduced in the following section.

Table 2.3: Decision support stages adopted in the constructive decision making processes, redrawn as table from Moallemi et al. (2020)

No.	Stage	Key Questions
I	Representation of the problem context	· Who are the stakeholders? · What are stakeholder initial assumptions? · What is their expected goal?
II	Problem framing	· What are potential candidate decisions? · What are the possible outcomes? · What are outcome indicators to measure performance?
III	Evaluation and implementation	· What evaluation model can be utilized? · What are the available technical choices?
IV	Decision recommendations and monitoring	· What are the results of the evaluation? · How can the results be utilized? · How will the decisions be dynamically used in changing circumstances?

The primary motivation of the collaborative decision making approach is in providing a trigger that can realize the results theory into reality (Palmer et al. 2013; Basco-Carrera et al. 2017). The needs and importance of collaborative approach has been addressed by multiple international organizations such as the United Nations Framework Convention for Climate Change (UNFCCC). Especially, inclusion of the stakeholders from different backgrounds during the key processes of decision making has been highlighted by the UNFCCC, in order to cope with the changing environment induced from climate change.

In particular, the collaborative decision making has been actively utilized with simulation-based models. Especially, the use of system dynamics models proved to be effective during the communication processes. Shared Vision Planning (Palmer et al. 2013) which includes the features from traditional water resources planning, structured public participation, and integrated computer model has been frequently adapted (Kim et al. 2019a) as a representative collaborative decision making technique. Since collectively uniting the theoretical results from the optimization models are vital in order to execute appropriate and optimal decisions derived from calculation, this dissertation aims to contribute by presenting an integrated approach that connects the theoretical world and the actual world by utilizing collaborative interactions throughout the entire study processes in the dissertation.

2.3 Decision Making under Climate Change in WRSA

Although the traditional decision making strategies have led the trend in most of the contemporary WRSA problems, climate change and the consequences primarily induced from climate change aroused the necessity of novel approaches. In specific, climate change refers to the change in the statistics (e.g., mean or variability) of climate features that persists for an substantial length of time. Since the change in the climate features will highly impact the hydrological cycle to be more intensified (Arnell 1999), water resources management problems will inevitably face changes due to climate change. Finally, given the fact that the change in water resources management schemes widely impact the fields of the society such as economy, ecology, and human activities, measures to minimize the adverse effects from climate change are essential.

In this context, studies by far have discovered the few major causes of behind climate change, such as natural processes or external forcings, or anthropogenic changes, and land uses (Field et al. 2012) and are highly human-driven. Since the results from both studies and the onsite applications also have proved that the resulting consequences from climate change to affect the lives of the general public, handling climate change and its effect on water resources management schemes have been considered as an influential subject in WRSA for the past few decades (Němec and Schaake 1982; Frederick and Major 1997; Lettenmaier et al. 1999; Christensen et al. 2004).

As briefly mentioned in Chapter 1, this dissertation aims to focus on the two key features of climate change: non-stationarity and Knightian uncertainty. In particular, the non-stationarity refers to the change in the historical statistical properties in the future, and the Knightian uncertainty refers to the condition in which the probabilistic

distributions are unknown and are subject to change. Ultimately, among the multiple measures from the humanity to cope with the above two features of climate change, this dissertation focuses on two of the approaches that were proved to be effective by past studies and applications: adaptive and robust decision making techniques (Herman et al. 2020) (Table 2.4).

Table 2.4: Key concepts behind adaptive and robust decision making approaches in WRSA under climate change

Type of Approach	Main Concepts & Key Ideas
Adaptive Decision Making	· Updates decisions in response to new information · Aligns well with various optimization algorithms (SDP, EMODPS) · Highly dependent on uncertainty characterization
Robust Decision Making	· Find solutions that perform well under a wider range of scenarios · Supplements the conventional vulnerability assessment frameworks · Not so dependent on uncertainty characterization

2.3.1 Adaptive Decision Making Strategy

Adaptive, or dynamic according to Herman et al. (2020), decisions refer to the decision making technique in which the decisions are iteratively updated by the results of its previous outcomes and new information (Kolokytha et al. 2017). Since climate change have augmented both the frequency and the intensity of hydro-climate disasters, the flexible decisions from adaptive decision making techniques enables the user to dynamically update their actions by predicting the following consequences. The adaptive decisions then enables the reduction of potential damage from the unsteady and deeply uncertain climate to the society. In this context, adaptive decision making strategies have been considered as one of the main pillars towards climate change adaptation and also in reservoir managements (Eum and Simonovic 2010).

The decision making techniques those are classified as adaptive should include the change in normative behaviors which result from the unexpected change from the expected conditions. Moreover, the modified actions should perform better than the normal actions. Adaptive decision making strategies are therefore selected as a mean of confronting climate change with their flexibility during the execution of actions. Few representative examples of adaptive decision making includes real option analysis (ROA) (Watkiss and Hunt 2013) in which the traditional cost-benefit-analysis action is extended, adaptive reservoir managements (Georgakakos et al. 2012), and natural resources management problems (Williams 2011). Dynamic Adaptive Policy Pathways (DAPP) (Haasnoot et al. 2013) provides an integrated framework which combines adaptive policy making and adaptation pathways during the water resources infrastructure planning process. Finally, Engineering Option Analysis (de Neufville

et al. 2019) has been applied to making decisions regarding multiple water resources infrastructure investments.

One of the distinctive characteristic of adaptive decision making strategy that is contrasting from the robust decision making is in the key assumption regarding uncertainty characterization (Table 2.4). Since the key concept of adaptive technique requires sequential decision making across the simulation period, assuming of the probable future uncertainty is required. As an example, the SDP model discretizes the possible range of uncertain state variables by assuming the distribution from historical observations. Other alternate methodologies which are utilized in adaptive strategies to outline the shape of uncertainty include simulations based on Monte Carlo method. Nevertheless, since the characterization of uncertainty is hardly possible under deeply uncertain conditions, which is another key factor of climate change, robust actions are less dependent on the uncertainty assumptions than the adaptive strategies.

2.3.2 Robust Decision Making Strategy

Contrasting from the adaptive measures, robust alternatives refers to the set of solutions that perform comparatively better than the traditional optimal solutions over a wide range of plausible future scenarios (Kolokytha et al. 2017). Robust decisions, although not optimal, perform reasonably well under various future states of the world and consequently are evaluated with distinctive measures. Ultimately, making robust decisions as a measure to cope with deep uncertainty of climate change in WRSA studies have been proven to be effective and utilized in multiple studies, which are to be explained.

An example in which the concept of robustness is incorporated well is the concept of robust decision making (RDM) (Kasprzyk et al. 2013). RDM is based on iterative

simulation process to evaluate the decisions against a wider range of future scenarios. In order to achieve this objective, RDM includes the following features (Kolokytha et al. 2017): (1) multiple scenarios of plausible future climates (Lempert 2003), (2) a robustness criterion, and (3) one of the following approaches (a) multi-criteria analysis or multi-criteria decision analysis under uncertainty, (b) assessed risk of policy, (c) vulnerability and robust response, and (d) decision scaling (Brown et al. 2012). Overall, robust concept primarily spreads the amplified risk from climate change over the range of scenarios. Consequently, the system which was run with a robust alternative is resulted in possibly more frequent, but with less probability of critical failures. This concept of robustness aligns well with the risk-averse nature of decision makers in water resources, and the intrinsic studies in the past have stressed the connection between these two underlying concepts (Hashimoto et al. 1982a).

Few of the approaches include theoretical robust optimization (RO) (Mulvey et al. 1995; Watkins and McKinney 1997) often coupled with the optimization models. In a broader context, many-objective robust decision making (MORDM) framework (Kasprzyk et al. 2013; Hadka et al. 2015) advances the conventional RDM approach by the utilization of diverse methodologies in generating alternative strategies. MORDM includes not only the concept of RDM, but also the key aspects from many objective evolutionary algorithms and interactive visual analytics in order to deliver the key findings more efficiently. Frequently, the primary optimization tool is selected as the MOEAs, and they are adapted as the key tool in generating optimal alternatives in the MORDM framework. Later uncertainty analysis and scenario discovery methods (Kasprzyk et al. 2013), also adapted in the methodology and analysis of this dissertation, are utilized to discover the crucial factors that most contribute to the adverse

results of the target system. Ultimately, the concept of robustness is also connected to water resources system planning procedures (Matrosov et al. 2013). Supplementary explanations regarding the concept of robustness adapted in the WRSA problems, especially RO, will be thoroughly explained in section 6.3 of this dissertation. This dissertation aims couple RO with an optimization model in order to provide both optimal and robust set of solutions that perform satisfactorily under wider range of uncertain future scenarios.

Table 2.5 provides a brief summary of literature regarding both adaptive and robust decision making strategies in the WRSA problems. In this dissertation, both the adaptive and robust decision making strategies are utilized in order to account for climate change. Chapter 5 presents an adaptive approach of reservoir operation accounting for multi-year drought, and Chapter 6 includes the concept of RO into the objective function and the constraints of SDP in order to improve robustness of reservoir operations.

Table 2.5: Literature summary of the adaptive and robust decision making strategies utilized in WRSA

Authors (Year)	Optimization Tool	Target Problem (Main Research Question)	Key Findings (Conclusions)
Williams (2011)	Adaptive decision making	Application of adaptive decision making in natural resources management problems	Stresses the difficulties presented by non-stationarity and organizational challenges
Georgakakos et al. (2012)	Adaptive decision model	Comparative approach between adaptive reservoir managements and traditional approaches	Adaptive management constitutes an effective mitigation measure to climate change
Mulvey et al. (1995)	Robust Optimization (RO)	Theoretical concept of robustness coupled with optimization models: comparison with stochastic LP models	Suggests the two different types of RO: feasibility robustness and solution robustness. Highlights the applicability to actual problems
Kasprzyk et al. (2013)	Robust Decision Making and Info-Gap	Application of two models in water resources system planning procedures	Different recommendations from the two models. Highlights the complementary relationship between the two
Hadka et al. (2015)	Many-objective Robust Decision Making (MORDM)	Provides an open source software implementation (OpenMORDM)	Provides the platform for constructive decision support by balancing the conflicting objectives

Chapter 3. Study Site and Case Study Overview

3.1 Study Site: Boryeong Multipurpose Reservoir

Throughout this dissertation, Boryeong Dam, a multipurpose reservoir located on the western coast of the Korean Peninsula, and the districts dependent on the reservoir for their water uses, are selected as the test case and the proposed methodologies in the following chapters are all applied. In general, Boryeong Dam supplies water for municipal and industrial (M&I), agricultural (A) and environmental (E) water primarily to the eight nearby administrative districts: Taean, Seosan, Dangjin, Yesan, Hongseong, Boryeong, Cheongyang, and Seochon (Figure 3.1). The drainage area of the Boryeong Dam is relatively small, 163.6 km² (0.163% of the area of Republic of Korea), and it receives an average of 1245.0 mm of precipitation annually with annual runoff of 236.84×10^6 m³/yr and dry season runoff of 64.65×10^6 m³/yr. However, the target reservoir was not able to experience this average precipitation and runoff recently, mainly due to the recent long-lasting intensified droughts aroused from climate change. Since one of the primary motivations behind the studies presented in this dissertation is in the adaptation to this drought, the effects from the drought and the society's key efforts towards the drought damage mitigation will be further investigated in this section.

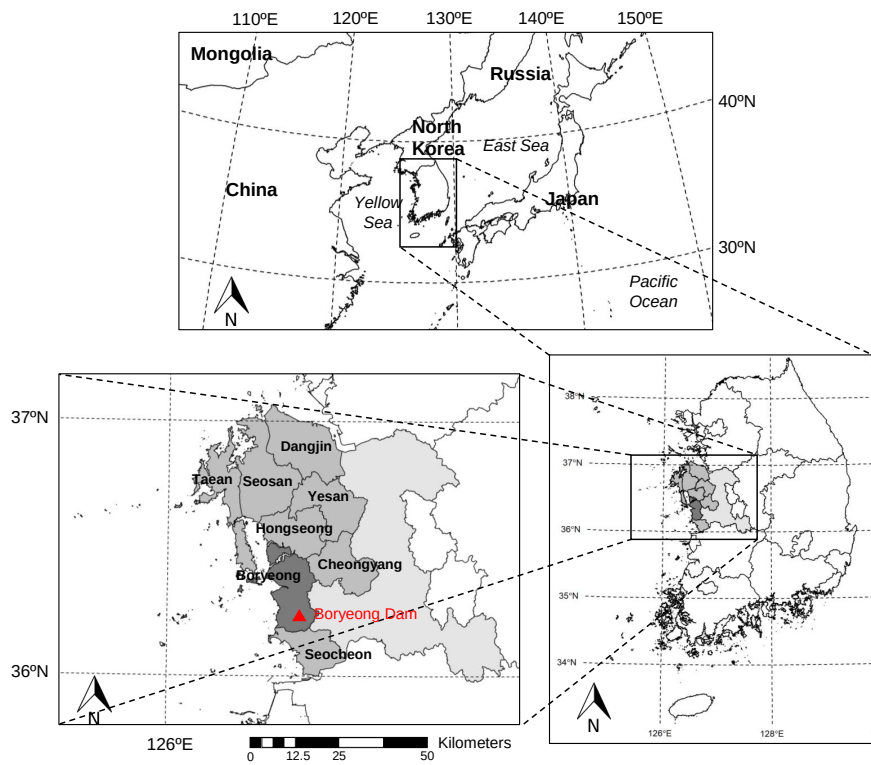


Figure 3.1: Map of the Boryeong Dam and the eight administrative districts that receive water from the reservoir

This target reservoir has a few distinctive features from other typical reservoirs in the Republic of Korea, which were all considered during the model formulation processes during all case studies. Primarily, due to the marginal difference between the capacity of the reservoir ($116.9 \times 10^6 \text{ m}^3$) and the aggregate annual demand of districts ($106.6 \times 10^6 \text{ m}^3$, 91.2% of reservoir capacity), mismanagement of the reservoir may lead to critical water supply failure which may affect the public water users. The low gap between the capacity and the aggregate water demand may lead to severe water deficit in case of droughts during the flooding period. Since the precipitation is mostly concentrated during the 3-month flooding season (July through September) in Korea, water stock management during the 9-month drawdown period (October through June of the following year) is extremely crucial. Moreover, differently from other typical reservoirs, the majority of the water demand from the Boryeong Dam is concentrated on M&I uses (83%) with relatively low ratio in A and E water uses (11% and 5%, respectively) (Jung et al. 2016) (Figure 3.2). This high ratio for M&I water demand consequently increases the extent of the potential exposure of the public to drought damage if stocks are mismanaged, resulting in extremely risk-averse citizens who are highly sensitive to a shortage in usable water.

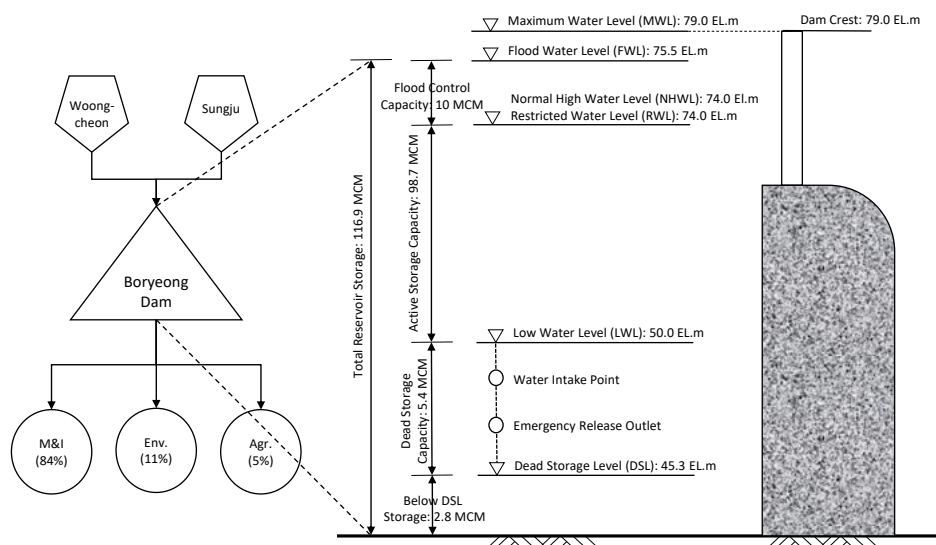


Figure 3.2: Schematic diagram and water levels of the Boryeong Dam

As previously mentioned, the recent multi-year drought from 2015 through 2019 significantly affected and damaged the water users depending on the Boryeong Dam (Kim et al. 2019a). The cause of the recent drought and consequent water deficit was both in the lack of cumulative rainfall for a long period of time (Figure 3.3), and the mismanagement of the reservoir in case of droughts, resulting in relatively low water levels (Figure 3.4). This dissertation aims to present a few reservoir operation alternatives which may prove effective against the climate change-induced disasters.

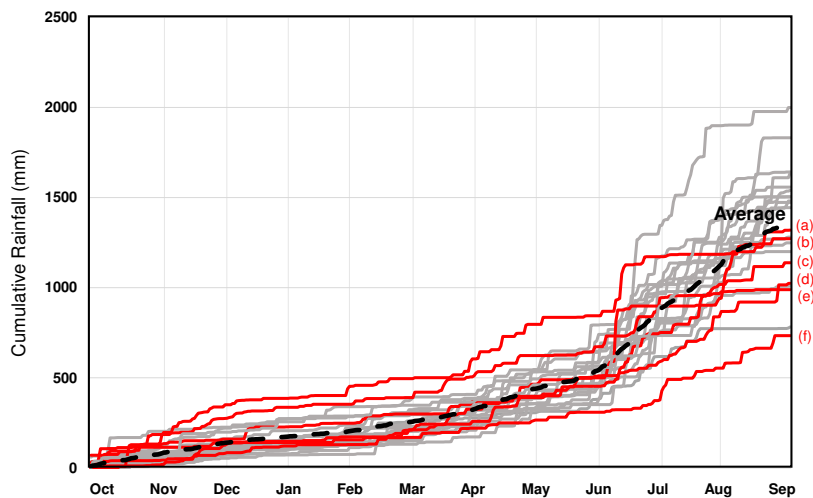
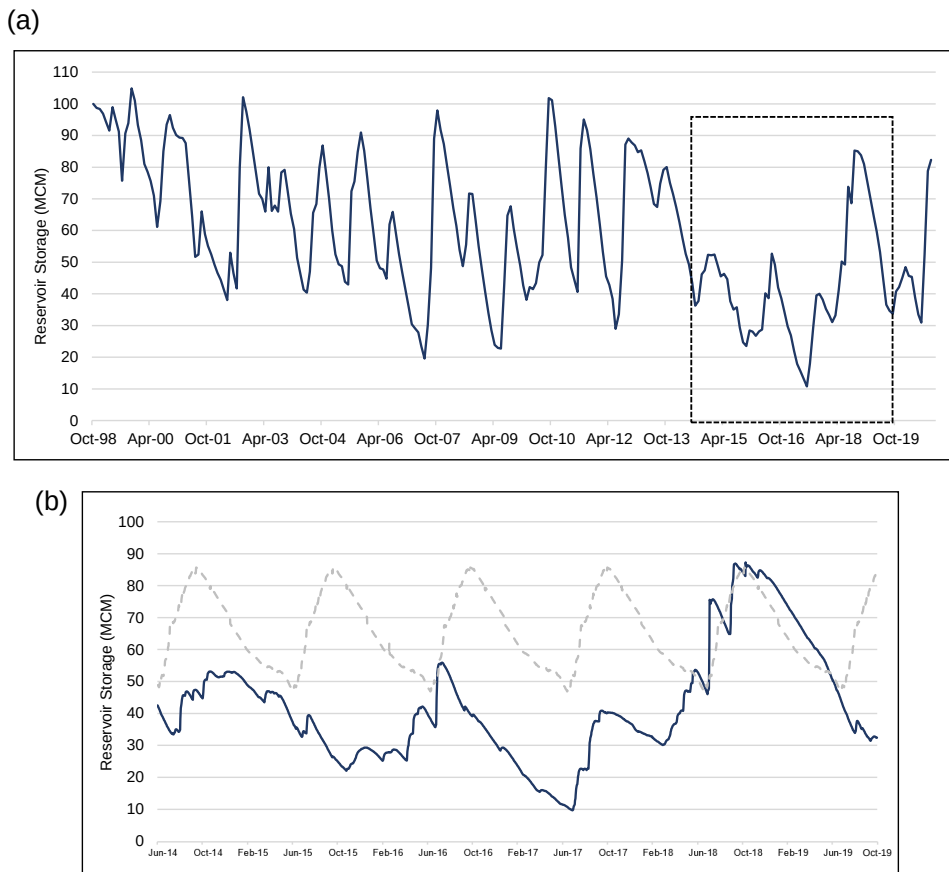


Figure 3.3: The cumulative rainfall plot of the Boryeong Dam basin for 22 water years: October 1998 through September 2020. Black dashed lines represent the annual average cumulative rainfall, and the red lines represent the cumulative rainfall during the recent 6-year drought: (a) 2015-2016, (b) 2017-2018, (c) 2016-2017, (d) 2013-2014, (e) 2014-2015, and (f) 2018-2019



Against the drought, acquirement of additional sources of freshwater through the construction of new infrastructure or an extension of the existing reservoir capacity are often considered as alternatives. However, they are limited, especially in the study area where stakeholders with different backgrounds compete with each other (Lee et al. 2019) such as the target basin considered in this dissertation. Thus, various efforts to mitigate the damage from the drought include the development of a water policy council (Kim et al. 2019a,b), construction of a conduit which acts as an additional water source (Ihm et al. 2019), and improvements of the operating rules of the reservoir (Yoon et al. 2018). Additionally, due to the national scale of the recent drought, Korea Water Resources Cooperation (K-water) (Lee et al. 2019) has computed the benefits from making use of the emergency storage, which is the difference in reservoir storage between the low water level (LWL) and the dead storage level (DSL) (Figure 3.2) in the case of extreme droughts. Detailed collaborative stakeholder elicitations (Kim et al. 2019b) informed the problem formulations explored in this study. Building on these stakeholder interactions, this dissertation aims to improve the efficacy of actual reservoir operations by contributing optimization techniques that allow applications to better take advantage of recent innovations in adaptive and robustness concepts.

Two of the key measures to be highlighted in this dissertation are the Boryeong Dam conduit construction project and the use of emergency storage below LWL. Firstly, completed in February 2016, the Boryeong Dam conduit enables the supply of freshwater to the reservoir in case of droughts from the downstream of Geum River. Since the capacity of the conduit ($115,000 \text{ m}^3/\text{day}$) is able to cover about 47% of the daily M&I demand, the conduit is a decisive factor that may alleviate the potential water deficit in the Boryeong Dam. By far, the conduit supplied a total of 66.63 MCM

(million cubic meters) (Table 3.1) to the Boryeong reservoir. Note that the storage graph in Figure 3.4 include the water supplied from the conduit after February 2016. Therefore, in order to provide realistic outcomes, the conduit was considered both during the alternative generation process and the simulation process in Chapter 5.

Table 3.1: Records of the Boryeong Dam conduit operations

No.	Start Date	End Date	Duration (days)	Supplied Water (MCM)
1	Feb-16, 2016	Mar-18, 2016	32	0.842
2	Mar-25, 2017	Apr-06, 2018	378	34.622
3	Aug-26, 2019	Jan-08, 2020	136	13.248
4	Aug-16, 2021	Aug-29, 2021	14	1.251
5	Sep-27, 2021	Nov-30, 2021	65	6.656

Another alternate measure to be highlighted regarding this dissertation is the use of emergency storage below LWL and above DSL. The agent primarily responsible for reservoir operation in the Republic of Korea, K-water, has considered the utilization of additional storage below LWL after the simulation with the emergency storage. The reservoir resulted in the increase in the evaluation metrics measured in terms of water supply (Lee et al. 2019). With these firm legal bases behind the potential policies, Chapter 6 proposes an alternative policy in which the storage between LWL and DSL is used, but with a penalty term proportional to the magnitude of violation.

3.2 Case Study Overview

Whereas the three studies in this dissertation all target the identical reservoir, each study's research objectives are all different under the common objective of the dissertation, presented in subsection 1.1.3. In the first case study which is to be presented in Chapter 4, a comparative analysis between two optimization algorithms is conducted. Afterwards, one of the optimally derived policies is then coupled with the onsite zone-based hedging rule utilized in actual operations by K-water. The key research questions to be answered in Chapter 4 is as follows:

- (4a) How does different optimization models or their formulations affect both the quality and the diversity of solutions in reservoir operation problems?
- (4b) How about the status quo (SQ) rules? How do they perform under both drought and normal conditions compared with optimal solutions?
- (4c) How does the implementation of SQ rules with the solutions from optimization algorithms improve (or degrade) the performance under extreme drought conditions?

Then in Chapter 5, few alternative hedging rules in which are more resilient against multi-year droughts than the current policy from K-water are presented. The substitute policies are utilized with adaptive operation techniques, first approach presented to confront climate change, and the performance of the reservoir from the adaptive operations is compared to non-adaptive operations in terms of supply and demand. The study in Chapter 5 aims to provide resolutions for the following questions:

- (5a) How does the inclusion of long memory (or multi-year droughts) during the operation policy generation process change the policy outcome?
- (5b) How will the adaptive operations enhance the results from the reservoir operation problems compared to non-adaptive reservoir operations?
- (5c) What are the possible trade-offs which will be discovered by adaptive reservoir operations?

Finally in Chapter 6, another pillar from decision making in climate change, robust decision making strategy is combined with the optimization scheme in reservoir operation problems. The proposed robust policy is then simulated under a wider range of inflow scenarios and evaluated in terms of both conventional and robust metrics. Research questions corresponding to the study in Chapter 6 is as follows:

- (6a) How can the traditional optimization algorithms (such as SDP) utilized in reservoir operation problems coupled with robust concept? (such as robust optimization)
- (6b) Under what conditions will the robust alternatives perform better than the conventional optimal solutions when measured in terms of frequency, duration, and magnitude?
- (6c) How robust are the robust alternatives when measured in terms of regret-based robustness metrics?

Comprehensively, Figure 3.5 summarizes the procedures and the connectivity from the case studies presented in this dissertation:

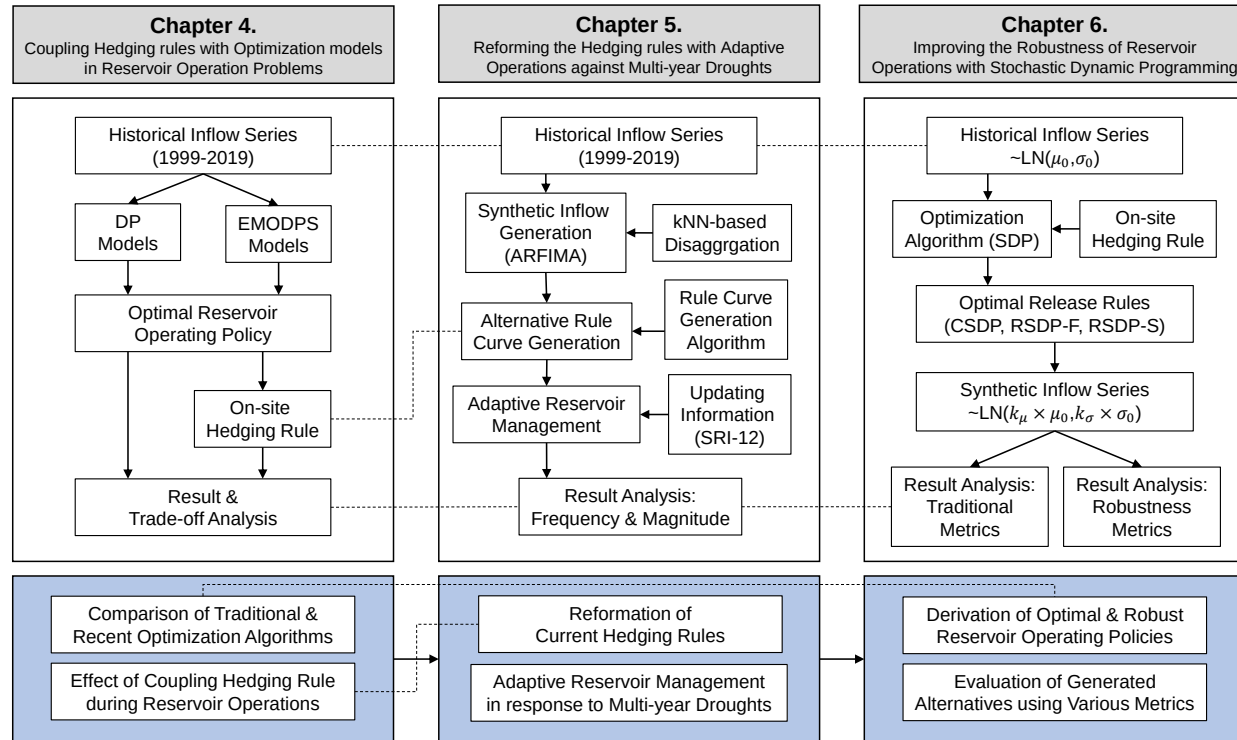


Figure 3.5: Flow chart of the dissertation. The relationship between Chapters 4, 5, and 6 are demonstrated in the flow chart

Chapter 4. Coupling Hedging Rules with Optimization Models in Reservoir Operation Problems

This chapter is partially drawn from the following peer-reviewed journal article:

*Kim, G.J. and Kim, Y.-O. "How does the coupling of real-world policies with optimization models expand the practicality of solutions in reservoir operation problems?" Water Resour Manage 35, 3121–3137 (2021).
<https://doi.org/10.1007/s11269-021-02862-y>*

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4.1 Abstract

This study first compares how different formulations of a reservoir operation problem with conflicting objectives affect the quality of the generated solution set. Six models were developed for comparative analysis: three using dynamic programming and three using the evolutionary multi-objective direct policy search (EMODPS) algorithm. Subsequently, to improve the quality of the generated solution set, an EMODPS model was selected and coupled with a zone-based hedging policy that has been currently applied in actual reservoir operations. The proposed methodology was applied to a multipurpose reservoir in South Korea. Among the different models, the EMODPS-Gaussian model with three parameters outperformed dynamic programming models. Moreover, coupling the status quo zone-based hedging rule with the optimization model improved the average duration of failure from the supply side by 37.4%. On the other hand, when measured from the demand side, the results indicated that the magnitude of failure also improved by 4.15% at the cost of frequency of water deficit. The overall results of this study suggest that the integrative use of optimization models with hedging rules is potentially applicable in future drought mitigation measures.

4.2 Introduction

During the last half-century, the derivation of optimal reservoir control policies through the use of emerging technological advancements gained considerable popularity and has become a key topic in the field of water resources management (Yakowitz 1982; Yeh 1985; Klemeš 1987; Labadie 2004). Optimization techniques include linear programming (Belaine et al. 1999), nonlinear programming (Birhanu et al. 2014), and dynamic programming (DP) (Stedinger et al. 1984). Thus far, these tools have been adopted as primary problem-solving tools for water resources engineers.

In particular, algorithms belonging to the DP family are prominent options applicable to the majority of actual optimization problems. Given that the system dynamics of the target problem is well known (Giuliani et al. 2016a), DP transforms the convoluted original problem into a sequence of relatively simpler problems. By approximating the state and decision space by discretization, DP provides the optimal set of decision variables corresponding to every possible combination of state variables. Because this procedure can be applied to all types of objective functions or constraints regardless of their shapes, DP and its variations (Stedinger et al. 1984; Saadat and Asghari 2019) have been continuously applied to numerous reservoir optimization problems.

Despite the wide applicability of DP, approximation procedures that are compulsory in the DP framework may cause decision biases that potentially limit optimal decision making (Kasprzyk et al. 2009). Thus, efforts have been extended to foster optimization techniques to derive more precise resolutions to multi-objective problems involving conflicting trade-offs (Reed et al. 2013). One of the main approaches in this context is the adaptation of multi-objective evolutionary algorithms (MOEAs)

(Javadi et al. 2015; Liu et al. 2019b), in which the algorithm aims to address the target problem by making the least possible simplifications. Specifically, the evolutionary multi-objective direct policy search (EMODPS) framework (Giuliani et al. 2016a), which combines direct policy search (DPS) (Rosenstein and Barto 2001), nonlinear approximating networks, and MOEAs into a simulation-based optimization algorithm, has been proven to be more effective in yielding solutions to certain problems (Salazar et al. 2017; Quinn et al. 2019). Because solutions generated from EMODPS may yield more flexible decisions under more varied circumstances, EMODPS has been employed in complex multi-objective applications in which obtaining reference optimal solutions is relatively demanding. As a reference, Giuliani et al. (2016a) demonstrated that the use of EMODPS with both artificial neural networks (ANNs) and radial basis functions (RBFs) and proved that EMODPS model with RBF outperformed the ANN models in terms of both 75% and 95% thresholds.

Although operating policies derived via optimization schemes such as DP and EMODPS may guarantee theoretically optimal alternatives, several factors have hindered their implementation in onsite reservoir operations (Quinn et al. 2019), including the extremely risk-averse nature of decision makers, administrative or legal constraints, and the lack of expert knowledge among related stakeholders. Alternately, decision makers have adopted various types of hedging rules onsite, such as two-point (Bayazit and Ünal 1990), continuous (Hashimoto et al. 1982b), and zone-based hedging (Shih and ReVelle 1995), which conserve a portion of reservoir storage in case of future droughts (You and Cai 2008a,b). In particular, zone-based (discrete) hedging rules are popular in onsite reservoir operations and have been commonly adopted owing to their simple methodology and relative ease of implementation for dam operators with

limited advanced expert knowledge. The zone-based hedging rule first prioritizes water according to its uses and sequentially enforces restrictions in the case of droughts. Consequently, these earlier restrictions would prevent a reservoir from suffering from severe water shortages in the future.

Owing to the discrepancy between the solutions derived from optimization algorithms and hedging rules, only a few studies have endeavored to merge the concepts of policies derived from optimization algorithms and policies being implemented on-site. As an extension of participatory or collaborative decision making efforts in water resources management, this study intends to link theoretical optimization and real operation through the following procedures: (1) generate optimal solution sets of a multi-objective reservoir operation problem solely using distinctive optimization algorithms based on K-fold cross-validation and compare the quality of the derived solutions, (2) compare their performances with those of actual operations executed based on status quo (SQ) rules, and (3) implement onsite hedging rules to the solutions from the optimization algorithms and explore the trade-off relationship between conflicting stakeholders based on risk, resiliency, and vulnerability. The authors intend to contribute both to the algorithmic component and to actual drought response measures by providing a thorough analysis of multiple optimization frameworks and relating them to actual operating policies.

4.3 Methodology

4.3.1 Dynamic Programming

DP is an optimization algorithm that searches for an optimal set of decision variables that maximizes the current and the expected benefits from future operations (Tejada-Guibert et al. 1995), and its variations have been applied to multiple water resources management problems (Yakowitz 1982; Yang et al. 2009; Fayaed et al. 2013; Feng et al. 2020). For the calculations, the DP first discretizes the entire range of states, decisions, and disturbance variables with appropriate intervals. Then, the objective functions and constraints are iteratively calculated in discrete time steps across the entire optimization time horizon. Consequently, a steady-state solution or an optimal set of decision variables corresponding to each value of the state variables is computed as the outcome of the optimization procedures.

For example, let us consider a multi-objective problem in which the aim is to maximize a series of objectives, $\mathbf{J} = (J^1, J^2, \dots, J^k)$ by deciding on the optimal policy of release decisions, R_t^* across the optimization horizon T . Assuming perfect knowledge of the future inflow Q_t , this optimization problem can be solved using a deterministic DP (DDP) model by solving the following functional equation (i.e., recursive equation) under the time horizon:

$$f_{opt}^n(S_t, Q_t) = \max_{R_t^*} [Z_t(S_t, Q_t, R_t) + \gamma f_{opt}^{n-1}(S_{t+1}, Q_t)] \quad (4.1)$$

Subject to

$$S_{t+1} = S_t + Q_t - R_t \quad (4.2)$$

$$S_{min} \leq S_t \leq S_{max} \quad (4.3)$$

$$R_{min} \leq R_t \leq R_{max} \quad (4.4)$$

where $t = 1, \dots, T$; T = final time step; n = the number of time periods remaining until the end of the optimization horizon; f_t^n = the value of the benefit function from the current period to the end of the time horizon T ; Z_t = the value of the aggregated objective function during time period t with prescribed weights (i.e. $Z_t = \sum_{i=1}^k w_i \times J_t^i$); S_t = the storage at the beginning of period t ; Q_t = the inflow during period t ; R_t = the actual release during period t ; γ = a discount factor; S_{min}, S_{max} = the minimum and maximum storage, respectively; and R_{min}, R_{max} = the minimum and maximum release, respectively.

Because this specific formulation of the problem includes two state variables (S_t and Q_t) and a single decision variable (R_t), solving Eq. 4.1 yields an optimal set of decision variables R_t^* with dimensions proportional to the discretization intervals of the state variables and the length of the optimization horizon. Because DP manages both nonlinear and multi-objective characteristics of complex reservoir operational design problems (Giuliani et al. 2016a) it has been considered one of the most prominent options for solving water resources management problems. In particular, the stochastic version of DP (SDP), which addresses future uncertainties with hydrological state variables or transition probabilities, has been adopted in multiple reservoir operation problems (Stedinger et al. 1984; Tejada-Guibert et al. 1995; Kim and Palmer 1997; Kim et al. 2021).

Despite its advantages, few scholars have argued for the application of DP to more complicated water resources problems due to its notorious curses: curse of dimension-

ality (Bellman 1957), curse of modeling (Tsitsiklis and Van Roy 1996), and the curse of multiple objectives (Powell 2007). Briefly, the curse of dimensionality refers to the exponential increase in the number of computations proportional to the number of state variables, the curse of modeling refers to the necessity of including all input variables in the model as supplementary state variables, and the curse of multiple objectives refers to the exponential increase in the number of computations proportional to the number of objectives. This has motivated recent technical advancements aimed primarily at overcoming these three curses by using novel approaches other than DP (Bertsekas 2005; Giuliani et al. 2016a).

4.3.2 Evolutionary Multi-Objective Direct Policy Search

EMODPS is an example of the parameterization-simulation-optimization (PSO) approach (Koutsoyiannis and Economou 2003), where the MOEAs are merged with DPS and nonlinear approximating networks to derive a set of optimal policies. Instead of implementing approximation during problem formulation processes, such as discretization in DP, EMODPS directly optimizes the parameters of the optimal policy through a simulation-based optimization scheme. It is argued that the unique features of EMODPS partially resolve the three curses of DP: (1) the curse of dimensionality by evading the iterative computation of the value of the benefit function; (2) the curse of modeling by using a simulation model that can include exogenous information directly during simulation; and (3) the curse of multiple objectives by attaining the entire Pareto front through the adoption of MOEAs during its search (Kasprzyk et al. 2009; Giuliani et al. 2016a).

The problem formulation and optimization conducted through the EMODPS frame-

work can also be illustrated through the same problem that maximizes a series of objectives, $\mathbf{J}$. The EMODPS aims to search for a set of parameters ($\Theta^* = (\theta_1^*, \dots, \theta_n^*)$) that constitute the optimal policies p_θ^* . The formulation of the optimal policy that maximizes $\mathbf{J}$ can be expressed by the following equation:

$$p_\theta^* = \arg \min_{p_\theta} \mathbf{J}_{p_\theta} \quad (4.5)$$

The policy parameter vector Θ that constitutes the operating policy p_θ through parameterization, is coupled with diverse types of networks of functions that are flexible in shape. Typical examples of network functions include ANNs and RBFs (Park and Sandberg 1991) and radial basis functions (RBF) (Park and Sandberg 1991). The following example demonstrates the RBF with cubical shape, where the decision variable R_t is decided according to both the shape of the functions and the values of the parameters Θ (Giuliani et al. 2016a):

$$R_t = \min \left(\max \left(\sum_{j=1}^n \theta_{j,1} \left| \frac{S_t - \theta_{j,2}}{\theta_{j,3}} \right|^3, R_{min} \right), R_{max} \right) \quad \forall t, i \quad (4.6)$$

where n = the number of RBFs; $\theta_{j,1}, \theta_{j,2}, \theta_{j,3}$ = a set of parameters that compose the j th RBF; S_t = the value of the state variable at time t (storage at time t in this example).

The last component in the EMODPS framework, MOEA, is adopted during the searching procedures of the optimal policy parameter set Θ^* from Eq. 4.6. Because MOEAs can discover the critical trade-offs facing water resources systems (Reed et al. 2013), non-dominated or Pareto optimal sets of solutions can be derived without aggregating multiple objectives into a single function. Consequently, policymakers are presented with a full trade-off and can express their preferences for a single or several

alternatives among the non-dominated solutions. To generate a more precise set of solutions, improvements in diverse MOEA schemes have been developed, including the niched Pareto genetic algorithm (Horn et al. 1994), and the non-dominated sorting genetic algorithm II (NSGA-II) (Deb et al. 2002). For detailed description on the performances of different MOEAs, refer to the papers by Nicklow et al. (2010), Hadka and Reed (2012), and Reed et al. (2013).

4.3.3 Zone-based Hedging Rules

Operating reservoirs with alternatives derived from optimization algorithms, such as DP or EMODPS, would theoretically lead to higher measures of superior performance with respect to optimized objectives. However, the onsite application of these policies may be limited by the lack of theoretical backgrounds among decision makers (Seo et al. 2019), both actual and perceived errors in the models/forecasts, and the problematic attributes of systems models that are difficult to explain or understand (Quinn et al. 2019). Instead, the adaptation of relatively simpler policies such as diverse variants of hedging rules, or drought contingency plans, is more prevalent in actual reservoir operations (Eum et al. 2012). Among various alternatives, zone-based (discrete) hedging rules (Shih and ReVelle 1995), in which the release decisions are determined according to the storage at a specific time period, are considered to be primarily due to their relatively simple and intuitive application procedures.

An explanation of zone-based hedging rule adopted by all 20 multipurpose dams in South Korea as their main operating policy is provided as follows. Operating a reservoir solely based on the zone-based hedging rule requires the following procedures: first, the reservoir's total demand is prioritized according to its use. In the case of South Korea,

water use from a multipurpose dam is mainly classified into M, I, A, and E, and priorities are granted to each category accordingly. For instance, municipal and industrial water uses have the highest priorities, followed by irrigation, and environmental uses. Finally, the release R_t is determined according to the storage zone, and can be expressed as Eq. 4.7-4.11, which are considered the key operating procedures for the zone-based hedging rules (specific procedures used to derive the storage values of each zones is to be explained in Chapter 5):

$$R_t = D_t \quad \text{when } S_{t-1} \in S_{t,Normal\ Zone} \quad (4.7)$$

$$R_t = \alpha_t \times D_t \quad \text{when } S_{t-1} \in S_{t,Zone\ 1} \quad (4.8)$$

$$R_t = \beta_t \times D_t \quad \text{when } S_{t-1} \in S_{t,Zone\ 2} \quad (4.9)$$

$$R_t = \gamma_t \times D_t \quad \text{when } S_{t-1} \in S_{t,Zone\ 3} \quad (4.10)$$

$$R_t = 0 \quad \text{when } S_{t-1} \leq S_{DSL} \quad (4.11)$$

where α_t , β_t , and γ_t ($1 > \alpha_t > \beta_t > \gamma_t > 0$) = weighting factors determined according to the reservoir and season; D_t = the aggregate demand during day t ; S_{DSL} = the dead level storage of the reservoir.

This study aims to use of the current zone-based SQ policy in two manners: (1) compare the performance of the solutions derived from optimization algorithms with the actual performance based on the SQ policy and (2) couple the SQ policy during the simulation step of the PSO algorithm to analyze the effect of the incorporation of the hedging rule into an optimization algorithm.

4.4 Model Development for Boryeong multipurpose Dam

4.4.1 Set up Conditions for Optimization

To demonstrate how different optimization algorithms or problem formulation methodologies may ameliorate the performance of the reservoir, this study derived the optimal monthly release sequences for 20 historical water years (October 1999 through September 2019) using K-fold cross-validation. In this study, the observed inflow series were grouped into four subgroups (K=4) each with 15 years of inflow series for optimization, and the remaining five years of inflow for validation. During validation, the starting storage values were set as the actual storage values recorded at the beginning of the validation period (Table 4.1). The performance of each model was then comparatively analyzed. The overall flow of this study is shown in Figure 4.1.

Table 4.1: Specific formulations for each subgroup for the simulation based on K-fold cross validation

K	Optimization Period (15 years)	Validation Period (5 years)	Starting Storage in Validation (MCM, date)
1	Oct 2004 - Sep 2019	Oct 1999 - Sep 2004	99.597 (Sep-30, 1999)
2	Oct 1999 - Sep 2004, Oct 2009 - Sep 2019	Oct 2004 - Sep 2009	89.978 (Sep-30, 2004)
3	Oct 1999 - Sep 2009, Oct 2014 - Sep 2019	Oct 2009 - Sep 2014	64.365 (Sep-30, 2009)
4	Oct 1999 - Sep 2014	Oct 2014 - Sep 2019	47.364 (Sep-30, 2014)

* Average Inflow during the validation period is 4.64, 3.65, 4.39, 2.72 m^3/sec , respectively for subgroups K=1, K=2, K=3, and K=4.

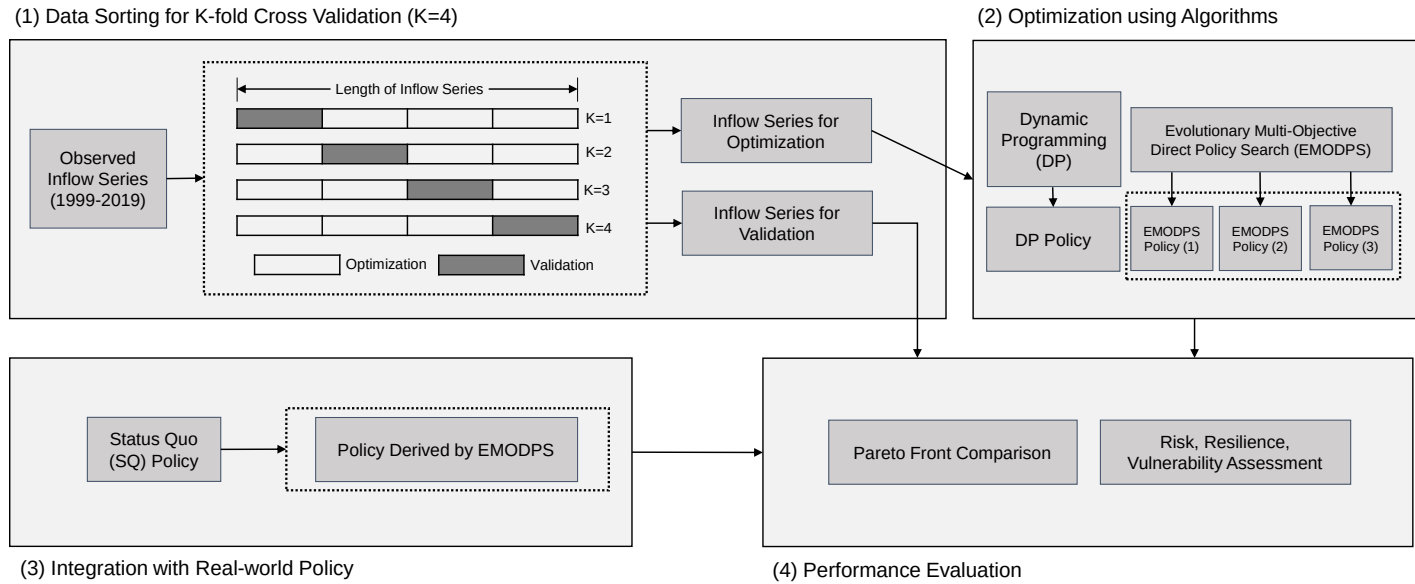


Figure 4.1: Research flow chart adopted in this study

4.4.2 Optimization and Reservoir Simulation

This study adopted the multi-objective optimal operations (M3O) toolbox developed by Giuliani et al. (2016b) to derive the optimal release rules with different alternatives. In total, six different models were used to derive six different optimal release rules for each subgroup: the DDP model with perfect forecast (DDP-Perfect), DDP model with average inflow forecast (DDP-Average), a stochastic DP model assuming a log-normal distribution of the inflow (SDP), and three EMODPS models with different approximating functions each with a distinct number of parameters (EMODPS-Linear, EMODPS-Root, and EMODPS-Gaussian).

In general, all models were operated mainly through the following mass balance equation Eq. 4.12:

$$S_{t+1} = S_t + Q_t - R_{n,t}^* \quad (4.12)$$

where S_t = the storage at the beginning of month t , Q_t = the inflow during month t , and $R_{n,t}^*$ = the optimal release decision during month t derived through the n th model ($n = 1, 2, \dots, 6$).

All models attempt to minimize the two objectives that reflect the preference of two conflicting groups of stakeholders: J^{supply} , which reflects the interests of reservoir managers and J^{demand} , which reflects the satisfaction of water users. Both objectives are measured in terms of magnitude and expressed as the average percentage of deficit over the entire optimization (or simulation) time horizon H . The first objective minimizes the deficit between the Zone 2 storage and the actual storage in each time step, and is expressed as a percentage. Among different zones, this study selected Zone 2 storage as the target storage since actual demand for A water use is first restricted when

the storage falls below Zone 2. Thus, this study considered Zone 2 as the threshold in which actual damage that affect the water users may happen, and included the objective term J^{supply} into the optimization model according to the following equation:

$$J^{supply} = \frac{1}{H} \sum_{t=1}^H \max \left(\frac{S_t^{target} - S_t}{S_t^{target}}, 0 \right) \times 100 (\%) \quad (4.13)$$

where S_t = the actual storage at the beginning of month t , and S_t^{target} = the predetermined Zone 2 storage during month t .

The second objective is to minimize the average deficits between releases and demands, which are expressed as a percentage:

$$J^{demand} = \frac{1}{H} \sum_{t=1}^H \max \left(\frac{D_t - R_t}{D_t}, 0 \right) \times 100 (\%) \quad (4.14)$$

where D_t = the monthly aggregate demand during month t , and R_t = the actual release during month t .

Ideally, both J^{supply} and J^{demand} should be zero, which indicates that the actual storage is at least above the target storage during the entire simulation period, and the release is higher than the demand. The two contrasting objectives in this study, J^{supply} and J^{demand} , were selected to reflect the actual conflicts that have historically have occurred (Kim et al. 2019a,b).

Policy Derived using DP

In this study, three models were developed using DP as their primary optimization algorithm with different forecast information: DDP-Perfect, DDP-Average, and SDP. Primarily, the DDP-Perfect model assumes perfect forecasting during the entire opti-

mization horizon of the future inflow. The solutions derived by the DDP-Perfect model were the reference set, in which the approximated solutions were not able to predominate. Conversely, the DDP-Average model was set to make release decisions that assumed the average inflow throughout the optimization horizon. In case of discretization of state variables, S_t was discretized into 55 values, while Q_t was discretized into 500 values. Finally, the SDP model developed in this study assumes a lognormal distribution of the inflow series and derives the optimal release decisions based on Eq. 4.15:

$$f_{opt}^n(S_t, Q_t) = \max_{R_t^*} \left[Z_t(S_t, Q_t, R_t) + E_{Q_{t+1}|Q_t} f_{opt}^{n-1}(S_{t+1}, Q_{t+1}) \right] \quad (4.15)$$

Because all DP-based models require a commensurate objective function, $Z_t(\cdot)$, for calculation in problems with two or more objectives, the aggregation was conducted as shown in Eq. 4.16 with combinations of weighting factors w_1 and w_2 :

$$Z_t(\cdot) = \omega_1 \times Z_{1,t} + \omega_2 \times Z_{2,t} \quad (4.16)$$

where ω_1 and ω_2 = weights multiplied by each of the objectives ($0 \leq \omega_1, \omega_2 \leq 1$); $Z_{1,t}$ and $Z_{2,t} = J^{demand}$ and J^{supply} calculated at month t , respectively.

Policies derived using EMODPS

Because the type of selected optimization algorithm and the formulation of the approximation function affect the overall quality of the generated solution sets, this study generated manifold alternatives with EMODPS by varying the shape and number of

parameters constituting the RBFs. Because it has been proven that the solution sets are sensitive to the shape of the parameterized function by Giuliani et al. (2016a), three shapes of RBFs with different numbers of parameters were adopted in this study: a 4-parameter linear RBF (EMODPS-Linear), a 6-parameter root-shaped RBF (EMODPS-Root), and a 3-parameter Gaussian RBF (EMODPS-Gaussian). All formulations adopted the NSGA-II for optimization because of its extensive use in water resources optimization problems and for equitable comparison.

The first alternative, the EMODPS-Linear model, was constructed using linear functions with four policy parameters (m_1, m_2, h_1 , and h_2) ($-0.5*\pi \leq m_1, m_2 \leq 0.5*\pi$, $-10 \leq h_1, h_2 \leq 10$). Consequently, the optimal release decision derived via the linear RBFs was determined according to Eq. 4.17-4.19:

$$L_1 = D_t + \tan(m_1) * (S_t - h_1) \quad (4.17)$$

$$L_2 = D_t + \tan(m_2) * (S_t - h_2) \quad (4.18)$$

$$R_t = \max(\min(L_1, D_t), L_2) \quad (4.19)$$

The second model adopted the Gaussian RBF, which is one of the most prominently adopted shapes of RBF in the field of scientific and engineering computing to determine the policy for the optimal release sequences. The three policy parameters that constitute the simplified Gaussian RBF (w_1, b_1 , and c_1) ($-1 \leq w_1 \leq 1$, $0 \leq b_1 \leq 10$, $-10 \leq c_1 \leq 10$) determine the decision R_t using the following equation:

$$R_t = \min \left(\max \left(\sum_{i=1}^j \left(D_t - \exp \left(w_i * \frac{S_t - c_i}{b_i} \right) \right), R_{max} \right), R_{min} \right) \quad (4.20)$$

The final shape of the RBF used in this study was a root-shaped RBF with six policy parameters (w_1, w_2, r_1, r_2, d_1 , and d_2) ($0 \leq w_1, w_2 \leq 1, 0 \leq r_1, r_2 \leq 10, -10 \leq d_1, d_2 \leq 10$), and the release decision was derived according to the following equation:

$$R_t = \min \left(\max \left(\sum_{i=1}^j \left(w_i * \left| D_t - \frac{S_t - d_i}{r_i} \right|^{0.5} \right), R_{max} \right), R_{min} \right) \quad (4.21)$$

It has been proven that both the population diversity and convergence of solutions to the Pareto optimal front are paramount in several studies that apply MOEAs. Thus, to ensure both diversity and convergence of optimal solution sets, all models in this study adopted 40 individuals each with 500 generations during optimization. The generated optimal solution sets were then analyzed to identify more effective and reliable operating schemes during drought periods.

Incorporation of SQ Policy during Simulation

To analyze the effect of incorporating the hedging rule, this study coupled the actual SQ policy during simulation in drought periods with the release decisions from the optimization algorithm. Among the types of hedging rules, zone-based hedging is used to operate the Boryeong dam. Under this rule, the actual release R_t is determined according to the zone to which storage S_t belongs. In this study, the SQ operating policy was included in the simulation step according to Eq.4.7-4.11 and actual monthly rationing factors were adopted for a realistic simulation. Finally, the efficacy of incorporating hedging rules was analyzed in terms of frequency, duration, and magnitude from the stances of both water suppliers and users.

4.5 Results and Discussion

4.5.1 Comparison in Performance of Models

The application of the proposed methodology at the reservoir on the study site resulted in diverse sets of solutions, each distinct in quality and diversity. For comparative analysis of their performance, the solutions generated by the DDP-Perfect, DDP-Average, SDP, EMODPS-Linear, EMODPS-Root, and EMODPS-Gaussian models were plotted and compared with the actual operating results of the same period, which include onsite, real-time decision making by experts in reservoir operations (Figure 4.2).

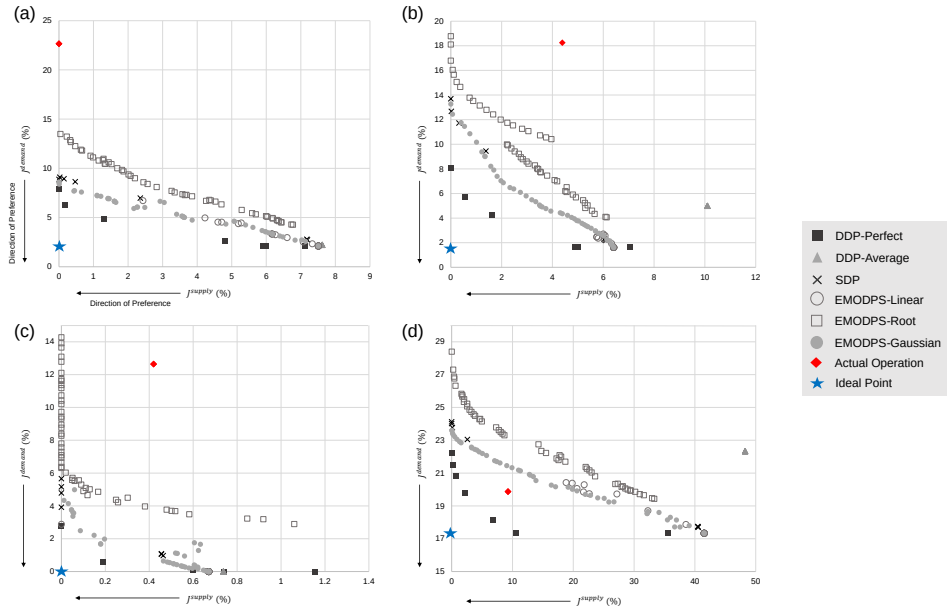


Figure 4.2: Pareto Fronts of the solution sets derived by each subgroup in K-fold cross-validation: (a) subgroup K=1, (b) subgroup K=2 (c) subgroup K=3, and (d) subgroup K=4

Multiple insights can be derived according to the Pareto fronts generated using different models. Primarily, solutions derived by the optimization algorithms outperformed the results from the actual operation, as indicated by the red diamonds in Figure 4.2, in three of the four subgroups. Although onsite release decisions were made with more information than those in the optimization algorithms, the results indicated that the adaptation of any type of optimization model to actual reservoir operation would lead to improved performance under normal conditions. However, the actual operation resulted in a better performance in the case of droughts (Figure 4.2d). This result indicates that the SQ policy, on which the actual operations are based, may yield more practical outcomes when adequately employed under suitable conditions.

Among the optimization models, the DDP-Perfect model, which assumed perfect forecasting of future inflow series, yielded the best performance, as initially expected in all subgroups. Of the three EMODPS models, the EMODPS-Gaussian model with three parameters yielded the results closest to the ideal point among all subgroups, represented as a blue star in Figure 4.2. Policies derived by the EMODPS-Linear model followed those derived from the EMODPS-Gaussian model regarding proximity to the reference set; however, they outperformed the other two models regarding solution diversity. Despite having more parameters than the other two, the EMODPS-Root model performed worst in terms of proximity to the ideal point. This indicates that the quality of the solutions generated by the EMODPS algorithm was more sensitive to the shape of the network functions than the number of parameters that constitute them, as proven in the past studies (Mongillo et al. 2011). Moreover, in cases where deriving both optimal and diverse solution sets is challenging, the overall trade-off relationship between the performance and the number of generated solutions, which

can be explored from the results of the EMODPS-Linear and EMODPS-Root models, stresses the importance of the decision makers choosing optimal operating policies depending on their circumstances.

Finally, from Figure 4.2, the alternatives derived from the DDP-Perfect model may be considered as the optimal set under which, in all circumstances, the reservoir should be operated because their performance is adjacent to the ideal point. However, deterministic models are types of open-loop control, where the sequence of actions is not updated as a function of new observations of state variables (Herman et al. 2020). Because of this attribute, the optimal release decisions made under the deterministic models would severely underperform in the case of unexpected states. This can be proved by the inferior performance of the DDP-Average model in two of the subgroups, $K=2$ and $K=4$ (Figure 4.2b and Figure 4.2d, respectively).

Conversely, the SDP model and all types of EMODPS models, which belong to the group of policy-searching closed-loop models, update their release decisions dynamically according to either the present or future states. This indicates that the closed-loop models would be more flexible to future uncertainties and would surpass the performance of deterministic models in cases of unanticipated incidents such as floods or droughts. Moreover, the release decisions from the EMODPS models are defined under the entire domain of state variables, whereas the decisions from the DP models are only defined in limited domains because of the discretization of the state variables. Thus, policies from adaptive optimization algorithms with approximating functions of various shapes should be considered when deriving optimal solutions for reservoir operation problems.

Regarding the computation time of the models from the two algorithms, the com-

putation time of the SDP model was compared with that from the EMODPS-Gaussian model. The target problem was manageable to be conducted on the local machine, without the parallelization of the cores. As a result, the computation for the EMODPS-Gaussian model took 18.2 times less than the computation of the SDP model in all subgroups. Finally, it can be concluded that the EMODPS-Gaussian model yielded in better set of solutions and at the same time, computationally efficient.

4.5.2 Effect of Coupling Zone-based Hedging Rules during Droughts

In this section, the effect of coupling the zone-based hedging SQ policy with optimal solutions from the optimization algorithm during droughts was investigated. Given the total precipitation during the optimization horizon and the conflicting nature of the objectives, the relocation of the Pareto optimal solution sets closer to the ideal point is hardly achievable. Hence, this portion of the study explored the expansion of the trade-off relationship between water users and suppliers that may arise from the coupling zone-based hedging policy during the simulation. Risk, resiliency, and vulnerability (Hashimoto et al. 1982b) were chosen as performance indices to measure the frequency, duration, and magnitude of predefined failures. Higher values of risk, resiliency, and vulnerability correspond respectively to more frequent, longer-lasting, and higher failures, respectively.

To quantify the effect of the hedging rule under drought conditions, this study selected the EMODPS-Gaussian model under subgroup K=4, which yielded both optimal and diverse solution sets. The same set of derived solutions was simulated twice under the historical inflow, once based on the actual hedging rule and once without it. Supply side failures were defined as those not meeting the target storage (i.e., $J^{supply} > 0$).

Conversely, failure from the demand side was split into different uses and defined as when the release was less than the target demand. Because in the ratio for M&I and A uses are dominant (94%) the risks and vulnerabilities for them were explored.

Figure 4.3 shows the performance indices from the supply side. Because the zone-based hedging rules restrict release from the reservoir in the case of relatively lower storage values, the average performance of the system measured from the supply side improved by 25.02% in risk, 37.38% in resiliency, and 24.23% in vulnerability. In particular, the maximum duration of failure, measured in terms of resiliency, was significantly reduced by 45.59% after incorporating hedging rules (Figure 4.3) from 17 months to 9.25 months. This indicates that the adaptation of hedging rules may potentially lead to less frequent, shorter, and smaller failures in case of droughts. Figure 4.4 depicts the comparative performances before and after coupling hedging that was calculated from the demand side. Owing to the conflicting nature of supply and demand, the frequency of failure measured from the demand side increased after hedging for both A and M&I uses (Figure 4.4a, 4.4c). However, the average magnitude of failure slightly improved in the stances of water users (1.68% for A and 6.68% for M&I) (Figure 4.4b, 4.4d). This result indicates that when hedging is applied, the users experience more frequent failures, but with a downsized magnitude of water deficit. Considering the extremely risk-averse nature of water users facing water deficiency, reducing the magnitude of failure by incorporating hedging seems to be a better strategy given the increased risk of droughts in the future. Overall, these results indicate that the coupling of theoretical algorithms with actual policies enlarges the range of choice with respect to vulnerability, although the previous studies primarily focused on the advancement of optimization techniques.

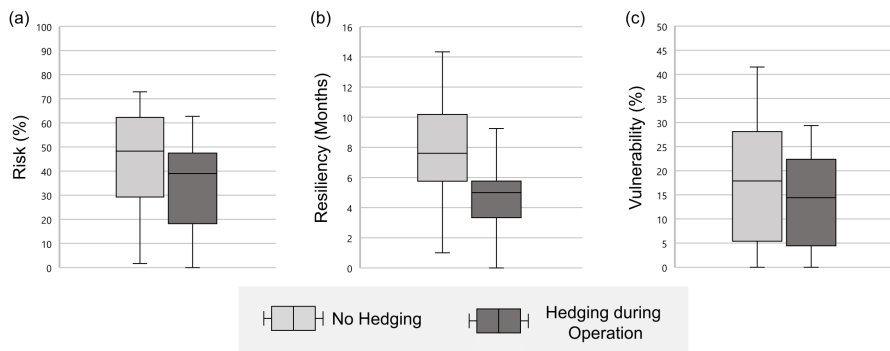


Figure 4.3: Performance indices measured from the supply-side: (a) risk, (b) resiliency, and (c) vulnerability

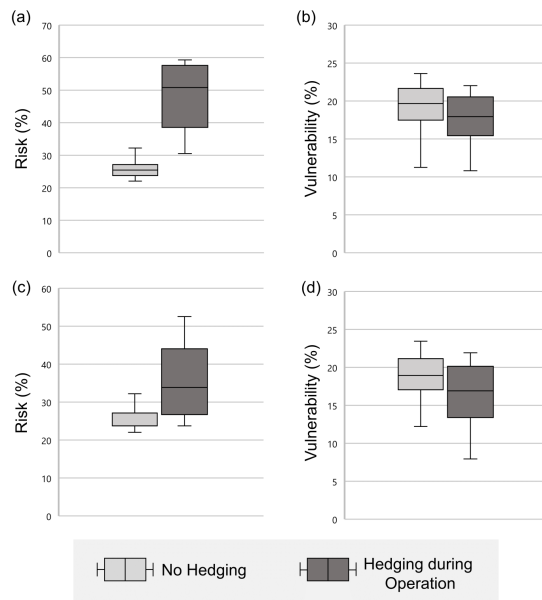


Figure 4.4: Performance indices measured from the demand-side: (a) risk of water for A uses, (b) vulnerability of water for A uses, (c) risk of water for M&I uses, and (d) vulnerability of water for M&I uses

4.6 Conclusion

The primary motivation for this study was to conduct a comparative analysis between distinct formulations of the optimization algorithms used in reservoir operation problems and with actual operation results. Consequently, six different models were formulated using two eminent optimization algorithms: DP and EMODPS. The performance of all optimization models, measured as the magnitude of the deficit compared to the target storage and demand, surpassed the results from the actual operation with real-time calculations and experience of related experts under normal inflow conditions. Among the optimization models, solutions from the EMODPS models dominated those from the DP models in their proximity to the ideal point and variety of solutions across the entire Pareto front.

Another objective of this study was to combine the actual zone-based hedging rule into the simulation process of optimally derived policy using an optimization algorithm, in which few studies have succeeded thus far. The solutions from the EMODPS-Gaussian model were simulated under a historical inflow series with and without coupling the actual hedging policy. Despite the expected improvement in supply indices, the magnitude of failure from the demand side also increased after the incorporation of the actual policy. Considering the highly risk-averse nature of water users, the decrease in the average size of damage at the expense of increased frequency may be worthwhile in actual operations.

Ultimately, the findings from this study is not limited to the Boryeong Dam and can be extended to other reservoirs that are highly vulnerable to droughts with conflicting objectives, and rely on relatively simple hedging rules. The analytical framework of

this study may also be expanded to include complex reservoir network systems with a cascade of reservoirs with different objectives, such as water-energy nexus (Barros et al. 2003; Liu et al. 2019a). Despite multiple findings from the application of the proposed methodology, improvements can be achieved in future studies. In particular, the values of different MOEAs could be explored by adopting various types of MOEAs during the optimization process. Conversely, the simulation under future inflow scenarios that reflect the increased risk of future climate variability, particularly from climate change, and analysis of the proposed alternatives under a wider climate range would enable the assessment of their performance under even more extreme climate conditions.

Chapter 5. Reforming the Hedging Rules with Adaptive Operations against Multi-year Droughts

This chapter is partially drawn from the following peer-reviewed journal article:

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5.1 Abstract

In this study, the zone-based hedging rule, which is the main operating policy adopted from multipurpose reservoirs in Korea is adjusted to reflect the multi-year droughts caused by climate change. Synthetic inflow series with different magnitudes of long memory were generated using the autoregressive fractional integrated moving average (ARFIMA) model. The generated inflow series were then disaggregated into shorter time steps and utilized as input variables to derive the hedging rules. The alternative hedging rules from this study were used in adaptive reservoir management by newly updated information. Finally, the performance of the suggested policy is measured in terms of frequency and magnitude under the historical inflow series. As a result, adaptive reservoir management demonstrated improvements in the following terms of the frequency of critical failures (water deficit ratio greater than 30%): 6.14% of the simulation period in the status quo (SQ) policy, and 2.99% in the adaptive management. However, the overall reliability of the reservoir during the simulation horizon was better when operated with the SQ policy (41.19%) than the results from adaptive management (26.42%). Because this result is in a good agreement with the original objective of the hedging rules, the adaptive policy suggested in this study holds promise and may be utilized in further reservoir management under drought conditions.

5.2 Introduction

Reservoirs have been widely adopted as the principal source of freshwater for various purposes including irrigation, municipal, industrial, hydropower generation, and flood mitigation (Labadie 2004). Nevertheless, the recent surge in the frequency and magnitude of abnormal hydro-climatic variabilities and their consequences (Loucks 2000; Salazar et al. 2016; Kim et al. 2021) have forced the traditional reservoir management strategies to be reformed. The primary factor that complicates the advancement in operating policies is the inherently sophisticated nature of water management problems caused by conflicting interests and distinctive attitudes toward the risk of involving stakeholder groups (Castelletti et al. 2008; Quinn et al. 2019). Furthermore, the global increase in unpredictable severe meteorological multi-year droughts and their consequences have also contributed to the intricacy of water resources management schemes.

To date, the development of reservoir operating policies has been led by either optimization models based on optimization algorithms or simulation-based models based on predefined rules (Stedinger 1984). A few optimization algorithms that have been adapted for solving reservoir operation problems include: nonlinear programming (Birhanu et al. 2014), dynamic programming (Stedinger et al. 1984), and genetic algorithms (Cheng et al. 2008). Therefore, since numerous applications have been made regarding the optimization algorithms in reservoir operation problems, the reader is referred to the prominent review papers by Yeh (1985) and Labadie (2004) for more details. So, this paper intends to focus on the noted shortcomings of these optimization models in actual applications that primarily arise because of the discrepancies between academia and onsite end users.

In contrast to the optimization models, actual operations in actual multipurpose reservoirs rely more heavily on simulation-based predefined operation rules. Two notable types that are affiliated with the rule-based predefined policies are the standard operating policy (SOP) (Klemeš 1977) (the S-shaped curve of operation) and various types of hedging rules (Shih and ReVelle 1994; Kim and Kim 2021). Because the SOP targets meeting the demand on all possible occasions and spills only when the capacity is exceeded, the practical application of SOP is largely limited and is only adapted in theoretical simulations (Shih and ReVelle 1994). Contrastingly, hedging rules focus on preventing damage to the target reservoir from potential droughts and consequential water deficits by demand reductions before water shortages (Neelakantan and Pundarikanthan 1999). Among the various candidate types of hedging rules, onsite reservoir operators prefer to manage their reservoirs with relatively straightforward policies (Mohammad Ashrafi 2021), which are implementable under wide circumstances.

Other than the aforementioned inherently complex nature of the field of water resources management, another key decisive component, climate change, is forcing the review of conventional water resources management strategies. The firm linkage between climate change and water resources management has been proven in multiple past studies (Frederick and Major 1997; Sowers et al. 2011). As such, this study aims to focus on the two distinctive characteristics of climate change: non-stationarity (Milly et al. 2008) and Knightian uncertainty. While the non-stationarity refers to the change in historical statistics, Knightian uncertainty refers to the property of climate change in which even the distribution of future variables is unknown. Therefore, due to these two attributes, climate change has induced a rapid surge in irregular hydro-climate events

such as extreme floods and droughts that last for multiple years. In response, measures that aim to minimize the damage from hydro-climate disasters have been developed including robust (Kim et al. 2021) and adaptive (dynamic) decision making strategies (Herman et al. 2020). Thus, this study focuses on the adaptive approach, in which the decisions are updated with the introduction of new information, and aims to apply adaptive reservoir management based on multiple status quo (SQ) rule curves.

In this study, the rule curve of the reservoir which is adapted as the primary operating strategy is reformed by considering the multi-year drought during the rule generation process. The reformation procedure is conducted by utilizing the identical algorithm that actual reservoir managers onsite have developed to generate policies that are applicable during actual operations under drought conditions. The alternative policies are then adaptively applied with the rule curve that is actually being dynamically adapted in the target reservoir by updating information based on the standardized runoff index (SRI). For evaluation, a simulation using historical inflow series was conducted, and the results from the simulation are compared with the results from the SQ policy.

5.3 Methodology

5.3.1 Reservoir Management Based on Hedging Rules

The fundamental concept behind diverse types of hedging rules in reservoir operations lies in reducing the risk of water deficits in the future in the expense of shortage in the current release (You and Cai 2008a). This action is possible by utilizing only partial release, although all or at least more of the target demand could be provided (Hashimoto et al. 1982b). The specifics of hedging policy, such as the portion to be deducted and the timing of reduction, are determined according to predefined hedging rules of various types (Kim and Kim 2021). Chiefly, hedging rules are a type of simulation model-based decision support tool commonly adopted in onsite reservoir management problems (Bayazit and Ünal 1990; You and Cai 2008b). Owing to their applicability and effectiveness, numerous studies on hedging rules linked with optimization models in reservoir operations (Neelakantan and Pundarikanthan 1999; Tu et al. 2008; Taghian et al. 2014; Eum et al. 2011) have been conducted. The results from the previous studies proved that optimally derived hedging rules enhanced the efficiency of reservoir performance under both normal and drought conditions.

Depending on the shape and the specifics of the hedging policy, hedging rules developed in previous studies can be classified into one of the following: one-point (SOP), two-point (Bayazit and Ünal 1990), continuous (Hashimoto et al. 1982b), and zone-based (Shih and ReVelle 1994, 1995). Starting from the simplest one-point hedging rule, which is a slight variation from the SOP, other alternative hedging rules were suggested that all aim to prevent failures with greater magnitude in future operations. More specifically, to determine the timing and amount of demand reduction, available

storage and demand at the time step are often selected as the trigger (Taghian et al. 2014).

Among the various types of hedging rules previously introduced, this study aims to focus on the specific type of hedging known as zone-based (discrete) hedging rules. This is because all the multipurpose reservoirs in the Republic of Korea have adopted this policy as their main reservoir management technique. The fundamental concept behind zone-based hedging rules is adopted in multipurpose reservoir management with the term, water supply adjustment standards (WSAS). Figure 5.1 displays the zones of the Boryeong multipurpose reservoir, in which the standards of the sequential reduction in water supply that is proportional to the demand at each time step is based on. For specific explanations regarding the concepts behind zone-based hedging rules and how they are being applied, refer to the descriptions and tables in subsection 4.3.3. Explanations regarding the shape of the zones in Figure 5.1 and how they are determined are explained in subsection 5.4.2.

5.3.2 Time Series Models with Long Memory

The concept of long persistence, the existence of significant dependence between observations separated by a long time span (Hosking 1984), initially started from Hurst (1951), in which the hydrological and geophysical data (annual minimum series of the Nile River) were heuristically identified. This long-range dependence of hydro-climatic variables is often linked with the concept of non-stationarity (Koutsoyiannis 2006), which is identified as a key characteristic of climate change. Particularly, Montanari et al. (1997) identified two main reasons behind long persistence, also referred to the Hurst phenomenon in the literature: (1) inherent long-term dependence (or long

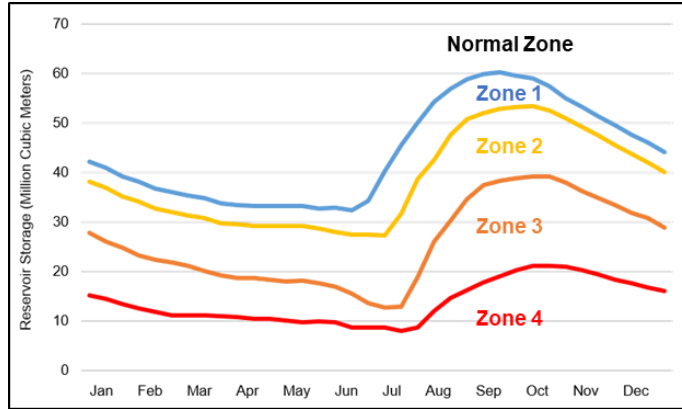


Figure 5.1: Actual rule curves of the Boryeong multipurpose reservoir in Korea. The storage of the reservoir is divided into five zones according to the four curves in the figure

memory) in the time series itself and (2) the non-stationarity of the time series. Although the principle behind the Hurst phenomenon is yet to be completely determined, multiple studies have discovered the existence of long memory in hydrological data (Koutsoyiannis 2002, 2003).

A widely known, heuristic approach often utilized in detecting long memory from time series data is often conducted using the Hurst exponent H ($0 < H < 1$) (Montanari et al. 1997). Specifically, a series with H less than or equal to 0.5 indicates no long-term persistence; moreover a series with exponent H greater than 0.5 indicates strong long persistence. Multiple models including the Hurst exponent have been developed and utilized so that the series generated from the model includes long memory, primarily because of the relative simplicity of the heuristic estimation methods of the Hurst exponent (Montanari et al. 1997). As an example, the fractional autoregressive integrated moving average (FARIMA) model (Hosking 1984) was designed to include long-term

persistence in the series with fractional differencing. This model is often referred to the autoregressive fractional integrated moving average (ARFIMA) model.

With a time series X_t with zero mean, the ARFIMA(p, d, q) model can be generally expressed using the following equation (Montanari et al. 1997):

$$\Phi_p(B)(1 - B)^d X_t = \Theta_q(B)\epsilon_t \quad (5.1)$$

where B = backward shift operator ($BX_t = X_{t-1}$); $\Phi_p(B)$ = p -order autoregressive polynomial; $\Theta_q(B)$ = q -order moving average polynomial; ϵ_t = independent, zero mean white noise; and d = order of differencing, not limited to integers but can include fractional values.

Specifically, the order term d in Eq. 5.1 is closely related to the Hurst exponent H with the following relationship:

$$d = H - 0.5 \quad (5.2)$$

Compared to other models, ARFIMA models have been widely adopted during the generation of long memory series primarily because of their strengths which allow the inclusion of both short- and long-term persistence. For instance, Mudelsee (2007) utilized the Hurst exponent during the spatial aggregation process across various rivers. More recently, Yang and Bowling (2014) applied the ARFIMA model to identify and estimate the daily streamflow series which were later analyzed for the effect of urbanization in the Great Lakes region. In this study, multiple ARFIMA models with different H values were used to generate inflow series which were used to create the standards for the hedging rules in reservoir operations.

5.4 Application on the Boryeong Multipurpose Reservoir

5.4.1 Application Overview

To derive alternative hedging rules with the consideration of multi-year droughts with long-term persistence time series models, this study first generated an annual inflow series while preserving the statistics from the historical inflow series (average and standard deviation) using the ARFIMA model. Then, a non-parametric disaggregation method based on the k-nearest neighbor (k-NN) resampling approach proposed by Nowak et al. (2010) was applied to transform the annual inflow series into a 10-day series. This disaggregation procedure was conducted because the "DrResOPT" software, a program utilized to derive the standards for the hedging rule used by onsite reservoir operators (K-water) is based on a time step of 10 days. While preserving other input variables such as monthly demand, reservoir capacity, and reliability, only the inflow series were assumed to vary so that the alternative rule curves were generated with inflow series with different magnitudes of long memory. A total of seven cases of alternative hedging rules (Cases 1 through 7) were generated with the inflow series with H from 0.6 to 0.9. After creating alternative rule curves, this study applied adaptive reservoir management, in which the alternative rule curves were selectively utilized according to updating information.

Ultimately, this study advances the previous work by Seo et al. (2019) in the following manner: (1) utilizes the synthetic inflow series during the zone-based hedging rule generation, (2) includes 100 sets of a 500 year series of synthetic inflow during the zone derivation process, and finally (3) applies an adaptive reservoir management scheme to cope with climate change-induced long-lasting droughts. Moreover, whereas

the previous study from Kim et al. (2021) only utilized the hedging rule by combining with optimization models, this study endeavors to reform the SQ policy against multi-year droughts and adaptively apply the alternatives with newly updated information according to each time step.

5.4.2 Procedures for Hedging Rule Generation

Disaggregation

Because the generated inflow series from the ARFIMA model is composed of an annual time step and the algorithm for hedging rule generation requires an inflow series with time steps of 10 days, a step for disaggregation is necessary. Among multiple candidates of disaggregation methods, a nonparametric stochastic approach based on the k-NN resampling method (Nowak et al. 2010) was selected in this study. Briefly, the disaggregation method adopted in this study identifies the vector of annual aggregate flows among the historical inflow series that is most identical to the inflow to be disaggregated. Then, the annual aggregate flow to be disaggregated is resampled according to the weight function shown in Eq. 5.3:

$$W(i) = \left(\frac{1}{i} \right) / \left(\sum_{i=1}^K \frac{1}{i} \right) \quad (5.3)$$

In this study, K , the number of nearest neighbors is selected according to the heuristic equation ($K = \sqrt{n} = 5$, where n is the number of years of observed data). Consequently, the outcome of this procedure, the disaggregated inflow series, can preserve the basic statistics (mean, variance, coefficient of skew, maximum, and minimum values) of the historical inflow series (Nowak et al. 2010). For detailed explanations

regarding the disaggregation method adopted in this study, the reader is referred to Nowak et al. (2010).

Rule Curve Generation

After the annual inflow series are disaggregated into a 10-day series, the rule curve of the reservoir, which is utilized as the standard for zone-based hedging rules, is determined according to the algorithm used onsite. Accordingly, three out of the four rule curves were established according to the results from the iterative reservoir simulation for 500 years, and one of them was calculated from the previous computation results. Figure 5.2 summarizes the algorithm used for the rule curve generation process.

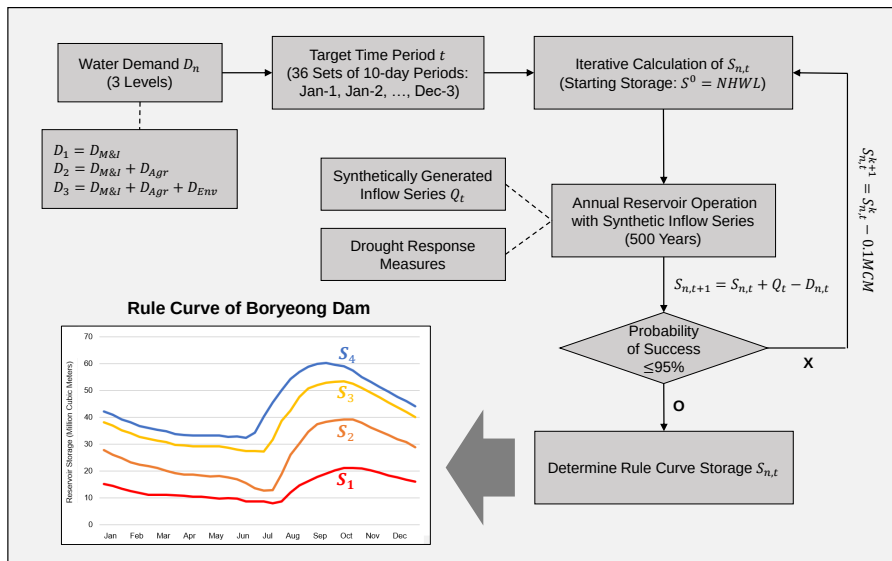


Figure 5.2: Procedures utilized to derive rule curves in hedging rules. The flow chart displays an example of the Boryeong multipurpose reservoir

For illustration, the sequence for deriving the rule curve for the caution stage among the four stages corresponding to attention, caution, alert, and critical zones is explained. First, the demand corresponding to each time step was determined. The demand for the caution stage is D_3 , which is the aggregate sum of M&I, A, and E water uses corresponding to each time step. Then, starting from the storage at the normal high water level, an iterative calculation of the reservoir using the continuity equation is conducted until the probability of success falls below 95% during the 500 years of simulation. During the simulation, a synthetically generated inflow series with long-term persistence was utilized along with the drought response measures available at the Boryeong reservoir. The lowest value of the storage level that meets 95% reliability is selected as the storage level corresponding to each stage. The rule curve for the attention stage is inferred from the rule curve for the caution stage by adding the amount of water corresponding to the 15 day demand. In this study, the procedures for the rule curve generation were improved as the follows: (1) by including multi-year drought in the rules by simulation using the synthetically generated inflow series with long memory, and (2) by generating multiple rule curves corresponding to each case to reduce the uncertainty during the synthetic inflow generations.

5.5 Results and Discussion

5.5.1 Inflow Series Generation

To remove the potential uncertainty arising from the synthetic inflow generation, this study utilized 100 cases of a 500 year series of synthetic inflow for each case with different Hurst coefficients ($H = 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9$). The statistics of the selected cases, which were utilized to derive the storage of the zones in the next step of the study, are shown in Figure 5.3. During generation, the average and standard deviation of the annual series aimed to replicate the values from the observed statistics. The Hurst coefficient, H , indicates the extent of the long memory incorporated into the generated time series. A time series with higher H indicates stronger long-term persistence, and consequently, a higher chance in containing a multi-year drought in the generated time series model.

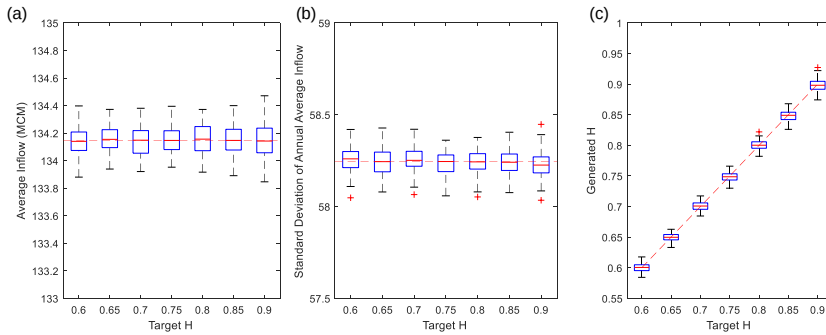


Figure 5.3: The statistics of the selected 100 cases of generated time series: (a) annual average (b) standard deviation, and (c) Hurst coefficient

5.5.2 Alternative Hedging Rule Generation

After the selection of 100 cases among the generated time series with different long-term persistence, the selected annual series were then disaggregated into 10-day intervals. Then, the 10-day time series were used during the hedging rule generation, with the algorithm that is currently being used in actual rule generation adapted onsite. Consequently, the rules generated in this study advance the SQ zone-based hedging rule with regard to the following points: (1) includes the effect of multi-year drought during the generation process, and (2) provides a range of zones by using 100 cases. Among the 100 cases, the median values were selected as representative values and utilized during adaptive reservoir operations. The generated rules are shown in Figure 5.4.

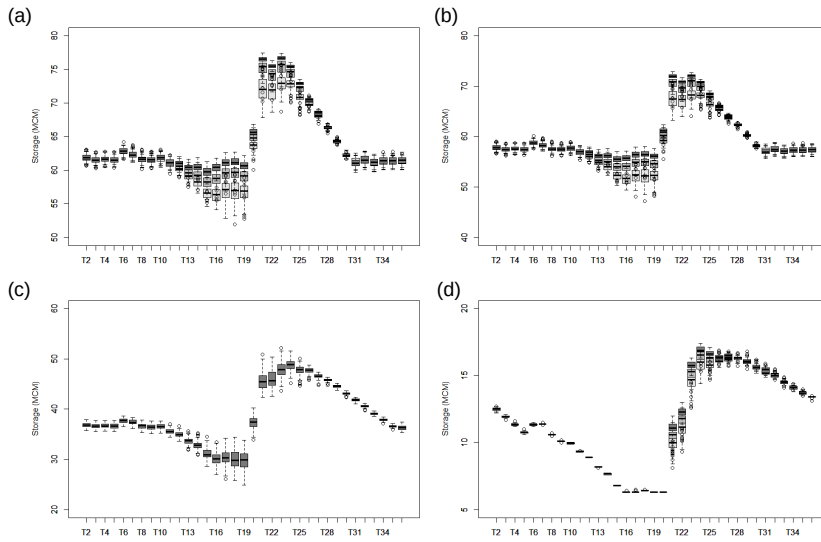


Figure 5.4: Alternative rule curves of the Boryeong Dam derived from synthetic inflow series with long memory (Cases 1, 4, and 7 or $H = 0.6, 0.75$, and 0.9): (a) zone 1 (b) zone 2 (c) zone 3, and (d) zone 4. Darker boxplots indicate rules derived using inflow series with stronger long-term persistence

Inclusion of the inflow series with long-term persistence during hedging rule generation yielded divergent policy outcomes. In general, the rule curves generated from the long memory inflow series resulted in higher rule curves than the SQ hedging rules. Higher values of rule curves indicate that the water supply reduction should take place earlier, even when there is enough water stock when operated under the SQ rules. Moreover, as shown in Figure 5.4, among the alternative rule curves, policies established using the inflow series with stronger long-term persistence yielded even higher rule curves. Given that the alternative rule curves aimed to cope with recent multi-year droughts, higher rule curves or preemptive actions against potential future droughts appear to be legitimate results.

5.5.3 Adaptive Reservoir Operations with Alternative Hedging Rules

Until this point of the study, alternative zone-based hedging rules were created by reflecting the multi-year drought through the generation of a time series with long persistence. However, as the procedures were utilized, the novel policies were created by only considering the reliability of the reservoir, and direct adoption of the alternative policy is hardly feasible. Thus, this study aims to dynamically employ the generated alternatives during reservoir operation under drought conditions with adaptive actions (Pahl-Wostl 2007). Particularly, adaptive strategies primarily utilize new information when updating decisions in response to new states. To trigger the alternative rule curve, the standard runoff index accumulated over 12 months (SRI12) was selected. The probability distribution employed during the fitting process for the SRI calculation was selected as the three-parameter log-logistic distribution.

Among the SRI values with different lengths, SRI12 was selected for the following

two reasons. Firstly, since the primary motivation behind this study was in deriving alternative reservoir operating policy based on adaptive decision making against multi-year droughts, including the information accumulated for 12 months is more reasonable than shorter periods. The second reason behind the selection of SRI12 was in considering policy inertia (Giuliani et al. 2014), which refers to low flexibility of water resources policy especially in laws related to water resources infrastructure unless either a dramatic failure or water conflict takes place. For comparison, this study performed simulations with SRI3, SRI6, SRI12, and SRI24 selected as triggers for adaptive operations. The simulation resulted in the change of policy for 47.10%, 42.39%, 26.09%, and 22.83% of the simulation period, respectively. Furthermore, the overall vulnerability of the reservoir, measured in terms of the average magnitude of failure during the simulation horizon, hardly improved when the system was operated with SRI6 as the trigger. Since the change of the operating policy for more than 40% may lead to extreme stress to the target system and the enhancement of the vulnerability can be considered insignificant, this study selected SRI12 as the trigger for adaptive reservoir management.

Ultimately, the results from the adaptive operations were analyzed, focusing on the trade-off relationship between supply and demand. The standards adopted in this study for adaptive reservoir operations are listed in Table 5.1.

Table 5.1: SRI triggers utilized in adaptive reservoir operation of the Boryeong multi-purpose reservoir

SRI Trigger Value	Cumulative Probability	Interval Probability	Utilized Policy
$0 < \text{SRI}$	100%	50%	K-water SQ Rule (Figure 5.1)
$-0.180 < \text{SRI} \leq 0$	50.00%	7.14%	Case 1 ($H = 0.6$)
$-0.366 < \text{SRI} \leq -0.180$	42.86%	7.14%	Case 2 ($H = 0.65$)
$-0.566 < \text{SRI} \leq -0.366$	35.71%	7.14%	Case 3 ($H = 0.7$)
$-0.792 < \text{SRI} \leq -0.566$	28.57%	7.14%	Case 4 ($H = 0.75$)
$-1.068 < \text{SRI} \leq -0.792$	21.43%	7.14%	Case 5 ($H = 0.8$)
$-1.465 < \text{SRI} \leq -1.068$	14.29%	7.14%	Case 6 ($H = 0.85$)
$\text{SRI} \leq -1.465$	7.14%	7.14%	Case 7 ($H = 0.9$)

To evaluate the proposed methodology, a reservoir simulation model based on a 10-day time period was developed, and the historical inflow series of 23 years after the construction of the reservoir (January 1998 through December 2020) was utilized. The starting storage for the simulation was observed storage at the beginning of 1998 (35.342 MCM). Finally, for the simulation to reflect the conditions of actual operations, the Boryeong conduit, an alternative operated in the case of extreme droughts was also included in the model according to its operating policies. The simulation results were finally evaluated using the reliability and vulnerability concept from Hashimoto et al. (1982b) by focusing on the trade-off relationship between the reservoir operators and water users.

Release Rule Comparison

The first advantage arising from the adaptive operations proposed in this study compared to the operations based solely on the SQ policy is the more flexible release decisions through the introduction of the SRI as the trigger. For instance, whereas the release decisions made from the SQ policy (Figure 5.5a) only considers the storage, the adaptive policy includes both the information from the storage and the SRI when making decisions. This indicates that even under the same storage value, the decision may vary according to the value of the SRI. The extensive range of decisions made from the adaptive policy is demonstrated in the contour plots drawn from the release decisions from the adaptive operations proposed in this study (Figure 5.5b). In addition to the improvements in adaptive operations in terms of reliability and vulnerability, the extension of the range of release decisions also presents another advantage over the SQ policy.

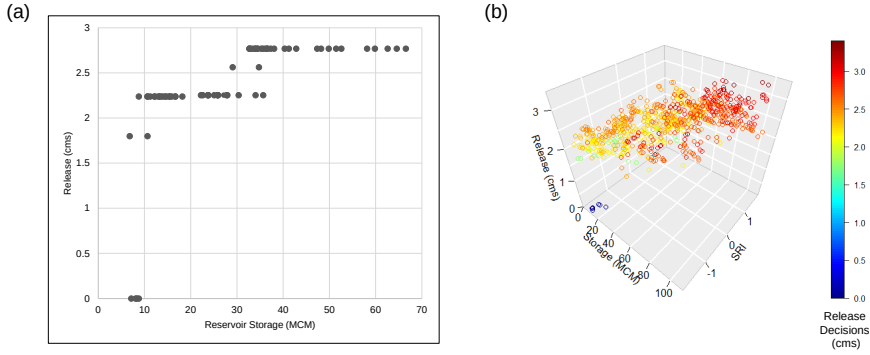


Figure 5.5: Release decisions from (a) SQ policy operations in April and (b) adaptive operations during all time periods proposed from this study. While the SQ policy only allows stepwise release decisions mainly based on the storage, adaptive operations enable a wider range of release decisions over time

Simulation Results Evaluation

Moreover, the simulation results of both operations were comparatively analyzed from the widely known concept of risk and vulnerability suggested by Hashimoto et al. (1982b) both from the reservoir operators and the end water users. Before the evaluation process, the criteria for evaluation were selected in the following manners: (1) the percentage of the reservoir storage for the supply side, and (2) water deficit ratio (i.e., $\max\left(\frac{D_t - R_t}{D_t}, 0\right)$) for the demand side. Figure 5.6 shows the distribution of either the storage (Figure 5.6a) or water deficit (Figure 5.6b) over the drought period (Jun-2014 through Sep-2019), and Figure 5.7 shows the same during normal periods. Darker bars in Figure 5.6 and Figure 5.7 indicate lower storage and greater magnitude of water deficit, and consequently, there are undesirable outcomes from the supply and demand, respectively.

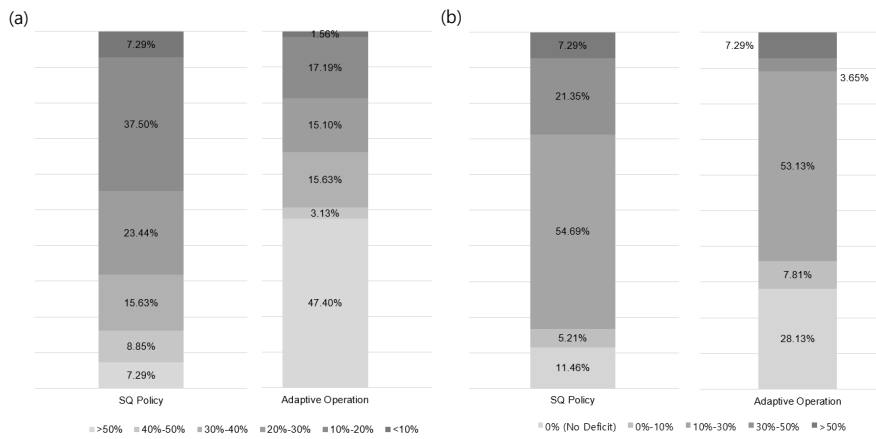


Figure 5.6: Frequency and magnitude (a) supply side (in terms of storage) and (b) the demand side (in terms of water deficit) during drought periods (Jun-2014 through Sep-2019)

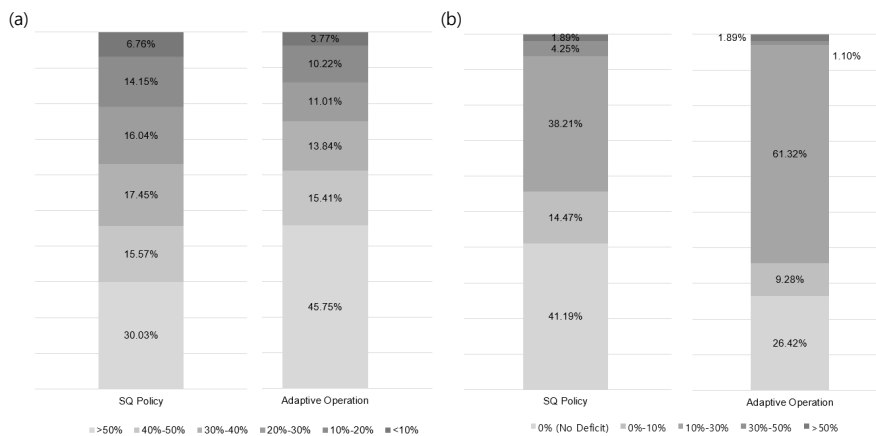


Figure 5.7: Frequency and magnitude (a) supply side (in terms of storage) and (b) the demand side (in terms of water deficit), during normal periods (Jan-1998 through May-2014 and Oct-2019 through Dec-2020)

Overall, the performance of the reservoir showed a trade-off relationship between supply and demand as initially expected. From the supply side, operating the reservoir with the alternative policies in an adaptive manner resulted in improved performance in terms of both magnitude and frequency in both normal and drought conditions. Because the reservoir storage below 20% conventionally indicates critical failure in terms of supply, the threshold of critical failure for the supply was selected as 20%. In terms of critical failure during the entire simulation horizon, the SQ policy yielded in average frequency, magnitude, and duration of 26.45%, 12.95%, and 12.88 10-day periods, respectively. Contrarily, adaptive operations suggested in this study resulted in 15.10%, 13.69%, and 13.89 10-day periods, respectively. The results indicate that the adaptive operations enhance the frequency of critical failures at the expense of magnitude and duration in terms of reservoir storage.

Contrastingly, the performance of the simulation measured from the water users in terms of water deficit ratio showed relatively similar, yet disparate results from the performance from the supply side. Figure 5.6b shows the distribution of the magnitude of the water deficit ratio during historical drought periods, whereas Figure 5.7b shows the same during normal periods (Jan-1998 through May-2014 and Oct-2019 through Dec-2020). In terms of overall reliability, SQ policy especially yielded in better results (41.19%) than adaptive operations (26.42%) under normal periods. Operation of the reservoir with the adaptive operating method proposed in this study largely improved the frequency of critical failures in terms of demand (incidents with water deficit ratio greater than 30%) during the drought period. Whereas the adaptive operations failed to improve the frequency of water deficit ratio greater than 50% than the SQ policy, adaptive operations were able to improve the frequency of the second worst failure

(incidents with water deficit ratio greater than 30% and less than 50%) by 17.7%. Finally, given that the fundamental objective of general hedging rules is to reduce the frequency of critical failures at the expense of overall performance (Eum 2007), adaptive operations proposed in this study better achieve this specific objective, and also hold promise under extreme multi-year drought incidents.

5.6 Conclusion

Because multiple reservoirs utilize various types of hedging rules as their main operating policy, this study first proposed seven alternative rule curves by considering the long-term persistence proven in historical multi-year droughts. To derive alternative rule curves against multi-year droughts, the inflow series reflecting long-term persistence were generated with ARFIMA models with diverse strengths of long memory and disaggregated using the nonparametric stochastic approach based on the k-NN resampling method. The synthetic inflow series were then used to derive alternative hedging rules, which were finally used in adaptive reservoir operations. The SQ policy and adaptive operations with alternative rule curves were simulated under the historical inflow series, which were ultimately evaluated in terms of risk and vulnerability.

The key findings of this study are organized and analyzed comparatively. First, the alternative rule curves created with inflow series containing long-term persistence yielded higher rule curve standards compared to the SQ policy. Thus, operating the reservoir according to the alternative rule curves suggests that water supply reduction occurs earlier. Given that the consequences of multi-year droughts may be detrimental and lead to severe water deficits, the alternative policies derived from this study can be concluded to yield reasonable results against multi-year droughts. Moreover, the results from the simulation using historical inflow series indicate that the adaptive operations reduce the frequency of most water deficit incidents at the expense of overall reliability. Because this result is the intention of the hedging rules, the results support the adaptive reservoir operations proposed in this study especially during the drought periods.

Despite the key findings from the results of this study, few possible improvements

remain for future research. Primarily, this study utilized the algorithm for hedging rule derivation which was utilized onsite (Figure 5.2) so that the alternative policies are not so distant from the SQ policy. However, a further analysis of the reliability of the algorithm should be conducted. Specifically, a sensitivity analysis with the input variables used in deriving hedging rules such as demand, synthetic inflow series, drought response measures, and the probability of success. Moreover, because many reservoir systems are complex and involve multiple water resources infrastructures, consideration of the system and expanding the model to include related structures also hold promise.

Chapter 6. Improving the Robustness of Reservoir Operations with Stochastic Dynamic Programming

This chapter is partially drawn from the following peer-reviewed journal article:

Kim, G. J., Kim, Y.-O., and Reed, P. M. (2021). "Improving the robustness of reservoir operations with stochastic dynamic programming." J Water Resour Plan Manag, 147(7), 04021030. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001381](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001381)

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6.1 Abstract

Reservoir operations should consider both adaptiveness and robustness to deal with two of the main characteristics of climate change: non-stationarity and deep uncertainty. In particular, robust operational strategies are distinguished from risk-neutral expected value optimization in the sense that they should be satisfactory over a wider range of uncertainty and improve the ability of a reservoir system to adapt to climate change. In this study, a new framework named robust stochastic dynamic programming (RSDP) is proposed that couples robust optimization (RO) with the formulations of objective function or constraints used in stochastic dynamic programming (SDP). Two main approaches of RO, namely feasibility robustness and solution robustness, are both considered in the optimization algorithm and consequently, this study uses the Boryeong multipurpose dam to evaluate three SDP framings: conventional-SDP (CSDP), RSDP-feasibility robustness (RSDP-F), and RSDP-solution robustness (RSDP-S). These three SDP formulations were used to derive optimal monthly release rules for the Boryeong Dam, and their relative performance were evaluated using simulations of a broader range of inflow scenarios. The simulation-based re-evaluations of the resulting reservoir operational policies were quantified using a wide range of metrics that include reliability, resiliency, and vulnerability, as well as regret-based robustness metrics. The results of this study suggest that the RSDP-S model not only increases the range of possible solutions, but also yields more desirable operation outcomes under extreme climate conditions with respect to both traditional and robustness metrics.

6.2 Introduction

Securing stable freshwater sources through water resource infrastructure, such as reservoirs, levees, and river regulators has been an important task for humanity since the beginning of civilization. Nevertheless, recently, increases in the frequency of extreme floods and droughts (Loucks 2000; Cosgrove and Loucks 2015), population growth and urbanization, and climate change (Salazar et al. 2016) have caused traditional planning and management frameworks for water resource infrastructure to evolve to be able to more effectively deal with these compounding challenges (Bouwer 2000; Li et al. 2010; Poff et al. 2016; Brown et al. 2015; Quinn et al. 2018). Climate change is difficult to address due to two of its main characteristics: non-stationarity (Milly et al. 2008) and deep (or Knightian) uncertainty (Lempert 2014; Dittrich et al. 2016). In the field of water resources, non-stationarity commonly refers to changes in the statistical properties of random variables with shifts in time (Razavi et al. 2015), while deep uncertainty accounts for a situation in which analysts do not know or cannot agree on probabilistic distributions of key random variables and parameters (Lempert 2003; Ben-Haim 2006; Hallegatte et al. 2012; Brown et al. 2020). Numerous studies conducted over the past two decades, which have been focused on planning and managing water resources systems during climate change also stress the importance of considering non-stationarity and deep uncertainty in adaptation strategies (Haasnoot et al. 2013; Steinschneider and Brown 2012; Herman et al. 2015, 2020; Moallemi et al. 2020).

Given challenging hydro-climatic variabilities and uncertain demands impacting actual water resources systems (Johnson et al. 1991; Trindade et al. 2017; Fletcher et al. 2019; Woodward et al. 2014), scholars have long agreed on the necessity of

seeking *robust* decisions that are "insensitive to errors, random or otherwise, in the estimates of those parameters affecting design choice," (Matalas and Fiering 1977; Herman et al. 2015; Giuliani and Castelletti 2016) when compared with traditional least cost optimization approaches. Given that there can be diverse concerns across the economic, engineering, and socio-ecological objectives of water systems (Botte and Schöbel 2019), recent studies have highlighted the importance of avoiding myopic definitions of robustness that fail to account for diverse stakeholder interests and needs (Kasprzyk et al. 2013; Herman et al. 2014; McPhail et al. 2018; Hadjimichael et al. 2020). Early conceptions of robustness in the optimization literature by Mulvey et al. (1995) and Watkins and McKinney (1997) sought to move beyond traditional deterministic least cost optimization by defining robust optimization (RO), which seeks to better address risk aversion by changing the formulations of mathematical programming models. Recently, Ray et al. (2014) applied RO to a city's water infrastructure planning example with two conflicting objectives by including both solution robustness (penalizing variance of the objective) and feasibility robustness terms (penalizing violation of the constraints) into the objective function of the model and demonstrated a trade-off relationship regarding stakeholder preferences. Ray et al. (2014) illustrated the necessity of adopting robust decisions despite the unlikelihood of a catastrophe with their experiment. Roach et al. (2016) compared the performance of RO with a different set of robust decision theory, info-gap (IG) (Ben-Haim 2006), and concluded that solutions derived using RO cover a wider range of varying target robustness levels, whereas solutions produced by IG yielded dominant solutions under a limited range. However, this study took only the financial aspects of the strategy into consideration when applying RO, which may result in a different set of solutions to be implemented in

an actual system. Yet, to date no study has coupled RO to an actual reservoir operation problem in which the primary concepts of RO can be incorporated into the problem formulation and optimization processes.

Over the last 50 years, reservoir operations have been one of the most active areas of research in the water systems literature (Yeh 1985; Labadie 2004; Quinn et al. 2019; Raso et al. 2019, 2020). Recent studies are focused on two main topics: forecast informed reservoir operations, (Tejada-Guibert et al. 1995; Kim and Palmer 1997; Zhao et al. 2011; Anghileri et al. 2016; Turner et al. 2017) and advancing optimization techniques that can broaden the array of decision relevant concerns that can be incorporated into the design of operations (Castelletti et al. 2010, 2013; Giuliani and Castelletti 2016; Quinn et al. 2019).

This study contributes a robust stochastic dynamic programming (RSDP) approach in which mathematical optimization concepts from RO are integrated into an SDP framework to identify reservoir control strategies that balance optimality and robustness. A conventional SDP (CSDP) baseline model is compared with two alternative robust SDP (RSDP) models that each emphasize two main concepts of robustness in mathematical programming as proposed by Mulvey et al. (1995): (1) solution robustness (nearness to optimality across all scenarios) and (2) feasibility robustness (nearness to feasibility across all scenarios). For more details on solution robustness and feasibility robustness, see section 6.3 of this dissertation.

The primary contribution of this paper is an extension of SDP to identify both optimal and robust reservoir operation rules to better confront the non-stationarity and deep uncertainty associated with climate change. This study proposes an integrated framework of SDP and RO for deriving optimal reservoir operating policies that are

then further evaluated with regret-based robustness evaluation metrics under a wide range of inflow scenarios. This work is an advancement of studies conducted by Ray et al. (2014) and Pan et al. (2015) in which only traditional evaluation metrics such as sustainability, reliability, and vulnerability were considered for evaluation, or in which only a limited concept of robustness was considered.

The remainder of this paper is organized as follows. In section 6.3, the key components and concepts of the main algorithms used in this study are introduced: SDP, RO, and the proposed RSDP. In section 6.4, the Boryeong Dam test case, as well as the specific mathematical formulations of its operations are described. In section 6.5, the results of optimization, sample operation under specific drought incidents, and metric evaluations are presented, highlighting the potential benefits as well as challenges that emerge with RSDP operational alternatives. Finally, in section 6.6, this study concludes with mentioning key findings, possible shortcomings of this research, and suggested future work.

6.3 Methodology

6.3.1 Stochastic Dynamic Programming

SDP, based on the concept of **Bellman's principle of optimality**, is definitely one of the most popular optimization algorithms used, by far, for the derivation of optimal reservoir operation policies (Stedinger et al. 1984; Kim et al. 2007; Haguma et al. 2018). Complex reservoir operational design problems are nonlinear, stochastic, and multiobjective (Giuliani and Castelletti 2016), which requires optimization algorithms that are capable of considering diverse types of objectives and uncertainties. Since SDP model can incorporate these complexities through discretization of the state space and use of sampling schemes based on statistical methods, they have been continuously used for solving reservoir operation problems. Recently, based on Butcher (1971), who first introduced the use of the discrete, lag-one Markov process to describe uncertainty in future streamflow, advanced variations, such as SDP models with multiple uncertainties (Shokri et al. 2013), stochastic dual dynamic programming (SDDP) (Gjelsvik et al. 2010; Raso et al. 2017) have been adopted for analyzing complex water resource systems. As SDP has been continuously used throughout the history of reservoir operations as previously cited, this paper will only briefly describe the few main components of the algorithm that are used in the study, instead of explaining all fundamental concepts behind SDP. Refer to review papers by Yakowitz (1982), Yeh (1985), Kim and Palmer (1997), Labadie (2004), and Castelletti et al. (2008) for more detailed explanations regarding SDP for reservoir operation and related applications.

In general, a steady-state monthly release policy, which is the result of optimization, is determined by solving the following backward recursive equation in a classical SDP

model (modified from a maximization problem into a minimization problem from Tejada-Guibert et al. (1995)):

$$f_{opt}^n(S_t, H_t) = E_{Q_t|H_t} \{ \min_{R_t^*} [B_t(S_t, Q_t, R_t) + E_{H_{t+1}|H_t, Q_t} [f_{opt}^{n-1}(S_{t+1}, Q_t)]] \} \quad (6.1)$$

where t = month index; $t = 1, \dots, T$; T = final time step; n = number of time periods remaining until the end of the planning horizon; S_t = storage at the beginning of the month t ; R_t = actual release during month t ; Q_t = inflow during month t ; H_t = hydrologic state variable for month t ; $B_t(\cdot)$ = value of cost (in a minimization problem) or benefit (in a maximization problem) function during month t ; f_{opt}^n = expected value of cost (benefit) function from current period to the end of the planning horizon; and $E_{Q_t|H_t}$ = conditional expectation operation for inflow of Q_t during month t , given H_t .

The decision variables of Eq. 6.1 are subject to the following feasibility constraint:

$$\max(R_{min}, S_t + Q_t - S_{max}) \leq R_t \leq \min(R_{max}, S_t + Q_t - S_{min}) \quad (6.2)$$

where R_{min} , R_{max} , S_{max} , and S_{min} denote the minimum release, maximum release, maximum storage, and minimum storage of the reservoir, respectively.

Tejada-Guibert et al. (1995) suggested that related stakeholders should choose the hydrologic state variable in SDP's recursive equation, H_t , mainly according to the characteristics of the system to be optimized. Among various choices, the most common choice for the hydrologic state variable was determined to be the current time step's inflow Q_t and the previous time step's inflow Q_{t-1} . In this study, the current month's inflow, Q_t , is selected and included as a hydrologic state variable in

the recursive equation, H_t in Eq. 6.1. Also, the inclusion of Q_t as a hydrologic state variable makes it possible for the correlation between two consecutive months to be included inside the recursive equation (Kim and Palmer 1997). Transition probability of inflow, $\lambda(Q_{t+1} = l|Q_t)$, which reflects the potential of the system to generate inflows with respect to the previous state (Haguma et al. 2018), is calculated according to the historical average monthly inflow records and included in the optimization process. Therefore, Eq. 6.1 can be modified as the following equation, which is the main recursive equation used in this paper in all models for the derivation of optimal monthly release rules.

$$f_{opt}^n(S_t, Q_t) = \min_{R_t^*} \left[B_t(S_t, Q_t, R_t) + \sum_{Q_{t+1}} (\lambda(Q_{t+1} = l|Q_t) \times f_{opt}^{n-1}(S_{t+1}, Q_{t+1} = l)) \right] \quad (6.3)$$

6.3.2 Robust Optimization

Robust optimization (RO) is a concept introduced by Mulvey et al. (1995) in which the optimization algorithm aims to integrate goal programming formulations with a scenario-based description of the problem data. RO has been applied in combination with various other methods in the field of water resources, including the use of stochastic optimization techniques (Mulvey et al. 1995; Ray et al. 2014), the use of the fuzzy approach (Li et al. 2006; Jia and Culver 2006; Xie et al. 2017), and to deal with uncertainties in scenarios (Xu et al. 2009; Kang and Lansey 2013).

For an illustration of RO, consider a typical stochastic programming model where scenario set of uncertain variables is defined as $\Omega = \{s_1, \dots, s_s\}$. The objective function

and corresponding constraints are formulated as:

$$\underset{\mathbf{x}, \mathbf{y}}{\text{Minimize}} \quad \mathbf{z}_S = \mathbf{c}^T \mathbf{y} + f(\mathbf{x}_S) \quad (6.4)$$

subject to

$$\mathbf{A}\mathbf{y} = \mathbf{b} \quad (6.5a)$$

$$\mathbf{g}(\mathbf{x}_S) + \mathbf{B}_S \mathbf{y} = \mathbf{d}_S, \quad s \in \Omega \quad (6.5b)$$

$$\mathbf{x}_S, \mathbf{y} \geq 0, \quad s \in \Omega \quad (6.5c)$$

where $\mathbf{z}_S$ = value of the objective function in scenario S ; $\mathbf{x}_S$ = a vector of control variables in scenario S ; $\mathbf{y}$ = a vector of decision variables; $\mathbf{c}$ = a vector of cost coefficients; $f(\mathbf{x}_S)$ = operation cost of the system in scenario S ; $\mathbf{A}$ = matrix for structural constraint of model; $\mathbf{b}$ = corresponding right-hand-side vector of structural constraint; $\mathbf{g}(\mathbf{x}_S)$ = function of response of the system to the value of control variables in scenario S ; $\mathbf{B}_S$ = matrix for effects of structural decisions on the system performance in scenario S ; and $\mathbf{d}_S$ = corresponding RHS vector in scenario S .

The typical stochastic programming model described above has been used in the past and yielded optimal solutions under circumstances, in which the status quo state of the world was guaranteed. However, the constraint in the control variables in Eq. 6.5b can strictly limit and consequently discard possible potential solutions that may be more robust than prespecified optimal solutions. To overcome this limitation of typical stochastic programming formulation, RO allows for potential violations of Eq. 6.5b by transforming it into a soft constraint, which is flexible over a scenario set at a cost,

$\rho(\alpha_1, \dots, \alpha_s)$. A new infeasibility variable set $\{\alpha_1, \dots, \alpha_s\}$ is introduced and included in the modified objective function which acts as a penalty function that is proportional to the magnitude of the constraint violation, as in the following equations:

$$\underset{\mathbf{x}, \mathbf{y}}{\text{Minimize}} \quad \sigma(\mathbf{z}_1, \dots, \mathbf{z}_S) + \omega \times \rho(\alpha_1, \dots, \alpha_s) \quad (6.6)$$

subject to

$$\mathbf{A}\mathbf{y} = \mathbf{b} \quad (6.7a)$$

$$\mathbf{g}(\mathbf{x}_S) + \mathbf{B}_S \mathbf{y} + \alpha_S = \mathbf{d}_S, s \in \Omega \quad (6.7b)$$

$$\mathbf{x}_S, \mathbf{y} \geq 0, s \in \Omega \quad (6.7c)$$

where $\sigma(\mathbf{z}_1, \dots, \mathbf{z}_S)$ = aggregate objective function for all scenarios; $\rho(\alpha_1, \dots, \alpha_S)$ = feasibility penalty function that penalizes the violation of the soft constraint in all scenarios; ω = weight indicating the degree of acceptance or rejection of infeasibilities in the solution set (Watkins and McKinney 1997).

By transforming the objective function and constraints of a typical stochastic programming model into those of the RO model, additional features that could not be included in the past can now be implemented. Mulvey et al. (1995) and Watkins and McKinney (1997) suggested various mathematical formulations of the aggregate objective function such as worst case, expected utility, and mean/variance models so that the risk-averse nature of decision-makers in the field of water resources can be captured during the optimization processes. There are two broad classifications of robustness that can be incorporated into the RO model through objective functions. The first and

more frequently applied is feasibility robustness in which the range of search for a potential solution is expanded by the use of soft constraints and corresponding penalty terms (Xu et al. 2009; Mortazavi-Naeini et al. 2015). Although less often applied, the second is solution robustness in which the variance of the initial objective, $\mathbf{z}_S$, is penalized according to the distance from either its best value ($\max_s \mathbf{z}_S$) or its most frequent value ($\arg \max_{\mathbf{z}_S} p(\mathbf{z}_S)$, where $p(\cdot)$ represents the probability of occurrence) through the use of higher moments in the objective function (Watkins and McKinney 1997; Kasprzyk et al. 2009; Ray et al. 2014). Consequently, this paper aims to contribute to the literature on water resources by introducing a novel methodology that modifies the objective function and the constraints of SDP through the inclusion of both, feasibility and solution robustness, which is to be explained in detail in the following subsection.

6.3.3 Robust Stochastic Dynamic Programming

RSDP, the optimization algorithm proposed in this paper, couples the key components of RO into the objective function and constraints of SDP. Although redefining the key components of the objective function may be limited in specific optimization algorithms, it was possible by selecting SDP as this study's main tool mainly due to its wide applicability over various types of objective functions. Moreover, when it comes to actual implementations of a theoretically optimized policy, Yeh (1985) and Wurbs (1993) argued the importance of considering the gap between the two. Thus, this study suggests a policy not so far from actual implementable policies through reflection of historical operation schemes.

Due to its nature, transforming the constraints of the typical SDP model given in Eq. 6.2 to a soft constraint, as suggested in the RO formulation in Eq. 6.7b, is feasible

but involves several steps. For instance, consider an example of a reservoir for which the former maximum storage, S_{max} is enlarged to $S_{max, ext}$ through the construction of additional infrastructure. Then, the feasibility constraint of SDP (Eq. 6.2) is primarily transformed into the following:

$$\max(R_{min}, S_t + Q_t - S_{max, ext}) \leq R_t \leq \min(R_{max}, S_t + Q_t - S_{min}) \quad (6.8)$$

Consequently, the benefit function of the SDP model originally defined as $B_t(\cdot)$ can be modified to include a penalty function so that it can include a penalty term of either violations of soft constraints in case of feasibility robustness or variance of the objective across different scenarios for solution robustness. Because specific mathematical formulas for both feasibility and solution RSDP models may vary according to the system, specific formulas adopted in this paper will be demonstrated through the derivation of monthly release rules in a multipurpose dam in Korea in the following case study.

Among other candidate algorithms, this study utilized SDP as the primary optimization tool to be coupled with RO owing to following advantages of SDP over other alternative optimization algorithms: (1) SDP enables the calculation of the value of each objective term at discrete time steps and (2) the evaluation of the value of the storage during the optimization horizon is possible without additional validation process. Especially, the first advantage of the SDP enables the calculation of one of the objective terms of the proposed RSDP. Therefore, despite the results from Chapter 4, this study adopted SDP as the main optimization algorithm and derived alternative optimization models based on the SDP.

6.4 Case Study: Boryeong Multipurpose Dam

The application of the proposed methodology is composed of three sequential steps: (1) optimization, (2) simulation, and (3) evaluation. Derivation of optimal monthly release rules according to each model, as well as the evaluation of the number of infeasible paths, is conducted in the optimization step. Then, the optimized rules are simulated under a wide range of inflow scenarios that are statistically generated to reflect the non-stationarity of climate change, thereby reflecting actual policy that is actually being implemented. Ultimately, the simulated results are evaluated in the last step using various metrics. The general flow chart of the case study is shown in Figure 6.1.

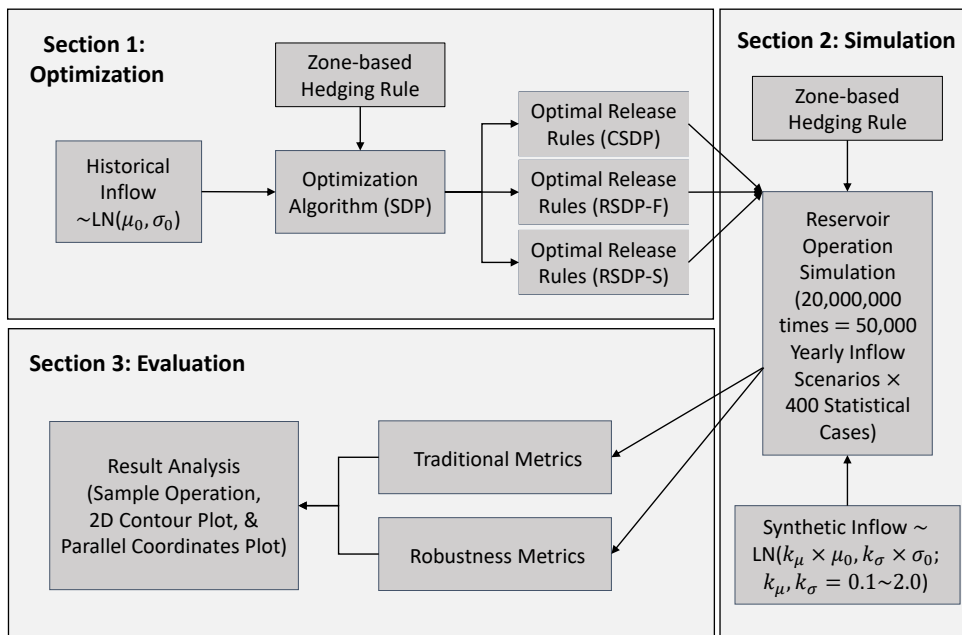


Figure 6.1: Overall flow of the study. section 6.4 is organized as in the steps in Figure 6.1: (1) optimization, (2) simulation, and (3) evaluation

6.4.1 Optimization

Model and Objective Function Formulation

For comparative evaluation, three models were selected and the objective function and constraints were formulated corresponding to each model type: CSDP, RSDP-F, and RSDP-S. The models differ from each other mainly according to the terms that are included inside the objective function and the type of constraint that is implemented. Since the primary target of the reservoir in the study basin is to meet aggregate water demand for all users, the main objective term f_1 , included in all the models, is to minimize the deficit in the monthly release, R_t , with respect to the demand, D_t , across all time steps:

$$f_1 = \max \left(\frac{D_t - R_t}{D_t}, 0 \right) \quad (6.9)$$

The second objective term, corresponding to the key concept of feasibility robustness, is introduced by advancing the research conducted by Lee et al. (2019), by modifying certain features so that the derived rules would be implementable to the application site. In this application, whereas the minimum storage for CSDP model is LWL, RSDP-F and RSDP-S models are allowed to use the storage below LWL and above DSL by transforming S_{min} into $S_{min, R}$. By allowing violation of constraint through the use of a soft constraint, a corresponding penalty term is included in the objective function to measure the magnitude of the violation. Thus, the second term, which is included in RSDP-F and RSDP-S models, is defined as f_2 :

$$f_2 = \max \left(\frac{S_{min} - S_t}{S_{min}}, 0 \right) \quad (6.10)$$

Due to this consideration, the former constraint given as Eq. 6.2 should be modified accordingly so that a fraction of all state variables in the RSDP models are able to pass through. As this application allows for the usage of storage below LWL and above DSL, the novel feasibility constraint applied in RSDP-F and RSDP-S models is written as Eq. 6.11:

$$\max(R_{min}, S_t + Q_t - S_{max}) \leq R_t \leq \min(R_{max}, S_t + Q_t - S_{min,R}) \quad (6.11)$$

As previously mentioned, solution robustness often adopts higher-order moments of the objective value either from its best value or from its most frequent value. Among these two candidates, this study defined solution robustness as the deviation from the best possible value across all scenarios due to the fact that the most frequent value was already considered in the recursive equation of SDP (Eq. 6.3) through the multiplication of transition probabilities. Consequently, the third term of the aggregate objective function, which corresponds to the key concept of solution robustness, measures the deviation of the main objective term of inflow scenarios Q_s ($f_{1, Q_s, t}$) from the best possible main objective across all inflow scenarios Q_S ($\min_{Q_s \in Q_S} f_{1, Q_s, t}$):

$$f_3 = \left(\min_{Q_s \in Q_S} f_{1, Q_s, t} - f_{1, Q_s, t} \right) \quad (6.12)$$

Finally, the last term in the aggregate objective function is included in order to prevent the optimization algorithm from finding solutions that may lead to depletion of remaining storage even after the first objective is satisfied. In actual simulations, the dam elevation at the end of the drawdown period, $S_{t, end}$ is required to meet a specific

standard elevation for the upcoming flooding season, S_{target} after releasing water for M&I uses. To enhance dominance on the first objective f_1 , the last objective f_4 is included with a relatively smaller weight ($w_1, w_2, w_3 = 10^6, w_4 = 1$) than the weights of other objective terms:

$$f_4 = \frac{S_{t, end} - S_{target}}{S_{target}} \quad (6.13)$$

Before determining the weights of the objective terms of each model, two cases of the RSDP-F model with different weights underwent a comparative analysis. After reservoir simulation utilizing the onsite operation rule based on the zone-based hedging rule with historical inflow series (October 1998 through June 2020) with various starting storage values (50%, 42.7%, and 30%), f_1 and f_2 were computed. The ratio of f_1 and f_2 ($f_1 : f_2$) during the simulation varied according to the starting storage values: 2.46:1 for the starting storage of 50%, 1.89:1 for 42.7%, and 1.39:1 for 30%. Therefore, the first case of the RSDP-F model was granted with the weights of w_1 and w_2 identically as 1×10^6 , and the second case of the RSDP-F model stressed the penalty from constraint violation by two times than the conventional RSDP-F model ($2 \times w_1 = w_2$). Figure 6.2 shows the difference between the monthly averaged release decisions between the two.

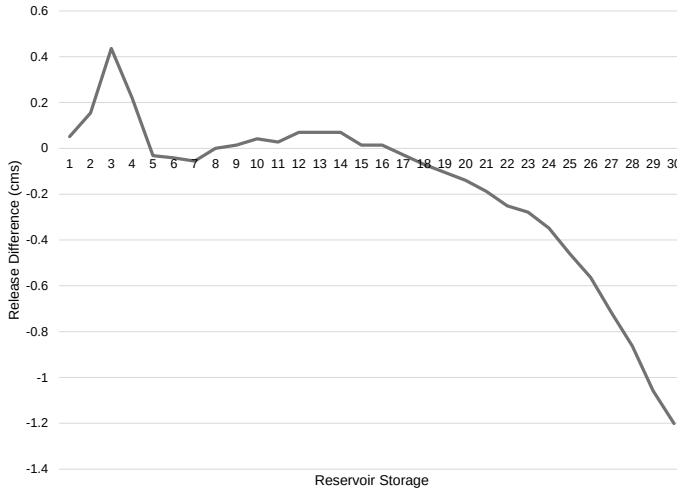


Figure 6.2: Release rule difference between two cases of RSDP-F models with different weights of the objective terms

Since the second case of the RSDP-F model stresses the penalty violation during the optimization process twice than the conventional RSDP-F model with identical weights, positive difference between the release decisions in low storage levels is reasonable. However, the positive difference is compensated with the negative difference (decrease in release decisions) in higher reservoir storage levels. This indicates that the conventional RSDP-F model allows more release in higher storage levels. Although a deeper analysis on how the weights on each objective determine the release rule formulations is indeed a promising research topic since the primary objective of this study is in the inclusion of either feasibility and solution robustness concept in the SDP formulation, analysis on weights is deferred to one of the potential future research topics.

Finally, by aggregating the four objective terms with predefined weights in the models, $\mathbf{w} = (w_1, w_2, w_3, w_4)$ (Table 6.1), the objective function, $B_t(\cdot)$ for each model can be written using the following equation:

$$B_t(S_t, Q_t, R_t) = w_1 f_1^2 + w_2 f_2^2 + w_3 f_3^2 + w_4 f_4^2 \quad (6.14)$$

Table 6.1: Weights of objective in each optimization model

Model Name	w_1	w_2	w_3	w_4
CSDP	1×10^6	0	0	1
RSDP-F	1×10^6	1×10^6	0	1
RSDP-S	1×10^6	1×10^6	1×10^6	1

Calculation

Prior to iteratively calculating Eq. 6.3 with Eq. 6.14, the state variables of the SDP model, S_t and Q_t , were discretized to k and l intervals, respectively. In this study, the authors selected the discretization intervals in SDP models to be $k = 30$ and $l = 11$ so that they can sufficiently cover the whole range of variables densely. Consequently, the total number of paths of the problem is calculated by multiplying the discretization intervals for $S(t)$, $S(t + 1)$, $Q(t)$, t , and n where t and n are the number of months and iterative calculations of the recursive equation, respectively. In this study, the total number of paths to be calculated for each model depends on the number of iterations for each model n_{model} : $k \times k \times l \times t \times n_{model} = 30 \times 30 \times 11 \times 9 \times n_{model}$, which is a relatively conservative number of calculation that could be conducted without parallelization, unless n_{model} increases exponentially.

The optimal policies derived from the three SDP models are coupled with the zone-based hedging rule, which is adopted as the main operating policy for all multipurpose dams in Korea, to derive policies that reflect how the actual reservoirs are operated. For comparative analysis, two optimal rules both derived by the CSDP model are simulated under a wide range of synthetic inflow series. While the first set of rules is derived directly according to the results of optimization, the second candidate rule set coupled actual zone-based hedging rule (Table 6.2) during all steps of optimization. Consequently, the second set yielded reduced release decisions especially in lower zones of storage (Zone 3 and 4 of Table 6.2). Then, both rules underwent 50,000 years of baseline simulation and the values of beginning-of-month storage were analyzed (Figure 6.3). The difference between the storage values of the two sets suggest that

the rule set that took hedging into consideration is more effective in terms of long-term response to drought since the release is regulated with respect to current storage. Since the difference between the two during March to May ($3.9 \times 10^6 \text{ m}^3$) is about half of average monthly demand ($8.7 \times 10^6 \text{ m}^3$), taking hedging into consideration during optimization procedures would result in significant difference in storage after simulation. This study simplified, yet endeavored to find a balanced operation rule between theoretically optimal solutions and actual policies in the optimization step. Indeed, a more delicate and detailed approach regarding integration is possible and remains as a potential future research topic.

Table 6.2: Ratio of hedging according to the zone-based hedging rule applied to all multipurpose dams in Korea

Zone Name	Municipal & Industrial	Irrigation	Environmental
Normal Zone & Zone 1 (Attention Zone)	100%*	100%*	100%*
Zone 2 (Caution Zone)	100%	100%	Up to 0%
Zone 3 (Alert Zone)	100%	Up to 70%	0%
Zone 4 (Critical Zone)	Up to 80%	70%	0%

* Release exceeding demand is possible in Normal Zones

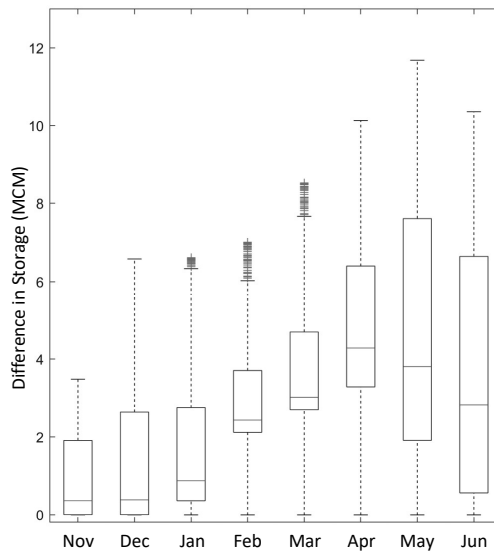


Figure 6.3: Boxplots of the difference in beginning-of-month storage after 50,000 years of simulation between optimized release rules and reduced release rules with hedging

Hence, all release decisions R_t were derived according to optimization models that were coupled with hedging rules before entering the continuity equation for the reservoir, Eq. 6.15:

$$S_{t+1} = S_t + Q_t - R_t \quad (6.15)$$

subject to

$$R_t = \min (\max [R_t, S_t + Q_t - S_{\max}], S_t + Q_t - S_{\min}) \quad (6.16a)$$

$$0 \leq R_t \leq R_{\max} \quad (6.16b)$$

$$S_{\min} \leq S_t \leq S_{\max} \quad (6.16c)$$

where $R_{\max}$ is the non-damaging outflow of the Boryeong multipurpose reservoir (200 m³/sec), set to consider operations during the flooding periods.

It was assumed that at the beginning of the drawdown period, the reservoir had been supplied with enough water from flooding and start the drawdown period from NHWL. Thus, only the rules for the 9-month drawdown period were considered, mainly due to the unique monsoonal climate of East Asia where precipitation is concentrated during the 3-month flooding season (Kim et al. 2007). The final outputs were tables of steady state release rules which are interpretable by stakeholders without expertise. These rules satisfy the following equation:

$$\begin{aligned} & ||f_{opt}(S_t^k, H_t)_{n_{model}} - f_{opt}(S_t^k, H_t)_{n_{model}-1}| - \\ & |f_{opt}(S_t^k, H_t)_{n_{model}-1} - f_{opt}(S_t^k, H_t)_{n_{model}-2}|| < 10^{-5} \quad \forall t, k \end{aligned} \quad (6.17)$$

6.4.2 Simulation

Inflow Series Generation

Recently, in the field of water systems operations, scenario-led approaches, in which the system performance is evaluated using global climate projection scenarios (Wilby and Dessai 2010) are being replaced by scenario-neutral approaches, in which the assessment is conducted under a wide range of uncertain scenarios sampled from deep uncertainty (Kwakkel et al. 2016; Biglarbeigi et al. 2018). In accordance, this study generated a wider range of inflow scenarios so that the envelop of the simulation sufficiently cover the scope of the projections.

Optimized release rules were tested under future inflow scenarios that capture the non-stationarity of climate change. Second order non-stationarity was assumed in future inflow scenarios and consequently, this study generated future inflow series by only modifying the mean and standard deviations of the historic inflow series. In order to select the range of parameters, the statistics of the 27 Global Climate Models (GCMs) under the Representative Concentration Pathways 4.5 for the Geum River Basin were computed. They were downscaled into the 60 Automated Synoptic Observing System (ASOS) stations using the inverse distance weighting method, and the systematic bias was corrected using the Quantile Delta Mapping technique (Eum and Cannon 2017). The specifics regarding the selected GCM scenarios is shown in Table 6.3, and their statistics compared with respect to the historical inflow series is shown in Figure 6.4.

Table 6.3: Specifics regarding the 27 GCMs scenarios utilized in this study

No.	GCMs	Resolution (°)	Institution
1	GMCC-CM	0.750×0.748	Centro Euro-Mediterraneo per I Cambiamenti Climatici
2	CCSM4	1.250×0.942	National Center for Atmospheric Research
3	CESM1-BGC		
4	CESM1-CAM5		
5	BCC-CSM1-1-M	1.125×1.122	Beijing Climate Center, China Meteorological Administration
6	MRI-CGCM-3		Meteorological Research Institute
7	CNRM-CM5	1.406×1.401	Center National de Recherches Meteorologiques
8	MIROC5		Atmosphere and Ocean Research Institute (The University of Tokyo)
9	HadGEM2-AO	1.875×1.250	Met Office Hadley Centre
10	HadGEM2-CC		
11	HadGEM2-ES		
12	INM-CM4	2.000×1.500	Institute for Numerical Mathematics
13	IPSL-CM5A-MR	2.500×1.268	Institut Pierre-Simon Laplace
14	CMCC-CMS	1.875×1.865	Centro Euro-Mediterraneo per I Cambiamenti Climatici
15	MPI-ESM-LR		Max Planck Institute of Meteorology (MPI-M)
16	MPI-ESM-MR		
17	FGOALS-s2	2.813×1.659	LASG, Institute of Atmospheric Physics, Chinese Academy of Sciences
18	NorESM1-M	2.500×1.895	Norwegian Climate Centre
19	GFDL-ESM2G	2.500×2.023	Geophysical Fluid Dynamics Laboratory
20	GFDL-ESM2M		
21	IPSL-CM5A-LR	3.750×1.895	Institut Pierre-Simon Laplace
22	IPSL-CM5B-LR		
23	BCC-CSM1-1	2.813×2.791	Beijing Climate Center, China Meteorological Administration
24	CanESM2		Canadian Centre for Climate Modelling and Analysis
25	GISS-E2-R	2.000×2.500	National Aeronautics and Space Administration
26	MIROC-ESM-CHEM	2.813×2.791	Japan Agency for Marine-Earth Science and Technology, Atmosphere and Ocean Research Institute
27	MIROC-ESM		

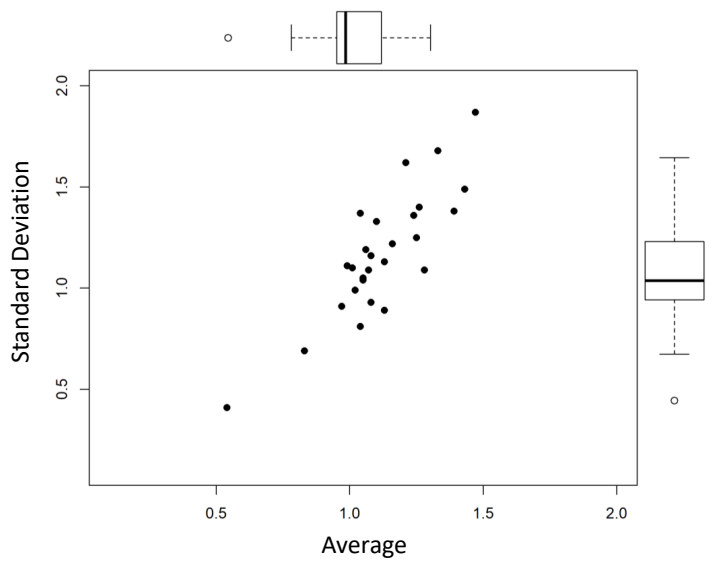
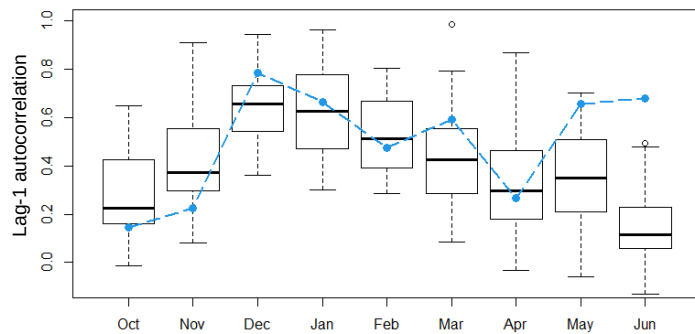


Figure 6.4: Range of average and standard deviations of selected GCM scenarios. This study generated a total of 400 cases of synthetic inflow series in which the average and standard deviations are varied with corresponding ratios (k_μ and k_σ), ranging from 0.1 to 2.0 so that the synthetic inflow would cover the entire range of GCM scenarios

When compared with the statistics from historical observations, the statistics of the GCM scenarios fell within a range of 0.5 through 1.5 times of the observations (Figure 6.4). Consequently, in this study, the authors expanded this range, so that the generated inflow scenarios would cover a wider range between 0.1 to 2.0, in order to account for possible extremes that may occur under climate change (Salman et al. 2018; Farjad et al. 2019).

Moreover, the range of the lag-1 autocorrelation of 27 GCM scenarios were also considered for additional simulations in which the correlation matrix of the synthetic inflow series varies (Figure 6.5). This study selected two more additional simulations in order to account for uncertain future conditions: median of the hindcast data, and the maximum average correlation scenario among the forecast data. Therefore, total of three sets of synthetic inflow series were generated with diverse lag-1 autocorrelation referred from the 27 GCM scenarios of the study basin.

(a) 1976-2005



(b) 2006-2045

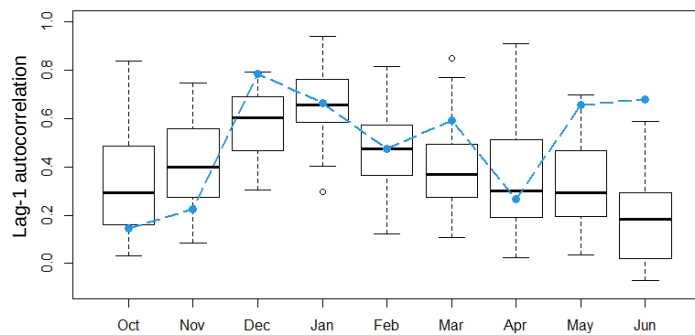


Figure 6.5: Range of lag-1 autocorrelation of selected GCM scenarios: (a) hindcast period (1976-2005) and (b) forecast period (2006-2045). Blue dotted lines indicate lag-1 autocorrelation of the historic inflow series

Specifically, for inflow series generation, this study first assumed that the historical monthly inflow series of the study basin, Q_i , follow a lognormal distribution and fitted each month's inflow to a lognormal distribution. Then, in order to synthetically generate inflow series X_i with varying mean and standard deviations, the parameters of the log transformed synthetic inflows ($Y_i = \ln(X_i + \epsilon)$), $\bar{Y}_i$ and s_{Y_i} were estimated using Eq. 6.18 and Eq. 6.19 (Yoon et al. 2018):

$$s_{Y_i} = \ln \left[1 + \frac{k_\sigma \times s_{Q_i}}{(k_\mu \times \bar{Q}_i + \epsilon)^2} \right] \quad (6.18)$$

$$\bar{Y}_i = \ln(k_\mu \times \bar{Q}_i + \epsilon) - \frac{1}{2} s_{Y_i}^2 \quad (6.19)$$

where i is the index of the month, and $\epsilon = 0.001$ is a small number added during log transformation, $\bar{Y}_i$ and s_{Y_i} are the mean and standard deviations of the synthetically generated inflow, $\bar{Q}_i$ and s_{Q_i} are the mean and standard deviation of the historical inflow, and k_μ and k_σ are the multipliers ranging from 0.1 to 2.0 each multiplied by the original parameters for synthetic generation.

Also, in order to maintain the lag-1 correlation between monthly inflow scenarios, the covariance matrix of the log transformed inflow between consecutive months i and $i + 1$, $\hat{\Sigma}_{Y_i, Y_{i+1}}$ was estimated using Eq. 6.20:

$$\hat{\Sigma}_{Y_i, Y_{i+1}} = \hat{\rho}_{\ln Q_i, \ln Q_{i+1}} \times s_{Y_i} \times s_{Y_{i+1}} \quad (6.20)$$

where $\hat{\rho}_{\ln Q_i, \ln Q_{i+1}}$ is the correlation matrix between the log transformed historical inflow of month i and $i + 1$.

As the synthetic log transformed inflow Y_i follows the multivariate normal distribu-

tion ($Y_i \sim MVN(\bar{Y}_i, \hat{\Sigma}_{Y_i, Y_{i+1}})$), this study adopted the "mvnrnd" function in MATLAB and created 50,000 continuous yearly inflow scenarios for each statistical scenario in which the average and standard deviation of the scenarios varied from those of historical observations. Finally, to compare all models fairly and to minimize the effects of outliers on the evaluation metrics, identical sets of inflow scenarios were used during the simulation of all models.

Reservoir Operation Simulation

For simulation, each optimization model underwent 50,000 years of simulation per each inflow scenario, in which the optimized release rules were applied. The main governing equation adopted during the simulation process is Eq. 6.15, and monthly averaged performance metrics were calculated for further analysis of simulation results.

6.4.3 Evaluation

After the simulation, the evaluation process of this paper was completed in two-steps. Step 1 focused on using traditional metrics originally proposed by Hashimoto et al. (1982b), and Step 2 focused on using metrics of the family of robustness metrics, in which a system's performance under deeply uncertain or less probable states of the world can be sufficiently evaluated (Maier et al. 2016). The primary motivation behind presenting the results using both traditional and robustness metrics is to possibly reduce human decision biases, such as cognitive myopia or short-sightedness (Hogarth 1981). This could also be used to broaden the understanding of the policies' operational performance and their ability to perform acceptably under challenging system states.

Traditional Metrics

In this paper, the concepts of reliability, resiliency, and vulnerability originally suggested by Hashimoto et al. (1982b) were adopted to measure the frequency, duration, and magnitude of the failure of the system, respectively. These three concepts were widely adopted to evaluate the performances of many engineering systems, and also provided the basis for variants of metrics. For instance, Onta et al. (1991) adopted the concept of reliability from Hashimoto et al. (1982b) for the assessment of alternative plans and policies for conjunctive use of surface and groundwater resources. Srdjevic et al. (2004) applied this methodology to evaluate simulations of multiyear water management scenarios, and El-Baroudy and Simonovic (2004) proposed a coupled measure that combined classic reliability and the fuzzy concept and used it to evaluate the performance of water resource systems.

To distinguish from another family of metrics to be introduced, this paper will refer to the reliability, resiliency, and vulnerability metrics as "traditional" metrics. As the main objective of the target reservoir is to supply water for the general public for M&I uses, this paper defined failure as all incidents in which the actual release, R_t , was not able to meet the target demand, D_t , of the specific month t (i.e., $R_t < D_t$).

For intuitive and consistent demonstration, all three metrics that were adopted in this study were devised based on Hashimoto et al. (1982b) with slight modification. Average deficit magnitude (ADM_y), average deficit frequency (ADF_y), and maximum recovery time (ReT_{max}) correspond to vulnerability, reliability, and resiliency from Hashimoto et al. (1982b), respectively. All metrics are annually-averaged values, and also standardized so that all metrics with higher values indicate events with greater,

more frequent, and longer water deficits. ADM_y represents the ratio of annual average water deficit magnitudes measured as a fraction of the total deficit over total demand; ADF_y indicates the annual average frequency of the water deficit, measured as a fraction of the number of failures over the total time step; and ReT_{Max} is the longest time step of each scenario for the system to come back from the state of failure to normal, measured in months. Mathematical formulas for the three traditional metrics adopted in this study are shown in Table 6.4.

Table 6.4: Selected traditional evaluation metrics in this study

Metric Name	Classification	Equation
ADM_y	Magnitude of Annual Water Deficit	$\frac{1}{T} \sum_{t=1}^T \max \left(\frac{D_t - R_t}{D_t}, 0 \right)$
ADF_y	Frequency of Annual Water Deficit	$\frac{1}{T} \sum_{t=1}^T I_t \left(\frac{D_t - R_t}{D_t} > 0 \right)$
ReT_{max}	Duration of Longest Failure	$\begin{aligned} & \max \{t_b - t_a\}, \forall t_a \\ \text{s.t. } & \frac{D_{t_a} - R_{t_a}}{D_{t_a}} < 0 \text{ and } \frac{D_{t_a-1} - R_{t_a-1}}{D_{t_a-1}} \geq 0 \\ & \text{and for first } t_b > t_a \\ & \frac{D_{t_b} - R_{t_b}}{D_{t_b}} < 0 \text{ and } \frac{D_{t_b-1} - R_{t_b-1}}{D_{t_b-1}} \geq 0 \end{aligned}$

* ADM : Average Deficit Magnitude, ADF : Average Deficit Frequency, ReT : Recovery Time (Duration)

** $I_t(\cdot)$: counts the number of occurrences inside the parenthesis

*** Units of the traditional evaluation metrics are unit-less, unit-less, and months, respectively

Robustness Metrics

Although RSDP-F and RSDP-S models were originally designed so that they would yield relatively ‘robust’ solutions than the CSDP model, this superiority may not be necessarily prominent when a system’s performance is solely measured using traditional metrics. Moreover, owing to the complexity of most water resources system problems, the first procedure in the evaluation of robustness starts with the meta-problem of deciding how to decide (Schneller and Spiccas 1983; Reed and Kasprzyk 2009). As the principal motivation of this paper was not in determining sophisticated robustness measures, but in the development of both optimal and robust operating policies and testing them under a wide range of future uncertainties, this paper adopted the existing concept of robustness in the field of decision making for water resources for its evaluation.

A myriad of robustness metrics in this specific field of study can be classified into different categories depending on the selected measures. Especially when the extent of risk aversion subsumed into the mathematical formula is considered, the metrics can be largely divided into either regret-based (Savage 1951) metrics or satisficing-based (Simon 1956) metrics. Regret-based metrics, in this specific context, measure the relative difference between the selected alternative performance and that of the best (or most probable) alternative inside the envelope of simulation (McPhail et al. 2018). In contrast, satisficing metrics evaluate the tendency of decision-makers to select alternatives that meet their requirements instead of being optimal according to other criteria (Giuliani and Castelletti 2016).

Thus, in this paper, with a focus on classical risk measures and risk aversion,

a regret-based robustness measure was analyzed. The regret metric defined in this study, R^i , measures the 90th percentile deviation from the average performance using a baseline scenario (Herman et al. 2015) with respect to each traditional metric D_i ($i = 1, 2, 3$). As regret metrics are separately calculated with respect to reliability, resiliency, and vulnerability, a total of three regret-based robustness measures for each model can be computed. The mathematical formula of the robustness metric primarily used in this study, R^i_{model} , is derived using Eq. 6.21 and Eq. 6.22:

$$R^i_{model} = [D_{i,90} : P(D_i \leq D_{i,90}) = 0.90] \quad (6.21)$$

$$D_{i,j} = \frac{|F(\bar{\mathbf{x}})_{i,j} - F(\bar{\mathbf{x}})^*_i|}{F(\bar{\mathbf{x}})^*_i} \quad (6.22)$$

where $F(\bar{\mathbf{x}})^*_i$ is the annual averaged value of metric i of the baseline state of the world, and $F(\bar{\mathbf{x}})_{i,j}$ is the annual averaged value of metric i calculated for the j th state of the world.

Specifically, the regret-based deviation metric adopted in this study (R^i) calculates the deviation of the 90th percentile performance from the baseline scenario in which the average and the standard deviation is kept constant from the historical statistics. The fundamental concept behind this metric is that if a robust model performs well under worst-case scenarios in this experiment, there is a high change of them performing well in deeply uncertain future scenarios as well (Kasprzyk et al. 2013). According to the definition of robustness metric in this study, the 90th percentile performance of robust model should not diverge much from the baseline performance.

6.5 Results and Discussion

6.5.1 Increase in Number of Feasible Paths

As previously mentioned, the first advantage of the introduction of soft constraints with RSDP models is in the expansion of the range of potential solutions to be considered. After calculation and derivation of optimal monthly operating policies, the total number of calculations, or possible "paths" of SDP, for each model can be summarized as shown in Table 6.5.

Table 6.5: Number of possible paths in each optimization model

Model	i	j	t	k	l	Number of Paths $= i \times j \times t \times k \times l$
CSDP	30	30	9	11	62	5,524,200
RSDP-F	30	30	9	11	65	5,791,500
RSDP-S	30	30	9	11	68	6,058,800

where i is the index for beginning month storage, j is the index for the end month storage, t is the number of months, k is the number of inflow discretization levels, and l is the number of iterations taken until the steady state release rule from each optimization models is computed.

For the CSDP model, 96.67% of the total number of paths passed through the feasibility constraint given in Eq. 6.2 and were considered as candidates for the optimal solution. Similarly, 96.67% of the state variables of RSDP-F and RSDP-S models passed through the feasibility constraint of RSDP (Eq. 6.11), and when the feasibility was checked by applying the CSDP standards, 93.33% of all state variables passed

the former constraint. This indicates that the differences between the two, 3.34% of all state variables in RSDP models, which were originally neglected when optimized under the CSDP feasibility constraint, are now considered as potential solutions in RSDP models. Although an enhancement of the simulation results measured by either traditional or robustness metrics was the primary motivation behind the introduction of the RSDP models, boosting in the range of possible solutions would potentially increase the number of candidate solutions. This enlargement in the range of solutions, regardless of their optimality, would ultimately allow both the simulation tool and the related decision-makers more flexibility under various circumstances.

6.5.2 Sample Operation under Specific Drought Incidents

The optimized release rules of CSDP, RSDP-F, and RSDP-S models were simulated under two sets of 10-year drought incidents both extracted from the baseline scenario. Historically during the 2014-2019 drought, the study reservoir had to undergo inflow conditions that correspond to the 87th percentile's worst case among 50,000 years of the baseline scenario. Thus, the first set of representative 10-year drought assumed the historical recorded drought conditions to last for 10 years, while the second set assumed the worst-case drought conditions extracted from the 50,000 years of baseline scenario. The release rules and the main objective of the three models were tested in order to compare their distinctive behaviors.

Release decisions of the three models varied the most when storage level was low. While the RSDP-S model yielded rules with significantly reduced release under dry conditions, CSDP and RSDP-F models lowered releases, yet not so significantly when compared with the decisions made under RSDP-S model. This difference consequently

led to greater magnitude in deficit near the end of drawdown periods where the demand was the highest among other months. When compared in terms of cumulative water deficit (f_1) (Figure 6.6), operating the reservoir under RSDP-S model's policy would potentially lead to better performance in the long-term.

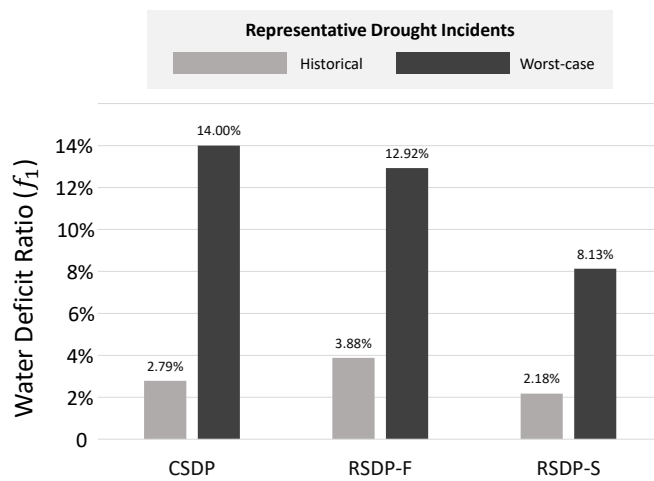


Figure 6.6: Water deficit Ratio (f_1) of three models over 10 years of representative drought incidents extracted from the baseline scenario: historical (2014-2019) and worst-case

6.5.3 Starting Storage Analysis

Given the impact of the starting storage in reservoir operation problems and the concentrated rainfall during the flooding season in the study area, how to set the starting storage values may significantly affect the performance of the optimization models. Therefore, this subsection of the study includes analysis for the reservoir using the proposed optimization models (CSDP, RSDP-F, and RSDP-S) under different starting storage values. For comparative analysis, the three cases for the starting storage values were set as 80%, 50%, and 30% (85.12 MCM, 53.2 MCM, and 31.92MCM, respectively) of the maximum storage value of the reservoir. Historically, the recorded starting storage values were 74.48 MCM during all periods (1998-2019) and 45.48 MCM during the recent drought (2014-2019). Therefore, the simulation in this study is able to cover the entire range of historical starting storage values. Finally, the drought inflow scenarios used in subsection 6.5.2 were utilized once again, and the performance of the models were conducted using the annual average water deficit ratio (f_1), similarly from subsection 6.5.2. Simulation results with different starting storage values are shown in Figure 6.7.

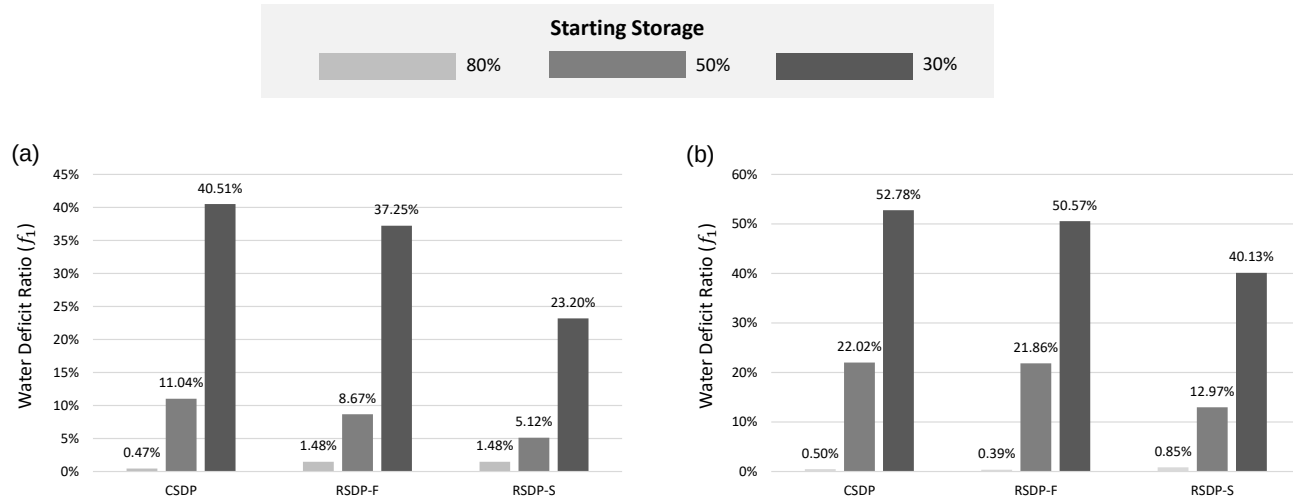


Figure 6.7: Water deficit Ratio (f_1) of three models over 10 years of representative drought incidents extracted from the baseline scenario with different starting storage values (80%, 50%, and 30%). Drought incidents are selected according to the annual cumulative inflow that are similar to: (a) historical (2014-2019) and (b) worst-case from the baseline scenario

As shown in Figure 6.7, starting storage values affected the performance of the reservoir significantly during the drawdown period. When the reservoir was supplied with sufficient amount of water during the flooding seasons (starting storage = 80%), the CSDP models resulted in better performance than the robust alternative models. However, when the reservoir is not supplied with sufficient water and consequently start low on October, the RSDP-F and RSDP-S models yielded better results in both drought incidents. Especially, the value of the primary objective f_1 improved by 42.7% when the reservoir was simulated under 30% of starting storage value in the historical drought incident (Figure 6.7a). Under the worst-case incident from the baseline scenario (Figure 6.7b), the RSDP-S model showed an improvement by 24.0% when compared with the performance of the CSDP model. In general, the robust alternative models (RSDP-F and RSDP-S) performed relatively well than the CSDP models under dry flooding seasons. This indicates that the RSDP models are able to cope with a wider range of possible drought scenarios. Moreover, additional simulations were conducted in order to discover the tipping point of the starting storage, which indicate the value of the storage where the RSDP models perform better than the CSDP models. By iterative calculation process, the starting storage values under the threshold of 61.1% (65.0 MCM) resulted the better performance in RSDP models than the CSDP model. Further study regarding the tipping point and the starting storage are to be conducted in the future with adequate numerical experiments.

Analysis on the Storage below LWL

In case of extremely low starting storage values, the two RSDP models proposed in this study allow the use of emergency storage below LWL of the reservoir. Although the target reservoir is equipped with infrastructures and policies which support the utilization of the emergency storage (the storage between LWL and DSL) (section 3.1), using the stock below LWL is different from the operation under normal conditions. Especially, the abnormal temperature and water quality of the emergency stock may be evaded from both the water users and the reservoir operators. Thus, the simulation of the reservoir focusing on the how much stock of the water below LWL is used is conducted and analyzed. For the drought inflow, two sets of 10 years of representative drought incidents extracted from the baseline scenario (historical (2014-2019) and the worst-case) is utilized. The starting storage of the simulation was set as 26.4% of the maximum storage (historical storage at Sep-21, 2015). Figure 6.8 shows the time series of the storage operated by the three models: CSDP, RSDP-F, and RSDP-S.

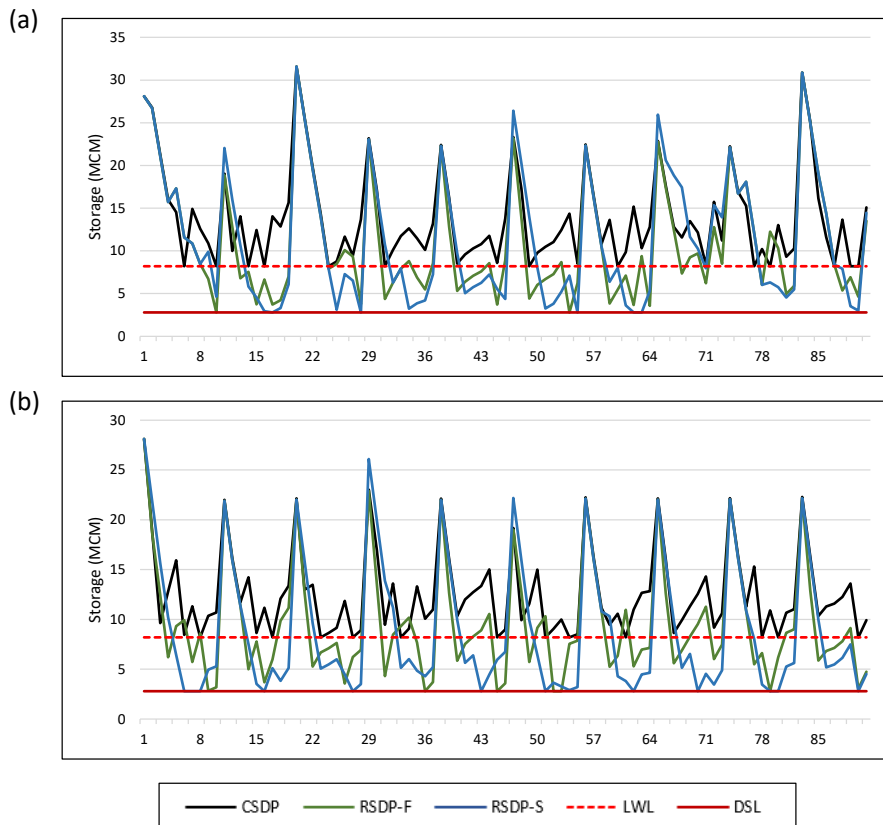


Figure 6.8: Storage of the three models over 10 years of representative drought incidents extracted from the baseline scenario: (a) historical (2014-2019) and (b) worst-case

The line plots in Figure 6.8 demonstrate the overall storage above and below LWL of the three models during the representative drought incidents for 10 years. As initially expected, in general, the RSDP models reduce the amount of water deficit at the expense of utilization of the storage below LWL (Figure 6.6 and Figure 6.7). On the other hand, since the constraints of the CSDP model strictly prohibits the use of the emergency storage below LWL, there is no risk in which the storage of the CSDP model falls below LWL. Among the RSDP models, the RSDP-S model actively utilized the emergency storage than the RSDP-F model (5.56% more in the historical drought, and 12.2% more in the worst-case drought). Moreover, whereas the RSDP-F model utilized 2.54 MCM and 2.62 MCM of the emergency storage during each drought incidents in average, during the same time, the RSDP-S model utilized 3.00 MCM and 3.60 MCM. Since the utilization of the emergency storage is can clearly be considered as a disadvantage, the CSDP model holds an advantage of not using the emergency storage over the RSDP models regardless of the evaluation results from the traditional and robustness metrics.

6.5.4 Evaluation through Traditional Metrics

The analysis of the system via traditional metrics aims to visualize the effects of adopting either the RSDP-F or RSDP-S model versus the CSDP model in terms of magnitude (ADM_y), frequency (ADF_y), and duration (ReT_{max}) of water supply deficit across the entire envelope of the simulation range. As traditional metrics or similar variations have been widely adopted in decision making in actual systems, a relatively intuitive display of the three selected metrics were visualized through contour plots for each model (Figure 6.9). Although the range of ratios of both parameters (k_μ and k_σ) was from 0.1 to 2.0, only the results under scenarios with decreasing average ($k_\mu = 0.1 \sim 1.0$)

are displayed mainly to reflect the severe multi-year drought the reservoir underwent recently and its consequences due to high ratio of M&I water use. In all plots in Figure 6.9, the change in colors in the horizontal directions indicates that the system is more sensitive to a change in k_μ than that of k_σ 's, except for specific cases where k_μ is extremely high or extremely low.

Both RSDP-F and RSDP-S models outperformed the CSDP model in different metrics, and also in different circumstances. Especially in extreme inflow cases with a decreased average and an increased standard deviation (bottom-right corner of Figure 6.9), both models outperformed the CSDP model in terms of magnitude (Figure 6.9a), and the RSDP-S model outperformed the CSDP model in terms of the duration of longest failure (Figure 6.9c). A significant trade-off relationship between frequency (Figure 6.9b) and the other two metrics also can be observed, where the RSDP-S model performed slightly worse than the other two models. Although the CSDP model appears to be the optimal policy in terms of magnitude, as the main objective term of the CSDP model is to minimize the magnitude of water deficit, it was proven that the system may undergo failures of greater magnitude in extreme cases.

To demonstrate how the elicitation of robust alternatives would affect the related stakeholders, this study addressed the problem in two ways. The first approach is aimed to analyze the values of traditional metrics to answer the following question: how extreme are the extremes? The second idea includes the use of robustness metrics and will be presented in the following subsection. For instance, in terms of the longest duration, the RSDP-S model outperformed both the CSDP and RSDP-F models when measured with the mean value across all scenarios with $k_\sigma = 1.5$ (dashed lines in Figure 6.9c). The difference in the ReT_{max} value of the RSDP-S and CSDP models,

0.435 (1.521 – 1.086) months indicates that the longest duration for water deficit may differ by more than 13 days. As the primary water use from this reservoir is for M&I use, failing to provide water for 13 days may result in critical consequences in real systems. Thus, despite the complexity and potential cost that may occur in transforming actual policies, in which robustness is not yet considered, shifting towards robust problem formulations appears to be necessary and may lead to better outcomes when measured with traditional metrics.

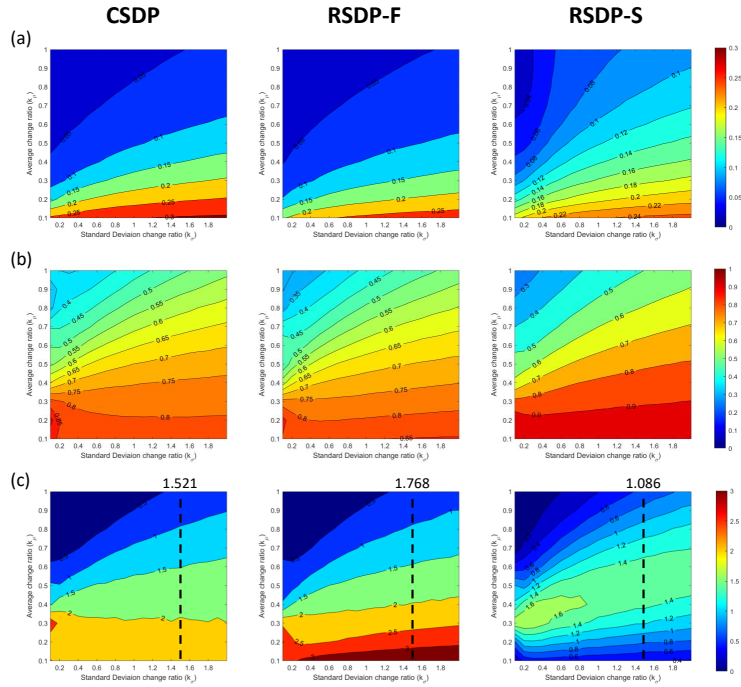


Figure 6.9: Contour plots of traditional metrics across synthetic inflow scenarios with scenarios with decreasing average: (a) ADM_y , (b) ADF_y , and (c) ReT_{max}

Additionally, traditional metrics of the two additional simulations based on the synthetic inflow series with varying lag-1 autocorrelation were also calculated. Figure 6.10 is the calculation results from the inflow series with median correlation of the hindcast data, and Figure 6.11 is the results from the inflow with the maximum average correlation scenario among the forecast data.

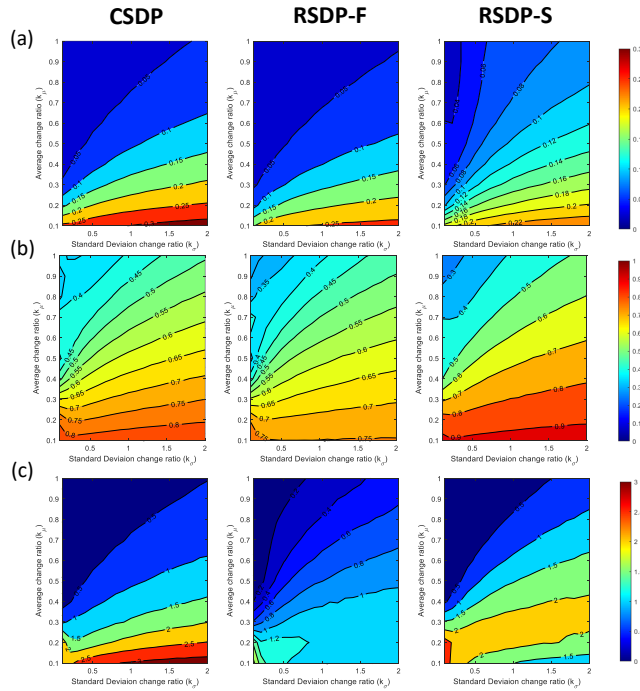


Figure 6.10: Contour plots of traditional metrics across synthetic inflow scenarios generated with the median correlation of the hindcast GCM scenarios (1976-2005): (a) ADM_y , (b) ADF_y , and (c) ReT_{max}

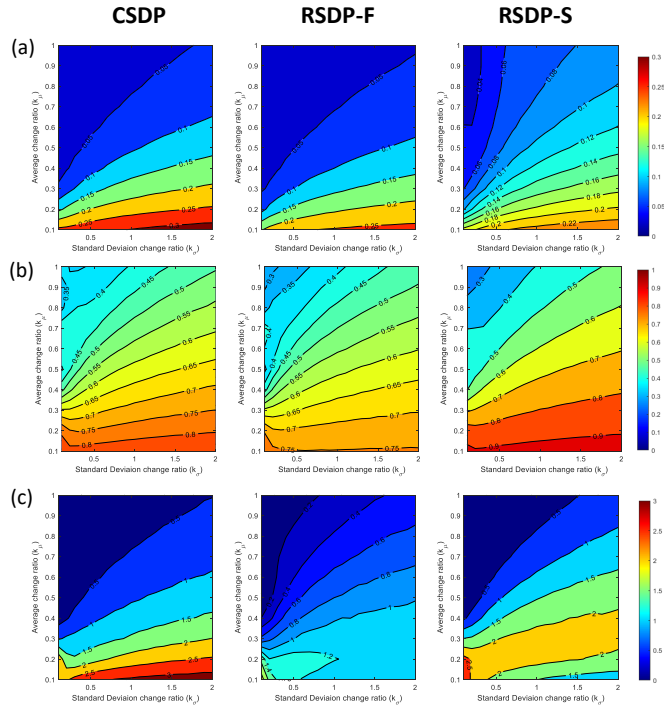


Figure 6.11: Contour plots of traditional metrics across synthetic inflow scenarios generated with the maximum average correlation scenario among the forecast GCM scenarios (2006-2045): (a) ADM_y , (b) ADF_y , and (c) ReT_{max}

Since the results from the two simulations in Figure 6.10 and Figure 6.11 yielded in similar results, their performance is compared with the results from Figure 6.9. Firstly, in terms of magnitude measured with ADM_y , the conclusions from Figure 6.9 is once again stressed: CSDP model performed worst under extreme cases whereas the RSDP-S model performed best among the models. Also, this minute differences among the plots imply that the change in average and standard deviation which affect the absolute amount of inflow into the reservoir affect the performance of the reservoir than the correlation among inflow series, as initially expected. In terms of frequency and resiliency, RSDP-F model yielded the best result among the models. Whereas the results from Figure 6.9 suggested that the RSDP-F model is the worst alternative in terms of ReT_{max} , the results from Figure 6.10 and Figure 6.11 imply that RSDP-F model can also be selected as an alternative under circumstances with varying autocorrelation of inflow series.

6.5.5 How Robust are Robust Policies?

A closer examination of the simulation results was made possible through the use of regret-based robustness metrics, R_i^{model} , which measure the relative difference between the performance in the baseline scenario and the performance under the 90th% percentile scenario. The parallel coordinates plot in Figure 6.12 illustrates the trade-off relationships between different objectives or metrics according to different policies.

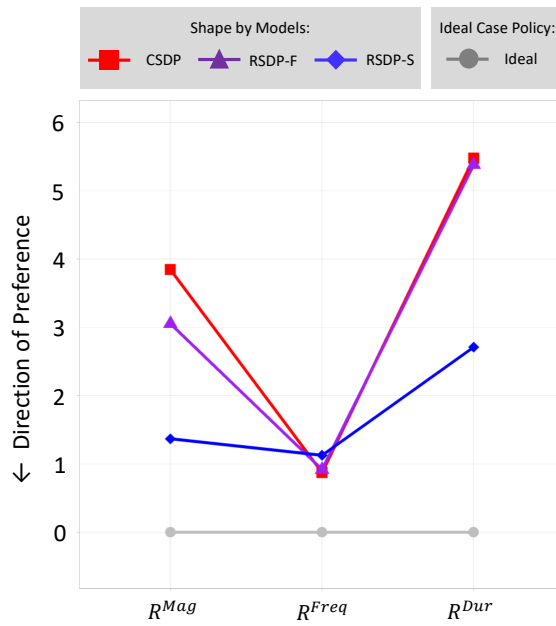


Figure 6.12: Parallel coordinates plot of robustness metrics by model. R^{Mag} is the robustness metric measured in terms of magnitude, R^{Freq} is the robustness metric measured in terms of frequency, and R^{Dur} is the robustness metric measured in terms of duration.

In Figure 6.12, each vertical axis on the plot reports on the robustness of each model in terms of magnitude (R^{Mag}), frequency (R^{Freq}), and duration (R^{Dur}). An ideally robust policy, consequently, would be a straight line at the bottom of the plot, as displayed in a grey line with circles in Figure 6.12. When compared with the models, the CSDP model and RSDP-F model appear to perform similarly in terms of R^{Freq} and R^{Dur} , whereas RSDP-F outperforms CSDP in terms of R^{Mag} . This advantage of the RSDP-F model mainly comes from the inclusion of feasibility robustness into the objective function and the constraint. In contrast, in the RSDP-S model, in which the magnitude of the variance in the main objective is penalized, robustness metrics measured in magnitude and duration unquestionably surpassed those of the other two models. Especially, the difference in R^{Dur} indicates that when compared with the baseline scenario, operating with CSDP policies may lead to two times longer extremes than the results of operating using the RSDP-S rules.

Although the simulation results evaluated via traditional metrics may seem explicit to determine which model performed better, the robustness metrics lead to the conclusion that operation under RSDP-S policies would yield far more robust outcomes than operating with conventional policies that do not include any concept of robustness into the problem formulation processes. Nevertheless, as the crossings in lines displayed in Figure 6.12 indicate a trade-off relationship between the three robustness metrics, the selection of models according to stakeholder preference is also possible. From another perspective, the results of the evaluation using the robustness metric also indicate that the current conventional policies can be extremely vulnerable to potential extremes, again highlighting the necessity of shifting away from conventional decision making strategies. Indeed, the fact that the two RSDP models are able to utilize more storage

stocks than the CSDP model should also be stressed once more since using storage below LWL can be risky depending on the available infrastructure of the reservoir. However, since the target reservoir of this study is equipped with concrete plans to utilize storage below LWL based on actual policies, it can be concluded that the results from the proposed methodology hold promise to be implemented under future droughts.

6.6 Conclusion

This study adds RO into the problem formulation process of deriving optimal and robust reservoir operation policies to confront the non-stationarity of climate change. Due to SDP's popularity and wide applicability in actual reservoir operation problems, key concepts of RO were incorporated into both the objective function and the constraints of SDP. The proposed methodology was applied to deriving release policies for a multipurpose dam located in Korea, the Boryeong Dam, of which the main objective is to supply M&I water to nearby cities and districts. Through relating the theoretical concepts with actual policies, this study was able to derive three sets of policies for comparative analysis. The simulation of conventional and robust policies was conducted under a wide range of climate conditions, and the results indicate that RSDP policies, especially RSDP-S model in which the solution robustness is incorporated into the model formulation, outperform CSDP policies in both traditional and robustness metrics in extreme climate conditions. Specifically, electing RSDP-S model would result in half duration of water deficit than that of CSDP model in terms of regret-based robustness metrics. Thus, the results of this study once again highlight the fundamental concept of decision making strategies under climate change: a separate set of strategies should be contemplated and adopted after the analysis of their strengths and weaknesses (Watkiss and Hunt 2013). Nevertheless, since the current analysis only reflects a constrained range of future with decreased inflow, the approach in this study could be further expanded to better reflect the projected future GCM scenarios where the mean inflow increases.

This study was the first attempt in which the concept of RO proposed by Mulvey

et al. (1995) was linked to SDP in a reservoir operation problem. Also, through the inclusion of actual discrete hedging policies into both optimization and simulation procedures, this study was able to suggest sets of candidate policies that are applicable to reservoirs in the actual. Nevertheless, multiple potential extensions of this research are possible, such as connections to finance or authority sides of the system, especially shadow prices in water finance (Molinos-Senante et al. 2010; Bierkens et al. 2019) or power asymmetries triggered by different interest groups (Denaro et al. 2018), which can be closely related to the proposed methodology and case study. Regarding the algorithmic implications of considering broader sources of problem complexity (e.g., additional state variables, exogenous information feedbacks, large numbers of conflicting objectives, stochastic simulation, etc.), more recent multi-objective policy-based control strategies hold promise (Giuliani et al. 2016a; Quinn et al. 2019).

Chapter 7. Conclusion & Future Research

7.1 Summary & Conclusion

This dissertation contributes to unravel three of the conundrums which remain unsolved in the field of water resources systems management: (1) intricacy in water resources systems analysis, (2) limited collaboration with stakeholders, and (3) climate change-induced stresses affecting WRSA. In response, the three discrete studies conducted in this dissertation specifically (1) comparatively analyzed how different algorithms or stochasticity of the conditions affect the quality of generated solutions, (2) proposed how to improve the onsite policy to perform under multi-year droughts with adaptive measures, and (3) coupled the robust optimization concept with the optimization model and evaluates their performance under a wider range of inflow scenarios with both traditional and regret-based robustness indices. All the proposed policies throughout the dissertation aimed to enhance the practicality in onsite reservoir operations. Thus, the proposed policies were linked with the actual conditions and drought response measures of the target reservoir. Ultimately, it is expected that the collective results from this dissertation to commit toward both in the field of academia and also in practice.

In the first experiment, the reservoir optimization problem with conflicting objectives was designed and six optimization models, three based on DP and EMODPS, were developed for a comparative analysis. The formulations of the DP models mainly varied in terms of the available information: DDP-Perfect model assumed perfect forecast during the optimization horizon, DDP-Average with average inflow, and SDP assumed

lognormal distribution if the inflow series. The EMODPS models utilized different shapes of the approximating policy functions (radial basis functions, RBFs) each with different number of parameters. Finally, one of the solutions from the EMODPS model which yielded the best result closest to the ideal policy was coupled with the on-site hedging policy of the Boryeong multipurpose reservoir. Among the optimization models, the EMODPS model with Gaussian shape of RBF outperformed the other optimization model in terms of solution diversity and optimality. The results suggest that the shape of the RBF is a more decisive factor than the number of parameters to be optimized. Moreover, the coupling of hedging rule with the EMODPS-Gaussian model revealed a trade-off relationship especially between magnitude and the frequency in terms of water deficit. The magnitude of failure measured from the demand side improved at the expense of degradation of frequency. Given that the objective of hedging is to reduce the risk of failures with immense magnitude, coupling of hedging rule with the optimization model may be utilized in future reservoir managements under drought conditions.

The second portion of this dissertation primarily aims to improve the zone-based hedging rule in response to recent multi-year droughts with adaptive measures. A total of seven synthetic inflow series each with different magnitude of long memory were generated using the ARFIMA model. Since the generated synthetic inflow series were composed of annual time series, they were disaggregated into 10-day by the k-NN disaggregation method. The algorithm for the hedging rule generation utilized onsite, which is based on iterative calculation of reservoir simulations, was used to derive seven cases of alternative hedging rules with the inflow series including long-term persistence. The alternative rules were simulated under adaptive measures, in which

this dissertation proposed as one of the pillars against climate change. The adaptive operations were triggered with the SRI values of the reservoir, which was continuously updated during the simulation horizon. When compared with the operations under the SQ policy, the proposed adaptive policy with alternative hedging rules yielded favorable results from the reservoir operators. On the other hand, the performance evaluated in terms of water deficit showed a similar result from the first case study, leading to more frequent failures with less magnitude. Since the original intention of hedging rules is to prevent critical failures, the adaptive measures proposed in this study hold promise under future reservoir operations with multi-year droughts.

Finally, the second pillar in water resources systems management against climate change, a robust strategy that perform well over a wider range of uncertainty, was coupled with the optimization algorithm, SDP. A novel framework named RSDP that integrates RO during the formulations of either the objective function or the constraints in SDP, is proposed. Whereas the first RSDP model, RSDP-F model only includes the concept of feasibility robustness, RSDP-S also includes the concept of solution robustness into the model. With the CSDP model, the three SDP framings were evaluated under a wide range of future inflow series assuming deep uncertainty. The evaluations of the three models were conducting using both traditional metrics based on the concept of risk, resiliency, and vulnerability, and regret-based robustness measures. The two RSDP models performed relatively well than the CSDP model under extreme future climate scenarios with decreasing average and increased variance, owing to the feasibility robustness concept which enables the use of emergency storage below the low water level. The results from this case study suggest that whereas the CSDP model may be preferable under normal or week drought conditions, RSDP models can also

be considered as possible candidate solutions under extreme climatic conditions when evaluated under traditional and robustness metrics.

The three case studies conducted in this dissertation all aim to present novel water resources management schemes applicable onsite under the deep uncertainty and non-stationarity of climate change. Since the direct application of the optimally derived policy from an optimization algorithm is hardly possible, the case studies presented in this dissertation aimed to target the onsite reservoir operations by continuously referring to the actual policies of the target reservoir. The first and second case studies included the actual zone-based hedging rules during the reservoir simulations, and the feasibility robustness concept in the RSDP model utilized the emergency storage both during optimization and simulation. The results from this study are to be summarized and organized so that even the non-experts may interpret, and to be delivered to onsite reservoir managers who are responsible in making related decisions. Given that the target reservoir, Boryeong multipurpose reservoir, is continuously experiencing water shortfalls, the results presented in this dissertation may serve as possible solutions for sustainable water managements. Ultimately, since the application of the problem framing process presented in this dissertation is not excluded into the Boryeong reservoir, it can also be extended to other water resources management problems globally.

7.2 Potential Future Research

Despite the notable contribution made from the results from this dissertation, multiple candidate topics exist for the subjects of future studies. In this section, potential topics for future research are presented by emphasizing their connections with the studies conducted in this dissertation.

The first and the second future research topics stem from the SQ zone-based hedging rule: (1) inclusion of hedging rules during the optimization process and (2) improvement of the hedging rule generation algorithm, respectively. Whereas the studies from this dissertation utilized the zone-based hedging rules either during simulation or tried to include long memory, the hedging rules may also be included more interactively especially during the optimization step. It is expected that the inclusion of hedging during optimization would yield in optimal solutions which meet the needs from the actual policy makers. Also, the current rule generation algorithm primarily based on iterative reservoir simulation can also be further reformed with novel optimization schemes.

Furthermore, the components from climate change adopted in this study can also be upgraded in future research. The studies from this dissertation tackled the two characteristics from climate change, non-stationarity and deep uncertainty, with robust and adaptive decision making techniques during reservoir operations. Especially, the climate change scenarios utilized during the simulation and evaluation processes in Chapter 6 can be further advanced in multiple ways. For instance, the information from the GCM scenarios may be used either in the objective function or input state variables of the optimization algorithm. On the other hand, since the GCMs are frequently

utilized in impact assessments from climate change at a regional scale (Eum and Cannon 2017), coupling the GCMs during the adaptive and robust water resources management techniques will enable more accurate impact assessment from climate change. In the future, expanding the climate change scenarios which include the information from the socioeconomic factors, such as Shared Socioeconomic Pathways (SSPs) or a combined scenario of the two such as Coupled Model Intercomparison Project version 6 (CMIP6), may also hold promise.

Nevertheless, the expansion of both the scope and the formulation process of the target problem so that the application of the MORDM framework (Kasprzyk et al. 2013; Hadka et al. 2015) is feasible and it will provide the related decision makers with a wider range of candidate policy actions in which they will be able to choose from. MORDM, in which the concept of robust decision making is coupled with *a posteriori* decision making supporting tools, will improve not only the quality of the solutions, but also will specify the problem without having to make unnecessary assumptions during the problem formulation and solving processes. Also, in terms of algorithmic approach of the problem, a deeper analysis exploring multiple MOEAs will theoretically yield a more thorough set of optimal solutions. Since the concept of robustness adopted in this study were based on the feasibility and solution robustness concept proposed by Mulvey et al. (1995), a further analysis regarding the adequate definition of robustness, such as the inclusion of future value function in SDP models, in reservoir operation problems also remains as a promising research topic.

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Appendix: Code for Hedging Rule Generation

DrResOPT Code (Code in R)

```
1 library(ramify)
2
3 nk = 100000
4 nt = 36
5 nstg = 3
6 ncol = 8
7 n365 = 365
8
9 iyear = 500
10 nmin = 15
11 reliability = 95
12 st_lwl = 6.2
13 st_flood = 108.7
14 st_nwl = 108.7
15
16 n = iyear*nt
17 demand = read.csv('BR_Demand.csv',header = F)
18 dem_dip = demand[,1]
19 dem_agr = demand[,2]
20 dem_riv = demand[,3]
21
22 nday = c(10,10,11,10,10,8,10,10,11,10,10,10,10,10,11,10,
23 10,10,10,10,11,10,10,11,10,10,10,10,10,11,10,10,10,10,10,11)
24
25 D2 = vector("double",length=iyear*nt)
```

```

26 D3 = vector("double",length=iyear*nt)
27 D4 = vector("double",length=iyear*nt)
28 i10d = vector("double",length=iyear*nt)
29
30 river_rdc = vector("double",length=nt)
31 river_rdc = 2.7*nday/100
32 agri_rdc_1 = vector("double",length=nt)
33 agri_rdc_1 = c(0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0.171,0.171,0.171,0,0,0,0,0,0,
34 0.006,0.006,0.006,0,0,0,0,0,0,0,0,0)
35
36 conduit = vector("double",length=nt)
37 conduit = 11.5*nday/100
38
39 agri_rdc_2 = vector("double",length=nt)
40 agri_rdc_2 = c
      (0,0,0,0,0,0,0,0,0,0.09,0.09,0.09,0.107,0.107,0.11177,0.338,0.338,0.338,
41      0.255,0.255,0.2805,0.305,0.305,0.3355,0.182,0.182,0.182,0,0,
42      0,0,0,0,0,0)
43
44 dip_rdc = vector("double",length=nt)
45 dip_rdc = 1.18*nday/100
46
47 for (i in (1:iyear)){
48   for (j in (1:nt)){
49     D2[(i-1)*nt+j]=dem_dip[j]+dem_agr[j]+dem_riv[j]
50     D3[(i-1)*nt+j]=dem_dip[j]+dem_agr[j]-river_rdc[j]-agri_rdc_1[j]
51     D4[(i-1)*nt+j]=dem_dip[j]-conduit[j]-agri_rdc_2[j]-dip_rdc[j]
52     i10d[(i-1)*nt+j]=j
53   }
54 }
55
56 source("DrResOPT_cal.R")
57 st_result = matrix(NA, 36,100*4)
58 inflow_case = read.csv('Input_Q/Q.csv',header = F)

```



```

59
60 for (i in (1:100)) {
61   flow = inflow_case[,i]
62
63   sto1 = rep(NA,n)
64   sto2 = rep(NA,n)
65   st_final = rep(NA,36)
66   demand=D2
67
68   st_result[, (i-1)*4+1] = DrResOPT_cal()
69
70   sto1 = rep(NA,n)
71   sto2 = rep(NA,n)
72   st_final = rep(NA,36)
73   demand=D3
74
75   st_result[, (i-1)*4+2] = DrResOPT_cal()
76
77   sto1 = rep(NA,n)
78   sto2 = rep(NA,n)
79   st_final = rep(NA,36)
80   demand=D4
81
82   st_result[, (i-1)*4+3] = DrResOPT_cal()
83
84   st_result[, (i-1)*4+4] = st_result[, (i-1)*4+1] + 15*(dem_dip+dem_agr+dem_riv)
      /nday
85 }
86
87 DrResOPT_cal <- function() {
88
89   for (i in (1:nt)){
90     sto_act = (st_nwl-st_lwl)*10
91     maxit = round(sto_act)

```

```

92
93     for (j in (1:maxit)){
94         sto1[i] = st_lwl + (j-1)/10
95         sto2[i] = sto1[i]
96
97         for (k in ((i+1):n)){
98             sto1[k] = sto2[k-1]+flow[k]-demand[k]
99             if(i10d[k]==i){sto1[k]=sto1[i]}
100
101             if ((i10d[k]>=18)&&(i10d[k]<=26)){
102                 if (sto1[k]<st_flood){sto2[k]=sto1[k]}
103                 else {sto2[k] = st_flood}}
104
105
106             if(sto1[k]<st_nwl){sto2[k]=sto1[k]}
107             else {sto2[k] = st_nwl}
108         }
109
110         icount=0
111         for (k in 1:(iyear-1)){
112             isum=0
113
114             for (kj in 1:nt){
115                 if(sto2[(k-1)*nt+kj+i-1]<=st_lwl){
116                     ick=1
117                     isum=(isum+1)
118                 } else {ick=0}
119             }
120             if(isum>=1) {icount=(icount+1)}
121         }
122
123         com_rel = (1-(icount/(iyear-1)))*100
124
125         if(com_rel>reliability){break}

```

```
126     }  
127     st_final[i]=sto1[i]  
128 }  
129 return(st_final)  
130 }
```


국문초록

기후변화의 비정상성과 깊은 불확실성 하에서 효과적이고 지속 가능한 수자원 관리는 전통적인 수자원 관리 기법으로는 달성하기 어렵다. 따라서, 성공적인 수자원 관리를 위해 최적화 알고리즘과 기존 정책을 포함한 새로운 의사결정 방식이 개발되어 기후변화 적응을 위해 적용되고 있다. 이러한 맥락에서 본 논문은 적응형 및 강건한 저수지 운영 기법을 한국의 보령 다목적댐에 적용하고, 새로운 저수지 운영 방법을 제안하고자 한다.

가장 먼저, 저수지 운영을 위한 최적화에 쓰이는 저명한 두 가지 알고리즘인 동적 프로그래밍(dynamic programming, DP) 모형과 진화적 다목적 직접 정책 검색(evolutionary multi-objective direct policy search, EMODPS) 모형의 비교 분석을 위한 연구가 진행되었다. 구성된 DP 모형들은 각각 다른 예보 정보를 제공받았으며, EMODPS 모형들은 이를 구성하는 근사함수의 형태와 추정이 필요한 변수의 개수를 차별화하였다. 그 후, 최적해의 실용성을 개선하기 위해 최적화 모형 중 하나를 현장에서 사용되는 헤징 룰과 결합하였다. 현장에서 사용되는 헤징 룰은 이후 장기 기억을 가지는 시계열을 바탕으로 대상 유역이 최근 경험한 다년 가뭄을 반영하여 개선되었다. 그런 다음 새로운 정보를 통해 다음 시점의 결정이 지속적으로 업데이트되는 적응형 정책을 새로 생성한 대안들을 활용하여 실현하였다. 마지막으로 ‘feasibility robustness’와 ‘solution robustness’의 개념이 추계학적 DP 모형과 결합되어 적용되었다. 새로 생성한 강건한 정책들은 기후변화의 깊은 불확실성에 대비하기 위해 임의로 생성된 더 넓은 범위의 유입량 시나리오 하에서 평가되었

다. 강건한 정책들은 횡수, 기간, 크기의 전통적인 평가 지표와 후회도 기반 강건성 지표의 측면에서 평가되었으며, 그 결과를 정량화시켰다.

최적화 모형 중 가우시안 형태의 근사함수를 사용하는 EMODPS 모형(EMODPS-Gaussian)이 최적화해야 할 매개변수의 수가 가장 적음에도 불구하고 가장 좋은 성능을 보였다. 또한 해징 물과 EMODPS-Gaussian 모형을 결합한 모형과 해징 물 생성 과정에서 장기 기억을 포함한 결과는 공통적으로 수요 측면에서의 더 빈번한 실패를 경험하는 대신 공급 측면에서 더욱 유리한 결과를 나타냈다. 특히, EMODPS-Gaussian 모형과 해징 물의 결합은 최대 실패 기간을 45.59% 감소시켰다. 새로운 해징 물을 바탕으로 적응형 저수지 운영도 수요자 측면에서의 치명적인 실패의 빈도를 17.7% 줄이는 결과를 보였다. 마지막으로, 솔루션 강건성 개념을 포함하는 강건한 정책은 후회도 기반 강건성 지표로 측정할 때 기존 모형과 가능성 강건함을 포함한 최적화 모형의 모의 결과를 모두 능가했다. 궁극적으로, 본 논문에서 제시하는 적응형 및 강건한 저수지 운영 계획은 특히 극한 기후 조건에서 기존 의사 결정 기술보다 더 나은 결과를 보이기 때문에 미래 기후변화에 대비하여 새로운 의사 결정 과정을 바탕으로 한 수자원 관리로의 전환이 필요함을 확인하였다.

주요어: 최적화, 저수지 운영, 실제 정책, 적응형 운영, 강건한 최적화, 기후변화

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